WEEKLY ROLLUP • PhD RESEARCH WATCH

Particulate Matter
Week in Review

Week ending 7 November 2026 | 1 November 2026 to 7 November 2026 | 151 records pooled from 7 daily issues

151
RECORDS POOLED
(151 UNIQUE DOIs)

40%
MEASUREMENT SIDE
(61 RECORDS)

11
LOW-COST SENSOR
RECORDS (7%)

53
POOLED EFFECT
ESTIMATES (53 WITH A CI)

24
ABSTRACT-FREE
(16% OF WEEK)

The week in one paragraph

Volume rose to 151, the measurement side recovered to 40 %, and the low-cost sensor stopped being the object of study and became an input: 33 rural nodes assimilated into WRF-Chem for Delhi's stubble-burning season, a statewide Bihar network carried to attributable mortality, a metrology-chain network calibration that worked for temperature and humidity and failed first on PM. Eleven records rest on a low-cost sensor, against 15 in the whole of October. On the health side, the week's strongest designs were a three-day controlled human exposure, a life-course constituent analysis in UK Biobank and a deeply phenotyped diabetes cohort — and none of their exposures came from a sensor.

Scale. 151 records over seven issues on daily totals of **7**, 19, **28**, 18, 25, 26 and **28**. Against recent rollups: 151 after 135, 119, 123, 87, 125 and 162. **All ten subtopics populated** for the seventh week running.
The measurement side rose to 40 % (61 records): *Exposure assessment & modelling* 34 → **40**, *Sensing* 16 → **21**. Share series 42, 44, 39, 39, 37, **40 %**.
Occupational & indoor led the health side's growth: 14 → **20** (coal-miner PMF and CWP, metro track maintenance, welding, cannabis grow rooms, facility-wide personal monitoring). *Cardiovascular* recovered 6 → 9; *Respiratory* 17 → 16, *Neuro* 8 → 7, *Other clinical* 8 → 6, *Reproductive* 7 → 6.
Estimates: 53, every one with an interval (seventh complete week at 100 %), one per 2.8 records, the densest week of the watch. 48 on ratio scales (OR 13, HR 10, PR 8, RR 7, aOR 6, GMR 4) and 5 on difference scales (3 mmHg, 2 % FEV_1).
Abstract gap 24 of 151 (16 %), from 17 %. **Tier mix 20 A, 52 B, 79 C** — tier A at **13 %**, from 12 %.

Subtopic trend and inflection

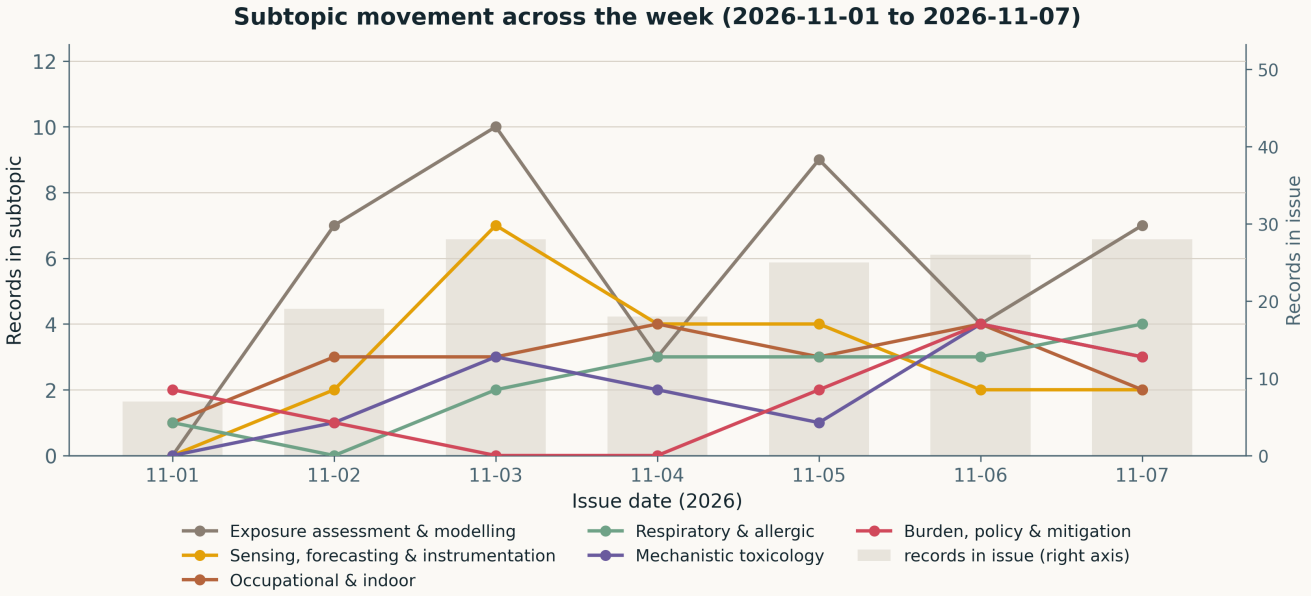


Fig. Per-issue subtopic counts across the week from `state/metrics.csv`, seven issues, daily totals 7, 19, 28, 18, 25, 26 and 28. Exposure assessment swings between 3 and 10 per issue; the Sunday (1 November) floor of 7 is the week's minimum. The 1 November daily was built on the 2 November run; 2–5 November on three late runs between 4 and 6 November; **6 and 7 November and this rollup on the 10 November recovery run**, after no scheduled run completed on 7–9 November.

Expanding. *Occupational & indoor* 14 → **20** is the largest gain and carries the week's largest ratios: in 30,319 US coal miners, progressive massive fibrosis prevalence ran 10 (5.7–17.5) times higher in Central Appalachia than elsewhere and 3 (2.3–4) times higher in continuous-miner operators. *Sensing* 16 → 21 is the other gain, and the composition of that band changed more than its size.

Contracting. Nothing fell by more than two records; *Other clinical* 8 → 6 is the largest fall.

The inflection worth naming. Last week the low-cost sensor was an instrument to be corrected; this week it is a data stream to be used. The 3 November CUPI-G preprint assimilated 33 rural nodes into WRF-Chem and cut Haryana mean fractional bias from −116.7 % to −56.8 %, recovering the Severe AQI category in about 95 % of Delhi cases where the control run called none. The 5 November *Nature Communications* paper carried a dense Bihar network through ML gap-filling to district attributable mortality (3.2-fold spread). The 4 November SensEURCity/SMEAR III network calibration found PM the hardest quantity to transfer node-to-node, because PM signals are localised and weakly correlated. Kinshasa (PurpleAir and asthma), Attica (smart benches), a 37-location school vape-detection network and a κ-Köhler humidity correction for the PA-II fill out the band. **The weakness is uniform**: none of these abstracts reports the propagated sensor uncertainty in the downstream quantity — assimilated fields, attributable deaths or exposure contrasts.

Last week's test (a smoke-specific estimate on measured network data, or a smoke-vs-non-smoke contrast with composition) **was not met**. The week's smoke evidence — All of Us smoke and adult asthma (OR 1.38, 1.32–1.44, at the highest lag), California smoke disparities, wildfire $PM_{0.1}$ in mice, the CMAQ evaluation showing 1,273 of 2,151 observed $>35\,\mu g/m^3$ smoke site-days missed — rests on models and regulatory monitors.

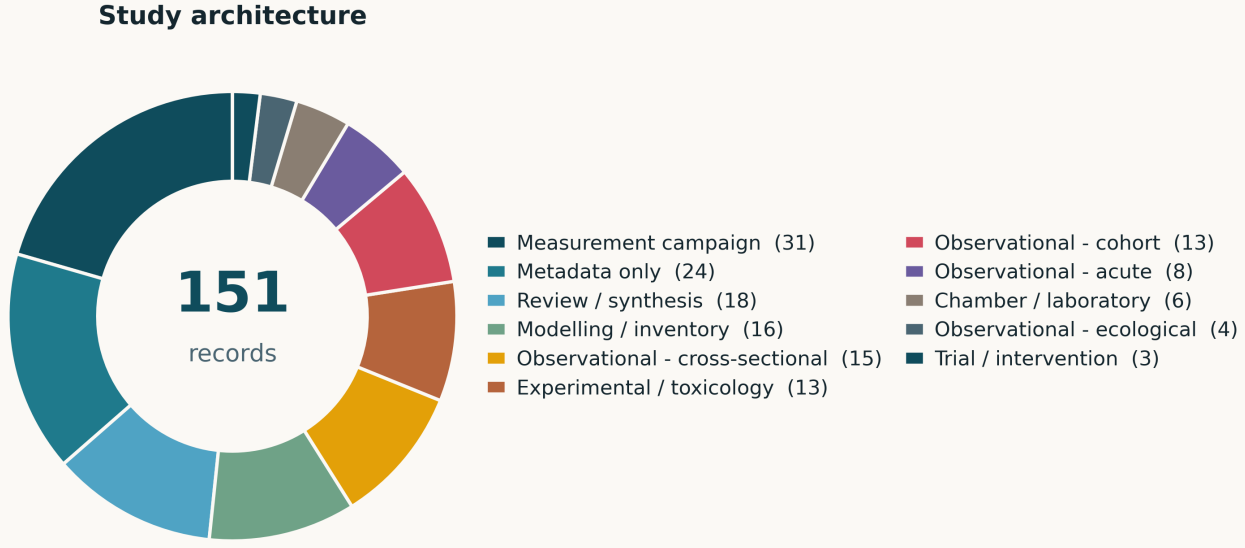


Fig. Study architecture over the pooled week. *Measurement campaign* 31 and *Metadata only* 24 lead, then *Review / synthesis* 18, *Modelling / inventory* 16, cross-sectional 15, cohort and experimental 13 each, acute 8, chamber 6, ecological 4. **Trial / intervention 3**: the EPA three-day controlled exposure (FEV_1 −1.9 % on day 2, −1.8 % on day 3), the Beijing clean-heating difference-in-differences and a paired intervention pilot.

Pooled analytics

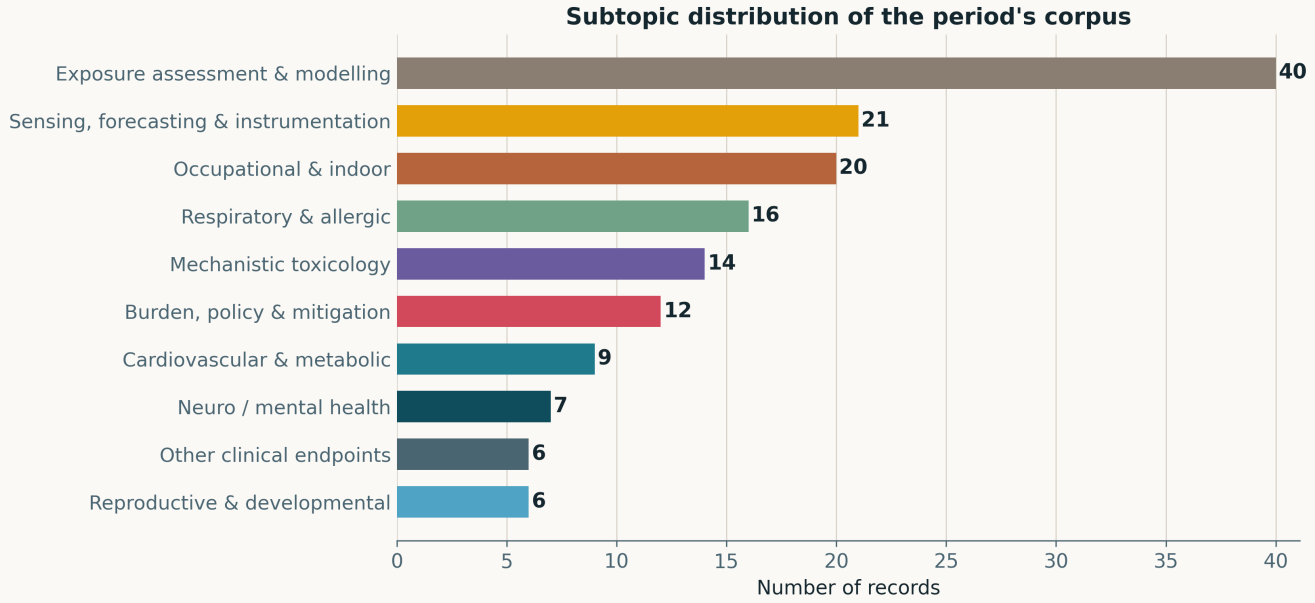


Fig. Pooled subtopic distribution, all ten bands populated: *Exposure* 40 and *Sensing* 21 put the measurement side at **61 of 151** (**40 %**). On the health side *Occupational & indoor* 20 leads, then *Respiratory* 16, *Mechanistic toxicology* 14 and *Burden* 12; *Cardiovascular* 9, *Neuro* 7, *Reproductive* and *Other clinical* 6 each. The endpoint chart (f2b) is omitted at rollup volume; pneumoconiosis prevalence, sinonasal malignancy, hospital-acquired pneumonia and ovarian hyperandrogenism are new endpoint classes for the watch.

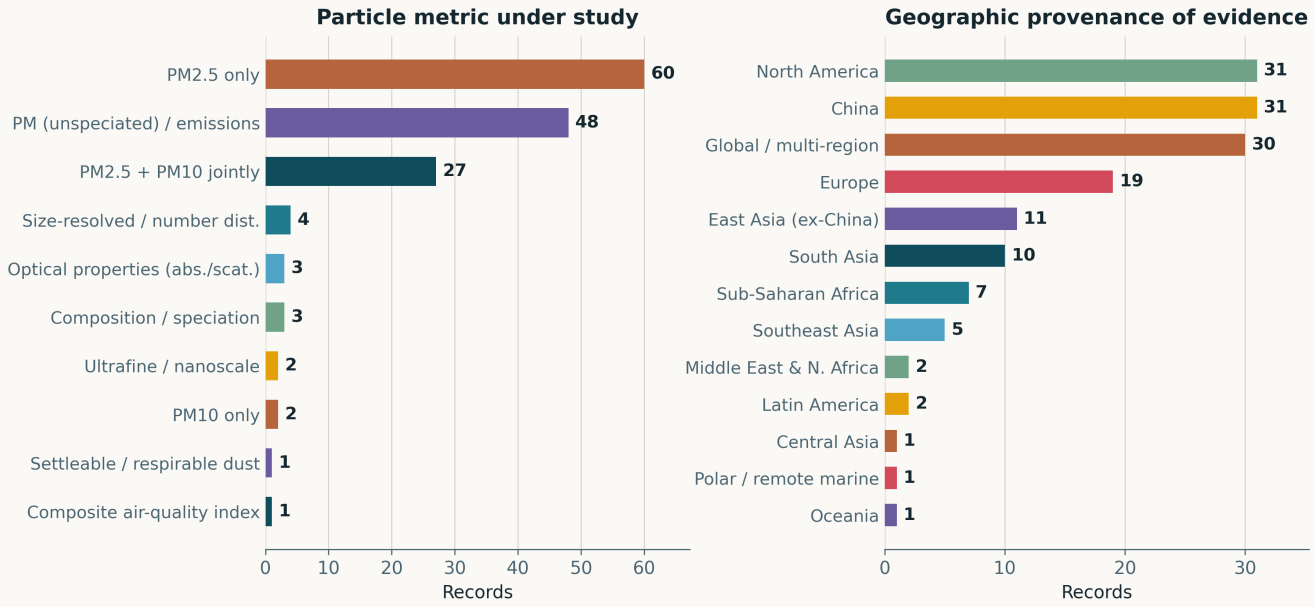


Fig. Left — particle metric: $PM_{2.5}$ alone 60, unspeciated PM or emissions 48, $PM_{2.5}$ and PM_{10} jointly 27; size-resolved, composition, optical and ultrafine metrics together 12. Right — geography: China and North America tie at 31, *Global / multi-region* 30, Europe 19, East Asia ex-China 11, South Asia 10, sub-Saharan Africa 7, Southeast Asia 5.

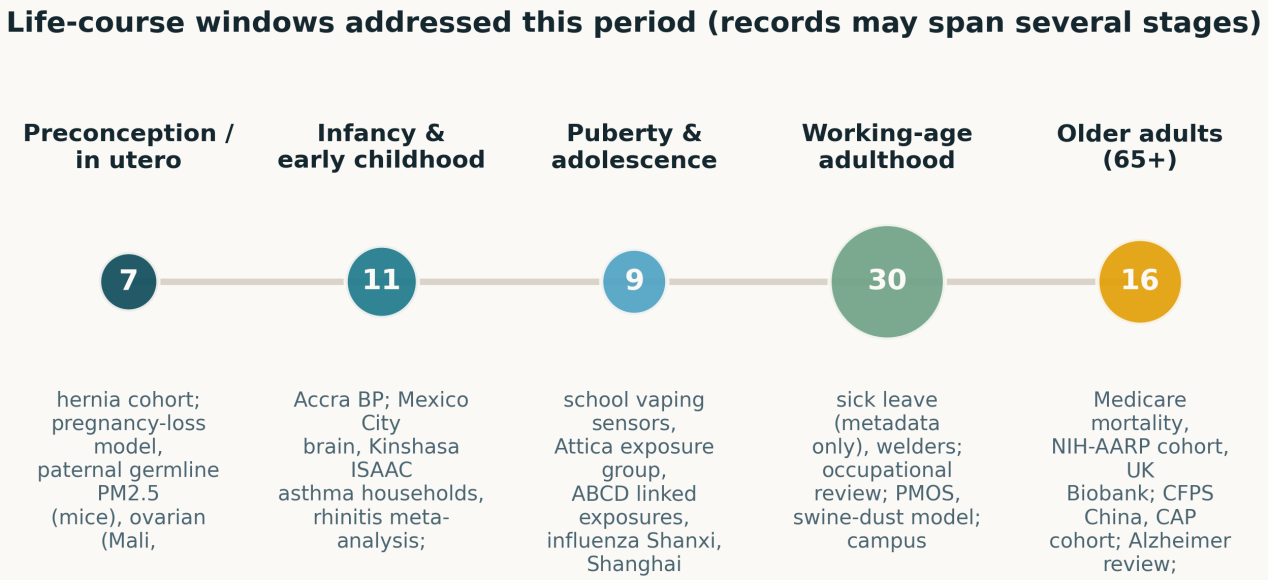


Fig. Life-course windows across the pooled week; counts non-exclusive. Working age leads, carried by the occupational cluster, the controlled exposure and the diabetes and NIH-AARP cohorts; preconception and in utero are carried by animal reproductive work, follicular-fluid metals, Korean offspring hernia/hydrocele and the Generation R epigenetic-age analysis.

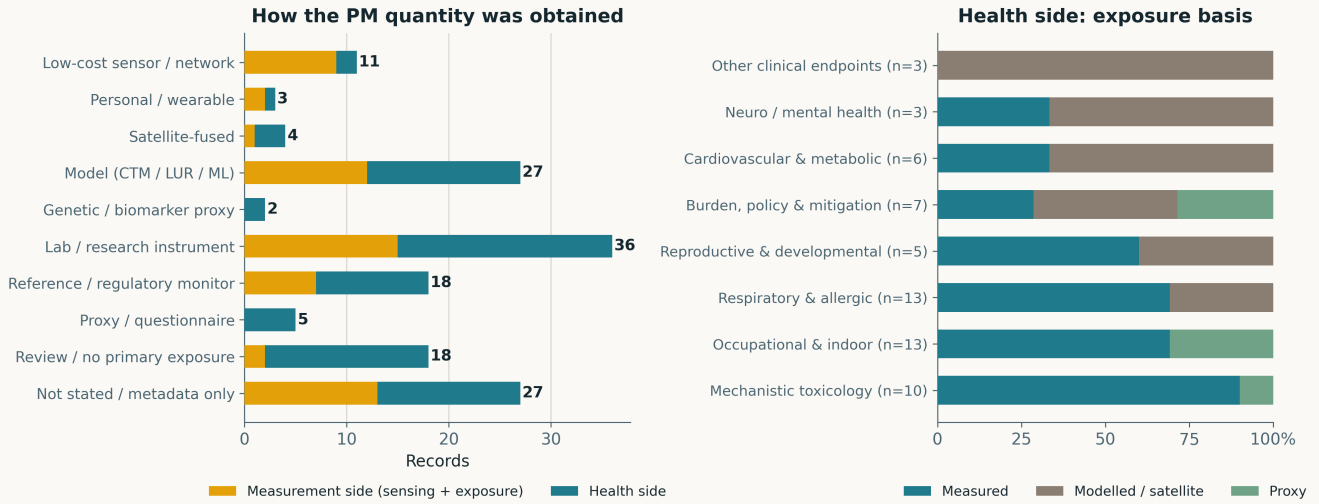


Fig. How each record obtained its PM quantity, classified from the exposure field. 11 of 151 records rest on a low-cost sensor or sensor network; 18 on reference or regulatory monitors. Health side, where an exposure basis is recoverable (n=60): measured 58 %, modelled or satellite-fused 30 %, proxy 12 %. Right panel sorted by modelled share: the subtopics at the bottom are where a measured low-cost network exposure would add most.

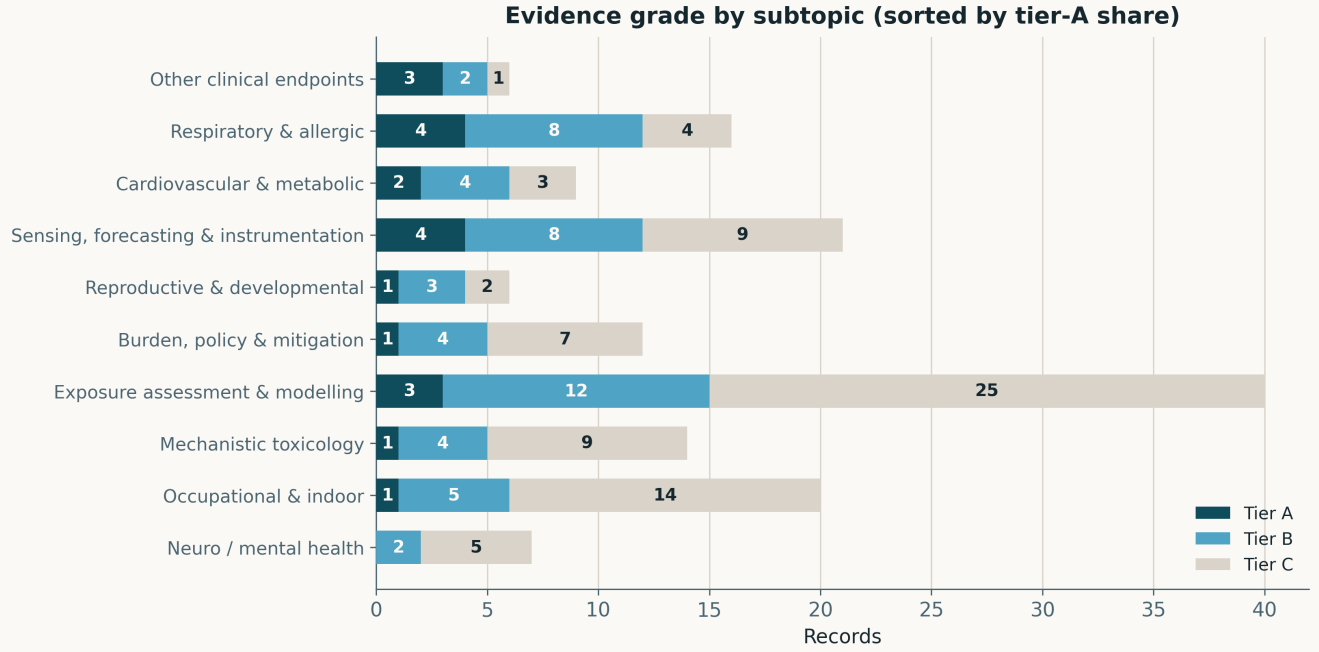


Fig. Tier mix per subtopic. 20 of 151 records are tier A (13 %). Highest tier-A share: *Other clinical endpoints* (3 of 6).

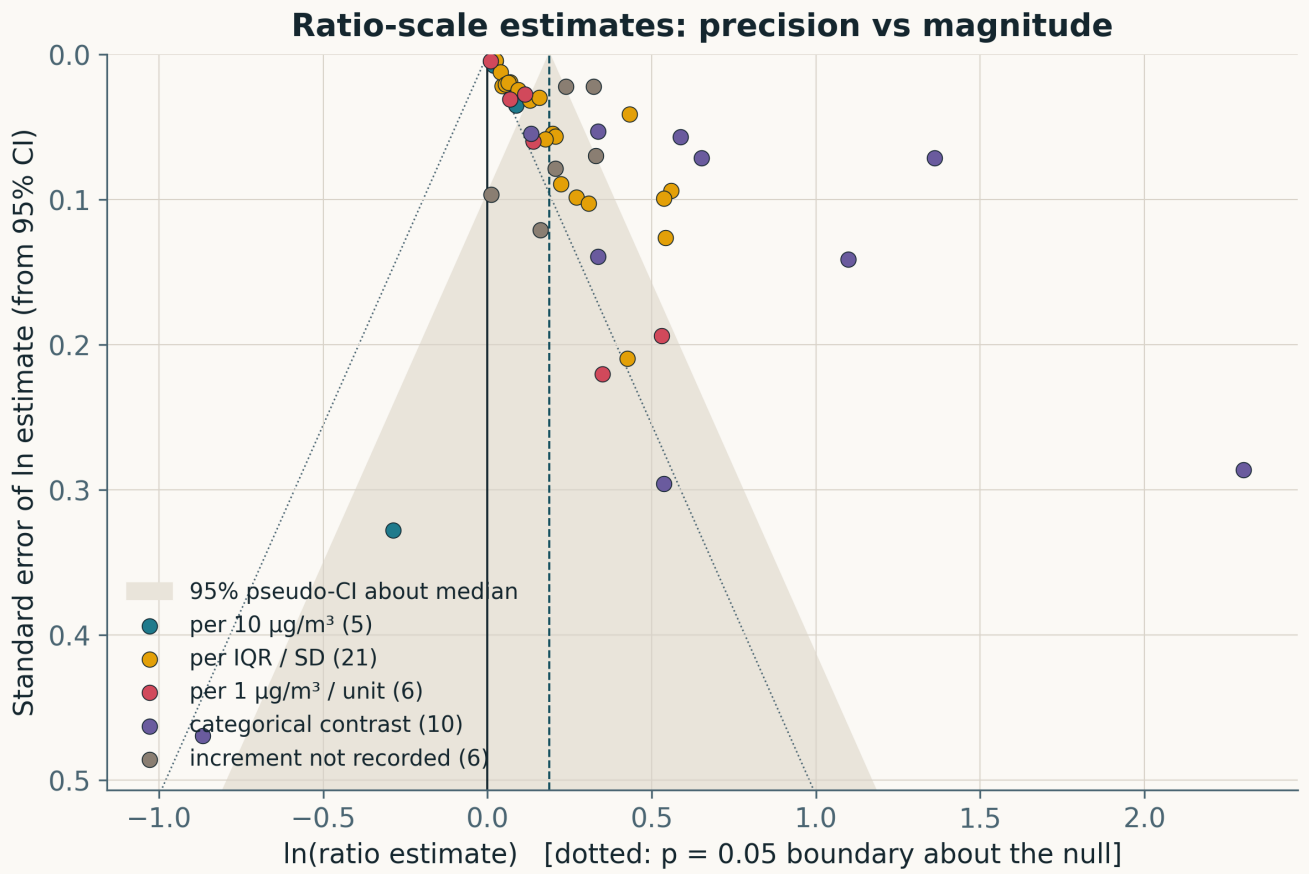


Fig. Every ratio-scale estimate (n=48) plotted as ln(estimate) against its standard error, coloured by the exposure increment the authors reported. 42 of 48 exclude the null. Increments: per IQR / SD 21, categorical contrast 10, per 1 $\mu g/m^3$ / unit 6, increment not recorded 6, per 10 $\mu g/m^3$ 5. Because increments differ (per 1 unit, per 10 $\mu g/m^3$, per IQR, categorical), the spread is not a heterogeneity estimate and the median line is not a pooled effect; the plot shows whether precision and magnitude co-vary. Egger regression intercept 4.38 (SE 0.66, normal-approx. p=0.000, n=48) — asymmetry: small imprecise estimates sit on the harmful side.

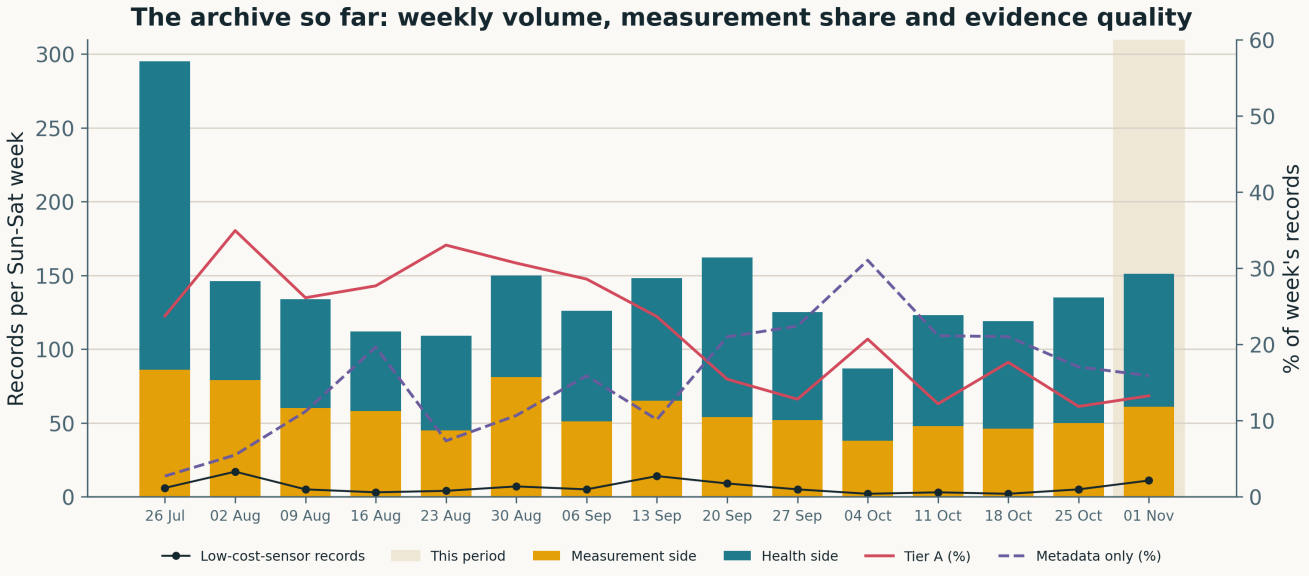


Fig. All 15 Sunday–Saturday weeks in the archive up to 2026-11-07 (2122 records). This period (shaded): measurement side 40 % of 151 records, versus 41 % across the archive. Tier-A and metadata-only shares on the right axis; partial first and last weeks are drawn as recorded.

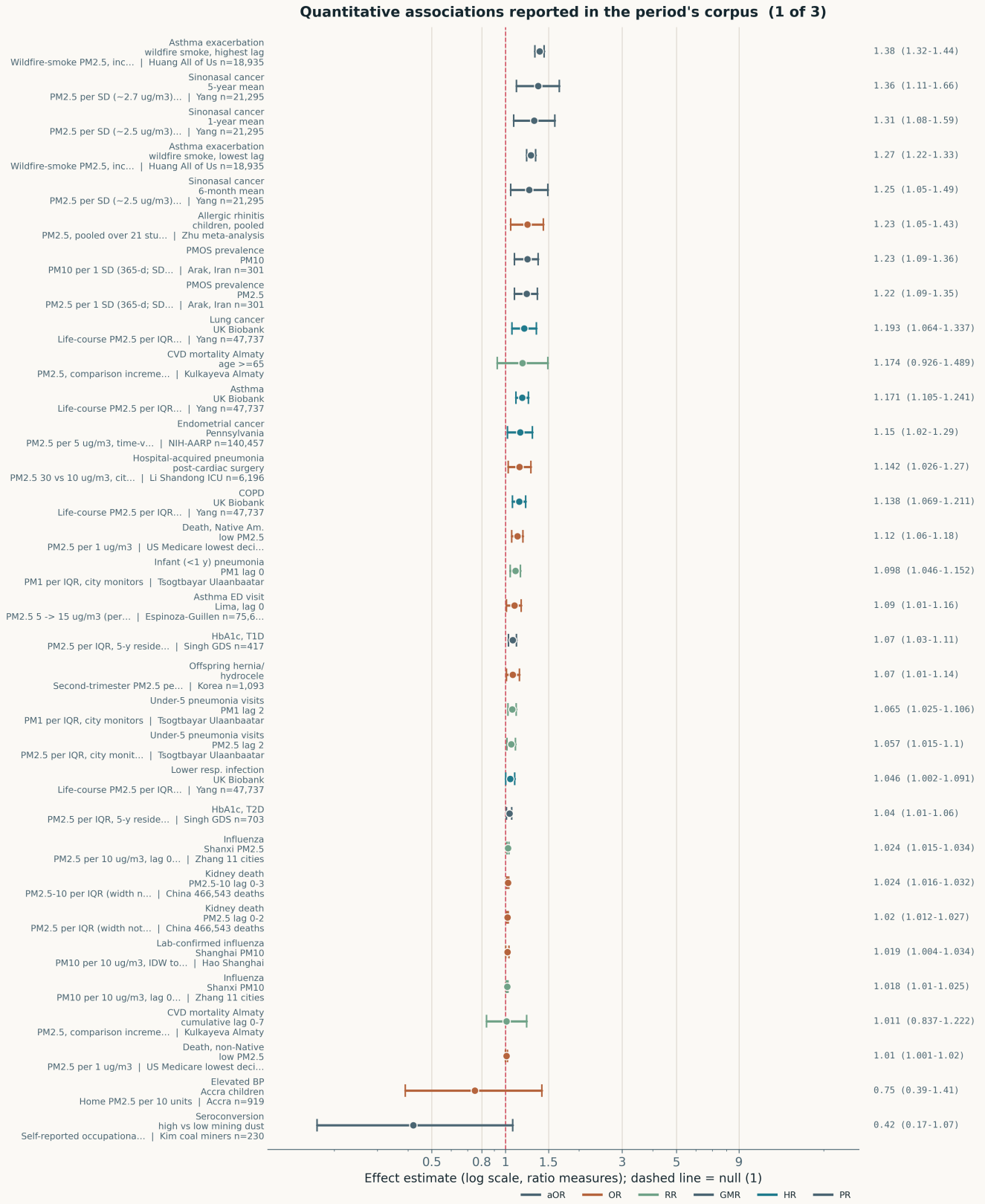


Fig. Ratio scale, part 1: the 32 estimates up to 1.38, from 0.42 (Kim coal-miner seroconversion, high vs low self-reported dust) to 1.38 (All of Us wildfire-smoke asthma exacerbation, highest lag). Nested sets share samples and are not independent: Yang sinonasal (3 averaging windows), Yang UK Biobank (4 endpoints), Singh diabetes (5 across both panels), China kidney deaths (2 pollutants), Tsogtbayar Ulaanbaatar (3 lags). Increments differ across sources; no pooling.

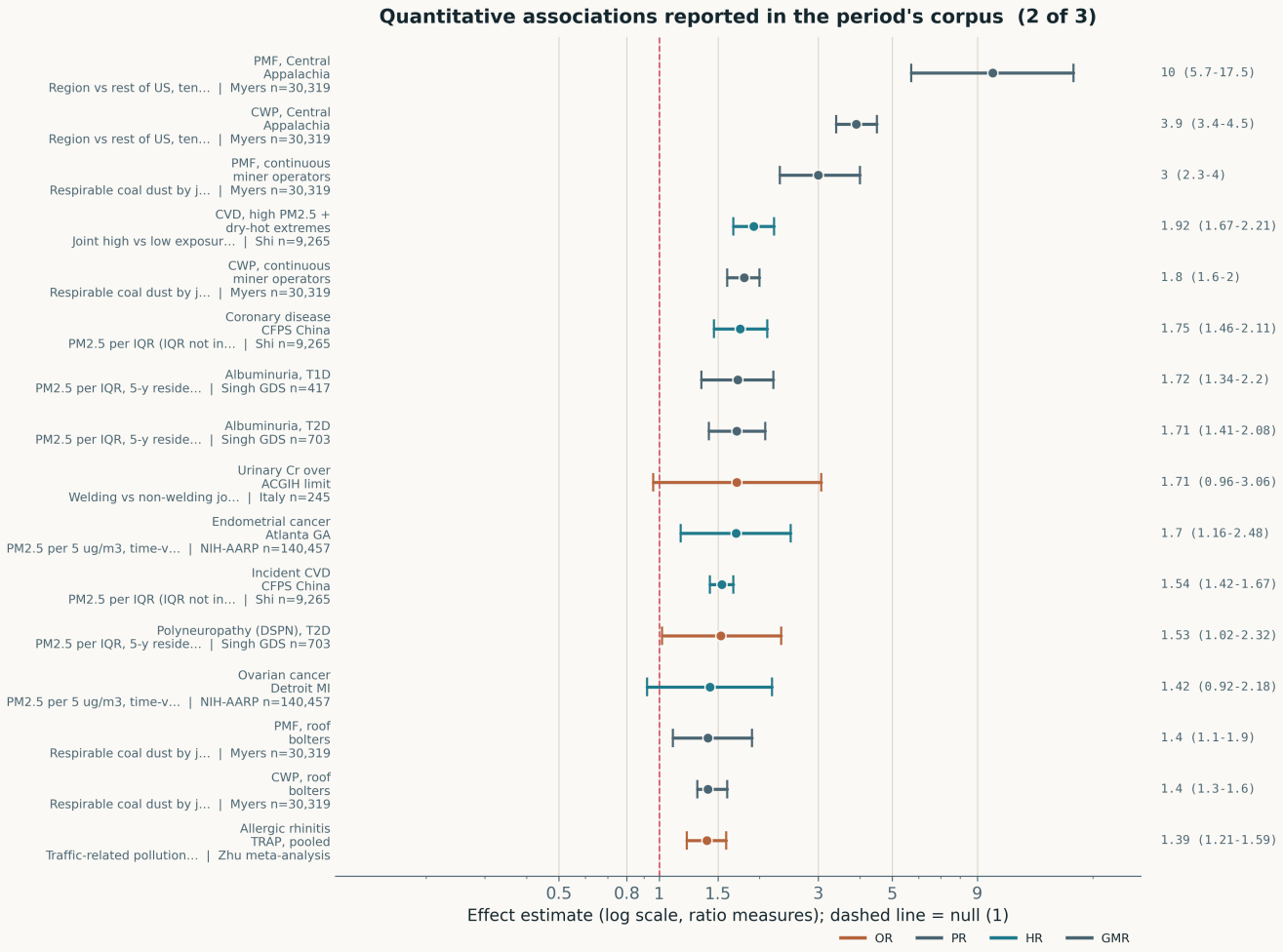


Fig. Ratio scale, part 2: the 16 estimates from 1.39 to 10. Myers' US coal-miner prevalence ratios supply 6 (PMF 10.0, 5.7–17.5, Central Appalachia vs rest of US; job-category contrasts for continuous-miner operators and roof bolters), Shi CFPS 3 ($PM_{2.5}$ with dry-hot extremes and CVD), Singh diabetes 3, NIH-AARP endometrial and ovarian cancer 2 (per 5 $\mu g/m^3$), Italian welders 1 and the TRAP allergic-rhinitis meta-analysis 1. Regional and occupational contrasts, not per-unit PM effects.

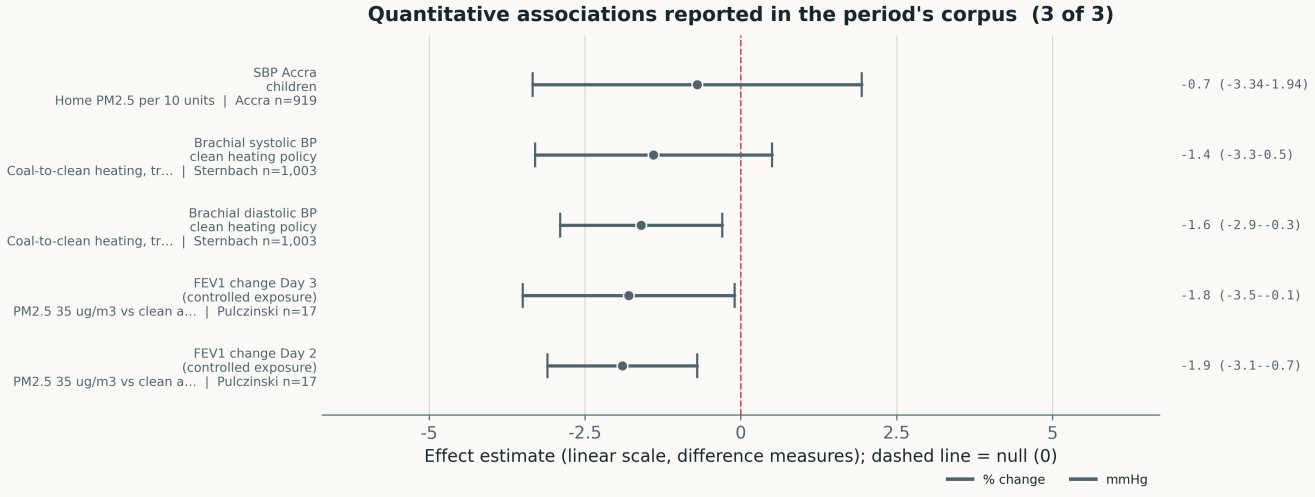


Fig. Difference scale, 5 estimates: Pulczinski EPA controlled exposure (35 $\mu g/m^3$ $PM_{2.5}$ vs clean air, n=17), FEV_1 −1.9 % (day 2) and −1.8 % (day 3); Sternbach Beijing clean-heating policy, brachial SBP −1.4 and DBP −1.6 mmHg; Accra children, SBP −0.7 mmHg per 10 units home $PM_{2.5}$ with an interval spanning zero.

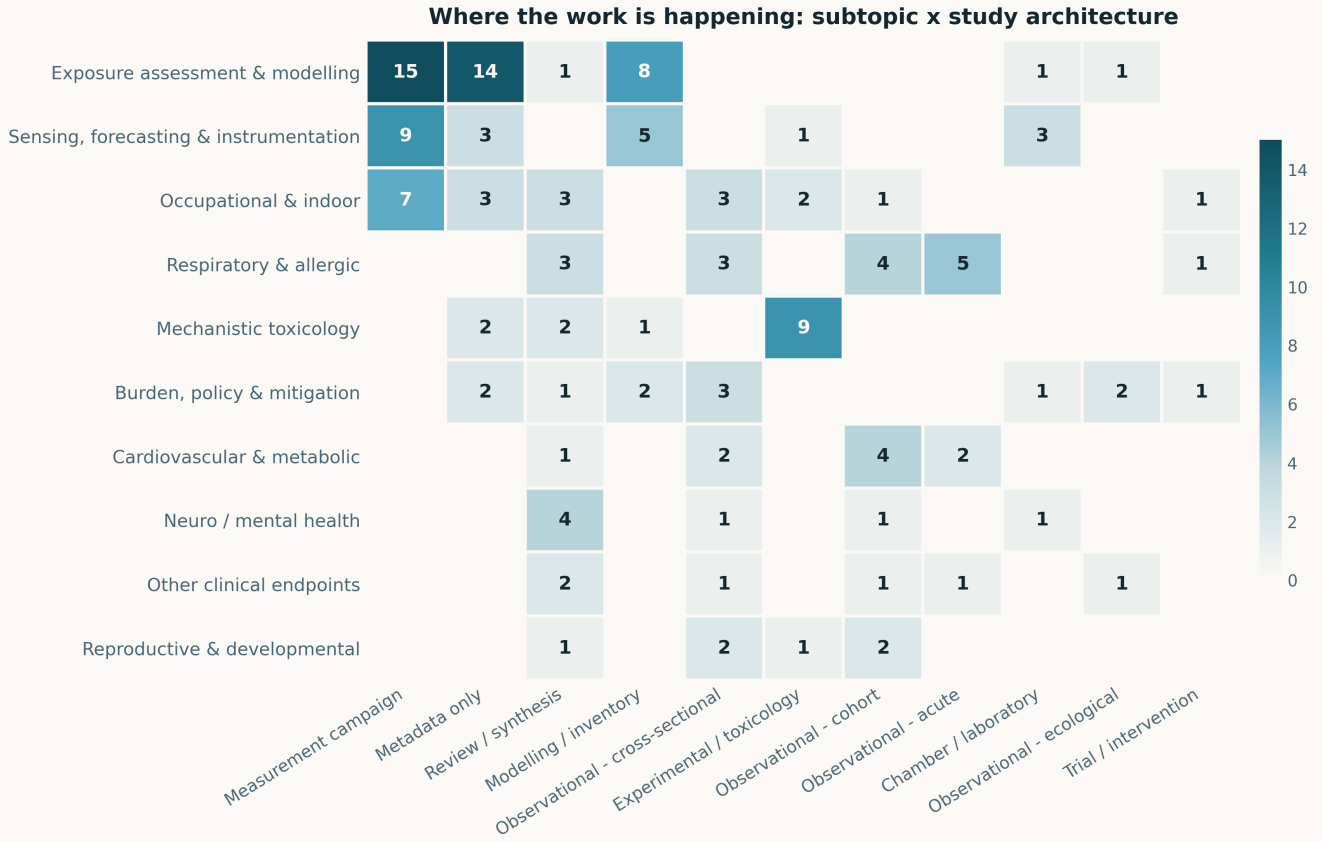


Fig. Subtopic × architecture, pooled. The densest cell is *Exposure* × *Measurement campaign*; occupational work is mostly measurement campaigns and cross-sectional prevalence; mechanistic toxicology is all experimental or chamber; sensing splits between instrument and calibration work, modelling and metadata-only deposits.

Exposure basis and the effect funnel

Exposure basis (x1). 11 of 151 records rest on a low-cost sensor or network (7 %), against 15 of 523 in October (3 %); 3 on a personal or wearable monitor. On the health side, where a basis is recoverable (n=60), **58 % measured, 30 % modelled or satellite-fused, 12 % proxy**, against the October baseline of 36/42/22. The shift is real but narrow: "measured" here is mostly regulatory monitors assigned at city scale and occupational filter campaigns. *Not one of the 53 effect estimates rests on a low-cost sensor exposure*; the only sensor-based health record (Kinshasa PurpleAir and asthma) reports no estimate with an interval in the pooled set.

Increments and precision (x2). Of the 48 ratio estimates, 21 are per IQR or SD, 10 categorical contrasts, 6 per 1 μg/m^3, 5 per 10 μg/m^3 and **6 carry no recorded increment** — 12 %, against 49 of 97 (51 %) in October, the clearest improvement in the record set. 42 of 48 exclude the null. The funnel is **not** a pooled estimate: increments differ and many estimates share a sample. The Egger intercept, 4.38 (SE 0.66), is close to October's 4.6 — the small-study asymmetry is a property of what gets published and indexed, not of one week.

Within-family heterogeneity. No two independent sources this week share metric, endpoint family and increment: the two influenza series (Shanxi RR per 10 μg/m^3 $PM_{2.5}$; Shanghai OR per 10 μg/m^3 PM_{10}) differ in pollutant and design, and the asthma estimates (All of Us smoke case-crossover; Lima $PM_{2.5}$ 5→15) differ in contrast. **No I^2 is reported.**

Venues, groups, trial watch, and what the pipeline did

Venues. *Environ Res* and Research Square lead at **7** each, then *Atmos Environ* 6, *Environ Sci Technol* and *Environ Health Perspect* 5, and *Atmosphere*, *Atmos Pollut Res*, *J Hazard Mater*, *Environ Pollut* and *Front Public Health* 4. Research Square supplied the CUPI-G assimilation, the California metals analysis and the CMAQ smoke evaluation — the sensing-relevant preprint stream is coming through Europe PMC, not arXiv.

Recurring groups. From the `affil` field (first- and last-author institutions, counted per mention): Sun Yat-sen University 6, Peking University 5 (plus 3 for its Shenzhen graduate school), and Environment and Climate Change Canada, Johns Hopkins, Colorado State and USC 4 each; IIT Bombay, Yonsei, AGH Krakow and CU Boulder 3. The Peking/Sun Yat-sen counts are spread over unrelated health records; the only coherent cluster is **ECCC**: the SWAPIT Toronto AQHI-representativeness campaign and a calibrated ventilation index for Lagrangian dispersion, both measurement-side and both from the federal air-quality research group. Scite citation counts were not retrieved (paid-plan refusal); for records days old they would be uninformative.

Trial watch. ClinicalTrials.gov `LastUpdatePostDate` windows 1–10 November: **3 new** registrations. **NCT07914010** REACT (UC Berkeley; air cleaners plus a PurpleAir indoor monitor with a visual indicator, asthma and COPD, n=100) on 4 November — the first registered trial in the set whose primary outcome is a *low-cost-sensor-measured* indoor $PM_{2.5}$ change. On 10 November: **NCT07925229** PINE (UBC; lodgepole-pine woodsmoke versus filtered air crossed with ibuprofen, phase 4, n=20) and **NCT07924839** (Michigan State; sham-filtration $PM_{2.5}$ × obesity × dietary antioxidant, n=20). Amass TrialCore (WHO ICTRP) carried ACTRN12626001238369 (HEPA RCT) and ChiCTR2600134305 (pollutant exposure and reproductive outcomes) from 4 November and nothing PM-related on 10 November. `trials.json` holds **26 trials** over 43 query windows. UBC now has two controlled woodsmoke chamber registrations (*Lungs on Fire* last week, PINE this week): the smoke question is being moved into the chamber, where composition is known.

What the harvesting legs did. Europe PMC re-surfaced records across days again: the 6–9 November backfill caught 33 cross-day duplicates before authoring, and returned zero records on 5 and 9 November (indexing lag). **Crossref-by-ISSN** 28–67/day (3 on Sunday 8 November). **PubMed connector** appended 2–18 records/day. **OpenAlex**: the connector's date filter worked once (3 November, 2 records admitted) and the connector was absent on every later session; the harvester's HTTP leg returned 429 throughout. **arXiv** no in-window entry. **Consensus** 10 hits/run, 1 new record (ARMIE). **Scite** refused (paid plan).

Open gaps

- **Sensor-derived quantities ship without propagated uncertainty.** Assimilated WRF-Chem fields, Bihar attributable deaths and the Jhansi compliance assessment all inherit sensor error that their abstracts do not quantify. The field now needs uncertainty propagation more than it needs another co-location.
- **Node-to-node PM calibration transfer is the weak link.** The SensEURCity result is the first in the watch to say directly that PM transfers worse than meteorology; any network that calibrates by chaining nodes to a single reference should expect it.
- **Smoke exposure is still modelled.** CMAQ missed most observed smoke-exceedance site-days; every smoke health estimate this week used a model or a monitor. The two UBC chamber registrations will supply composition-known dose–response, not population exposure.
- **Indoor endpoints on outdoor monitors.** ICU hospital-acquired pneumonia in filtered post-surgical air was fitted to city-wide ambient $PM_{2.5}$; an indoor sensor is the obvious test of the pathway.
- **The abstract gap is 16 %**, concentrated in day-0 Elsevier and ACS deposits.
- **Pipeline defects carried forward**: scheduled runs for 7–9 November did not complete (recovered 10 November); Europe PMC re-surfacing makes per-day dedup order-dependent; the OpenAlex connector is present only intermittently; `.git` lock debris on the delete-restricted mount.

Research direction. The sensing literature is moving from *is the sensor right* to *what does the sensor change downstream* — forecasts, burden estimates, inequity maps. The epidemiology has not followed: its best designs this week (controlled exposure, deep phenotyping, life-course constituents) still use monitors or models. The next step that matters is a health estimate whose exposure comes from a calibrated network with its uncertainty carried through.

Paper-level digest — the week by subtopic

Every record that appeared in a daily issue keeps that issue's critical summary verbatim; nothing here is re-written or re-assessed. Records surfaced without an abstract are shown as such. Ordered by subtopic, then by the date of the issue that first carried the record.

Sensing, forecasting & instrumentation — 21 records

Carried from the daily issues — full summaries (21 of 21 records in this subtopic)

Agrawal Bihar low-cost sensor network ML exposure inequity short-term mortality 2026 Nat Commun

Unveiling hidden air pollution exposure and impacts with low-cost sensor network-based frameworks.

Dense low-cost $PM_{2.5}$ network across Bihar, fused with ML surfaces, NFHS socioeconomic data and India-specific relative risks. Northern districts carried 1.2 times the mean exposure of southern ones; district mortality per 100,000 varied up to 1.7-fold and daily attributable mortality 3.2-fold. State-level exposure inequity was limited, intra-regional inequity substantial. The abstract gives no sensor count, calibration R2/RMSE, or uncertainty on the attributable deaths, and the mortality is borrowed risk functions applied to sensor-derived surfaces, not an observed association; network calibration error propagates straight into the burden. Relevance: direct - the clearest flagship case for low-cost networks as a burden-estimation input outside cities.

doi: 10.1038/s41467-026-77148-1

Ainslie ECCC calibrated ventilation index Lagrangian stochastic Richardson number 2026 J Air Waste Manag Assoc

The scientific development of Environment and Climate Change Canada's new ventilation index.

A Lagrangian stochastic model defines ventilation, then is parameterized on 10 m wind speed, friction velocity and Richardson number into an index from 0 to 100 where 33 units correspond to a doubling or halving. Development spanned stabilities, wind speeds and land-use classes against several turbulence parameterizations and a Gaussian plume model. The index depends strongly on Ri in the -0.1 to 0.1 transition, with residual uncertainty above Ri 0.25. Over one Quebec winter, hindcast CVI separated advisory from non-advisory evenings under highly stable conditions (CV 1.51) where the legacy index compressed against its lower bound (CV 0.56). Limitation: one winter, qualitative evaluation, no PM concentrations verified. Relevance: direct - an operational 2.5 km dispersion field is a covariate a sparse network can borrow for siting and for smoke-advisory timing.

doi: 10.1080/10962247.2026.2740009

Angita Jhansi semi-arid campus automatic sensor $PM_{2.5}$ PM_{10} annual cycle 2023 Research Square

Year-Round Seasonal Variation of $PM_{2.5}$ and PM_{10} at a Semi-Arid Institutional Campus Site in Jhansi, Uttar Pradesh, India

Preprint. One automatic sensor at a semi-arid campus in Jhansi, Bundelkhand, January-December 2023. Post-monsoon means were highest ($PM_{2.5}$ 63.55, PM_{10} 190.63 $\mu g\,m^{-3}$, November peak), monsoon lowest (33.17 and 99.52); annual $PM_{2.5}$ 37.45 $\mu g\,m^{-3}$, under the Indian NAAQS of 40 but 7.5 times the WHO guideline, and PM_{10} 113.00 against a standard of 60. ANOVA on 12 monthly means: F(3,8) 4.66 and 4.83. The instrument is unnamed, with no co-location, humidity correction or data-capture statistics, so the monsoon minimum and the $PM_{2.5}$ standard compliance both sit inside unquantified sensor error. Relevance: direct - a single uncalibrated sensor reported as a compliance assessment, exactly the use the low-cost field warns against.

doi: 10.21203/rs.3.rs-11210953/v1

Chen TRACER Houston micropulse lidar aerosol vertical profile boundary layer parameterisation 2026 Atmos Chem Phys

Aerosol vertical distributions shaped by boundary layer dynamics in a coastal urban environment: insights from the TRACER campaign

Six months of TRACER (Houston, 2022) micropulse lidar retrievals anchored to ground aerosol data. An S-shaped fitting function yields an aerosol-layer depth and a stratification-strength parameter. Concentrations drop sharply at strong capping inversions, layer depth tracks boundary-layer growth and collapse, and sea-breeze passage swaps continental for maritime aerosol. Relationships with BLH and the potential-temperature gradient are statistically significant but weak (no r given). No health endpoint. Relevance: direct for satellite-surface fusion - the AOD-to-surface $PM_{2.5}$ ratio depends on exactly this profile shape, and the parameterisation offers a physical predictor in place of BLH alone.

doi: 10.5194/acp-26-15543-2026

Glienke HOLODROPS holographic droplet sensor event-based PIV characterization 2026 Atmos Meas Tech

Development and characterization of an advanced holographic instrument for atmospheric research

Two configurations: HOLODROPS for long-term droplet size distributions and HOLODROPS-TRACK for short-interval 3D particle tracking of turbulence and collision-coalescence. Both recover size, shape and three-dimensional position of multiple particles above 8.8 um in one sample volume, replacing several probes. Hologram reconstruction is the binding constraint, so a complementary event-based PIV system identifies when to trigger the tracking mode at far lower data volume. All three were characterized in the laboratory, with one outdoor drizzle test of the prototype. Limitation: 8.8 um is far above the accumulation mode, so this is cloud microphysics instrumentation, not a $PM_{2.5}$ method. Relevance: low directly, but the trigger architecture - a cheap continuous channel deciding when to run the expensive one - is the design pattern dense networks need for event capture.

doi: 10.5194/amt-19-6973-2026

Goyal PurpleAir PA-II kappa-Kohler Lorentz-Lorenz random forest moisture artefact 2026 Discover Electronics

A physics-aware digital twin for correcting moisture-induced material artifacts in $PM_{2.5}$ sensors

A pair of PA-II units beside one EPA FEM monitor for 150 days in an industrial area. kappa-Kohler growth and a Lorentz-Lorenz refractive index mixing rule motivate features for a polynomial-enhanced random forest. At RH 76.16% optical intensity rose 59.5%, summarised as a mass-intensity divergence factor of 4.33. Raw bias of +17.55% fell to 0.13% NMB, R2 0.9010. Limitation: two sensors at one site, single author, and the abstract gives no train/test separation by time, sensor or site, so 0.13% cannot be read as field skill. Calling FEM 'gravimetric' is loose. Relevance: moderate - the physics is standard and testable; the transferability is untested. BACKFILL, Crossref 26 September 2026.

doi: 10.1007/s44291-026-00286-9

Kloppenborg ARMIE SPS30 node laboratory DustTrak SMPS APS source-out calibration 2026 Sensors

Laboratory Evaluation of ARMIE, a Portable SPS30-Based Low-Cost Sensor Node for $PM_{2.5}$ Monitoring

SPS30 nodes were co-located in the laboratory with a DustTrak and an SMPS+APS across candle, cooking, cigarette smoke and clean air, at regular and high humidity. Agreement with the DustTrak was r 0.93-0.98; with SMPS+APS it was moderate, r 0.74-0.94, with systematic underestimation at high concentration. Leave-one-sensor-out and leave-one-source-out calibration reduced mean error and narrowed limits of agreement. Limitation: a DustTrak is itself a photometer, so the high r partly reflects shared optical bias; the sizing reference is the honest comparator and shows the gap. No gravimetric or field arm, and r is not accuracy. Relevance: direct - leave-one-source-out is the right test for epidemiology nodes. BACKFILL, Crossref 2 January 2026.

doi: 10.3390/s26010280

Li Nanjing jute cellulose filter $PM_{2.5}$ sampling recovery oxidative potential 2026 Environ Res

An All-Natural Cellulose Fiber Filter Developed for High Recovery Sampling and In Vitro Toxicity Assessment of PM(2.5).

Freeze-dried jute-pulp filter: filtration efficiency 81.55-99.41% across particle sizes at 216.8 Pa pressure drop; $PM_{2.5}$ recovery 94.25-96.37%, higher than PTFE. Recovered $PM_{2.5}$ showed higher oxidative potential and dose-dependent macrophage cytotoxicity than PTFE-recovered PM, and the filter degrades in soil. Higher OP from the new medium may mean better recovery or contamination by the cellulose itself; the abstract reports no blank OP or endotoxin control, and lower efficiency at the low end breaks mass closure. Relevance: indirect - relevant only if gravimetric co-location uses it; 81.55% minimum efficiency would bias reference mass low.

doi: 10.1016/j.envres.2026.126128

Liu CNN-xLSTM attention SHAP $PM_{2.5}$ forecasting Beijing hourly 2026 Atmosphere

Hybrid Network Models Based on SHAP Interpretability for $PM_{2.5}$ Concentration Prediction: A Case Study of Beijing

A 1D CNN, extended LSTM and attention stack on Beijing hourly data for 2020-21 reached RMSE 9.845 $\mu g\,m^{-3}$ and R2 0.977 one step ahead, cutting RMSE by about 8.5% against CNN-xLSTM at one step and by 8.9% and 12.3% at the 2 h and 10 h horizons. SHAP put 57.09% of predictive power on the pollutant block, carbon monoxide alone at 32.791%, with non-linear dependence and candidate turning points for CO, humidity, wind speed and direction; the authors warn these are model-derived, not physical thresholds. Limitation: two years at one city, no external-year test, and a forecast whose dominant feature is another measured pollutant is unavailable to a PM-only node. Relevance: moderate.

doi: 10.3390/atmos17111083

Mao wavelet attention LSTM indoor aerosol concentration prediction 2026 J Aerosol Sci

A Wavelet-Based Attention LSTM Framework for Aerosol Concentration Prediction in Indoor Environments

METADATA ONLY. Metadata only; no abstract in Crossref at issue time, so no finding is asserted. The title describes a wavelet decomposition feeding an attention LSTM to predict indoor aerosol concentration. Carried because short-horizon indoor prediction is one of the candidate edge-inference tasks for a sensor node, and J Aerosol Sci is one of the venues where such work is tied to a measured aerosol. What decides its value is unknown until the abstract deposits: the measuring instrument, whether validation is by held-out episode rather than random split, and the prediction horizon. Relevance: possible, unassessed.

doi: 10.1016/j.jaerosci.2026.106908

Park micro-3D-printed liquid-metal coil inductive ferrous aerosol impactor sensor 2026 Lab Chip

Micro-3D-printed integrated system with liquid-metal coils for simultaneous size-selective sampling and concentration sensing of respirable ferrous metal aerosols.

Monolithic micro-3D-printed platform: two-stage inertial impactor (aerodynamic cut-offs 5.28 and 0.99 um), aqueous particle transfer, 100 um sensing channels, embedded EGaIn solenoids (50 turns, 907.1 nH, maximum Q 9.1 at 30 MHz). Fe-only aerosol: peak-rate calibration R2 0.900 and 0.910, time-resolved nRMSE 6.72% and 8.95% for the two fractions. Arizona test dust background: R2 0.9428 and 0.9348 against APS total particle increments. Responses persisted over a 60-min mixed-aerosol challenge. Limitation: laboratory-generated aerosol only, no field, humidity, drift or unit-to-unit data, and the dust-background R2 is against total count, not ferrous mass. Relevance: moderate - composition-selective inductive sensing is an orthogonal channel to optical scattering, but far from a deployable node.

doi: 10.1039/d6lc00748a

Peletidou fluorescence Raman depolarization lidar aerosol typing Thessaloniki 2026 Atmos Chem Phys

Vertical profiling and aerosol typing using fluorescence Raman and depolarization lidar measurements over Thessaloniki, Greece

First fluorescence lidar observations over the Eastern Mediterranean: 108 aerosol layers from 50 cases, April 2024 to December 2025, classified into smoke, dust, polluted continental, clean continental and mixed pollen. Spectral fluorescence capacity separated the types cleanly, highest for smoke (3.5e-6 to 12.7e-6 /nm), then mixed pollen, polluted continental, clean continental and dust (0.4e-6 to 1.3e-6 /nm), and a subset was compared against the EARLINET Mahalanobis typing algorithm. Pollen depolarization differed from values reported at more northern latitudes, possibly reflecting pollen type. Limitation: one site, one instrument, and typing is validated against another algorithm rather than against in-situ composition. Relevance: moderate - a column method that distinguishes smoke from dust is the external constraint a surface mass network cannot supply for itself.

doi: 10.5194/acp-26-15461-2026

Rogers VUV-AMS siloxane radical mass calibration source aerosol mass spectrometry 2026 J Phys Chem A

Siloxane-Radical Species: Chemical Characteristics and Use for Mass Calibration in Aerosol Mass Spectrometry.

Synchrotron VUV photoionisation aerosol mass spectrometry: heating conductive silicone tubing (150-230 C) inline with a soot source deposits siloxane oligomers on particles, producing three substituted-siloxane radical series and 21 recurring peaks between 73 and 799 Da. Calibration was verified on five PAHs (178-300 Da), 2,5-dimethylfuran and ethylene-pyrolysis hydrocarbons (165-374 Da). The radical-formation mechanism on the CuO target is hypothesised, not shown, and the method is specific to one research-grade platform. Relevance: indirect - siloxane from silicone sample lines is also a known artefact; the paper doubles as a warning for any sensor or sampler plumbed with heated silicone tubing.

doi: 10.1021/acs.jpca.6c04791

Sharon IoT vape detection school $PM_{2.5}$ PM_{10} 37 locations 18 months 2026 Int J Environ Res Public Health

Smart Vape Detection in Schools for Mitigating Student E-Cigarette Use

BACKFILL (entry 4 May 2026). Low-cost IoT PM nodes at 37 restroom and change-room locations in one Australian independent school, one-minute data for 18 months, threshold alerts with school-led CCTV verification. More than 300 vaping-related incidents, clustered by time of day and among senior students; alerts showed rapid concurrent $PM_{2.5}$ and PM_{10} excursions. Staff response declined through alert fatigue, and a regional smoke event produced false positives. No sensitivity, specificity or denominator of unverified alerts in the abstract. Limitation: first-author affiliation is the IoT vendor; 'highly effective' is claimed without a confusion matrix. Relevance: high - event detection from PM transients on low-cost nodes, with the human-response failure and the need for context-aware logic documented.

doi: 10.3390/ijerph23040501

Soderblom Laplace-domain in situ network calibration Helsinki Antwerp SensEURCity 2026 Meas Sci Technol

Laplace domain dynamic calibration in city-wide sensor networks

A Laplace-domain method estimates pairwise dynamic relationships between sensor signals, separates a medium-related spatial effect, and solves an overdetermined calibration against one known network reference, with Monte Carlo uncertainty. Tested on Helsinki (linked to SMEAR III) and Antwerp (SensEURCity). Coefficients were obtained for temperature, RH and pressure in both networks, but PM calibration was feasible only on restricted sensor subsets because PM signals are localized and weakly correlated. Injected shifts could be flagged when geometry and correlations sufficed. No PM R2 or RMSE in the abstract, so the size of the PM failure cannot be graded. Relevance: direct - a metrology-chain calibration transfer fails first on the measurand networks are built for.

doi: 10.1088/1361-6501/aebdb6

Sotiropoulou smart bench low-cost sensors Attica exposure AirQ+ FAIRCITY 2026 J Environ Health Sci Eng

Smart city developments: Pilot case study of smart benches usage for impact assessment due to exposure to air pollution in the urban environment.

A single IoT smart-bench node in a dense Attica district fed exposure and dose estimates for two population groups, with short-term respiratory admissions quantified in WHO AirQ+. O_3 exceeded $PM_{2.5}$, PM_{10} and NO_2 by more than a factor of two; NO_2 doubled in the cold season; PM_{10} breached the EU limit value in both seasons. Adolescents carried roughly double the general-population exposure through longer time in the district, while attributable admissions were an order of magnitude higher for the working-age group on population size alone. No CI, and no reference co-location is reported for the bench. Limitation: one node standing for a district, and AirQ+ coefficients imported wholesale. Relevance: direct, as a worked example of how far a single low-cost node is pushed in municipal practice.

doi: 10.1007/s40201-026-01010-1

Suter LDSA lung-deposited surface area summer measurements spatiotemporal modelling 2026 Atmos Environ

Measurements and exploratory spatiotemporal modelling of LDSA in summer

METADATA ONLY. Metadata only; no abstract in Crossref at issue time, so no finding is asserted. The title describes summer measurements of lung-deposited surface area and an exploratory spatiotemporal model of it. LDSA is the dose-relevant ultrafine metric that diffusion-charger sensors report and optical sensors cannot; a spatiotemporal surface would let health studies use it at scale. To be assessed when the Elsevier abstract deposits: instrument (diffusion charger model), number of sites or mobile route, model predictors and cross-validated skill. Relevance: possible, likely direct - LDSA surfaces would be the natural target for diffusion-charger networks.

doi: 10.1016/j.atmosenv.2026.122459

Tirpitz UV aerosol absorption retrieval actinic flux FIREX-AQ plume age 2026 Atmos Chem Phys

Retrieval of ultraviolet aerosol absorption from radiation measurements in young wildfire plumes

A VLIDORT-based algorithm retrieves the imaginary refractive index from airborne actinic flux over three FIREX-AQ flights, giving $k(387\ \mathrm{nm})$ of 0.02-0.03 and absorption Angstrom exponents of 3-5. Absorption decayed with physical plume age, with half-lives of 13+/-3 h for k, 16+/-4 h for the volume absorption cross section and 17+/-4 h for single-scattering co-albedo, and the authors publish a parameterization of absorptivity as a function of wavelength and plume age. Agreement with aircraft observations is good but less variable, which they attribute to spatial averaging in the method. Limitation: three flights, one fire regime. Relevance: direct - a measured decay timescale for optical properties is exactly the term that makes an optical sensor's response to smoke change with distance from the fire.

doi: 10.5194/acp-26-15423-2026

Wang $PM_{2.5}$ PM_{10} forecasting dual-data MWS-optimised ensemble learning 2026 Atmos Environ

$PM_{2.5}$ and PM_{10} Concentration Forecasting Using a Dual-Data Strategy and MWS-Optimized Ensemble Learning

METADATA ONLY. Metadata only; no abstract in Crossref at issue time, so no finding is asserted. The title describes $PM_{2.5}$ and PM_{10} forecasting from a dual-data input strategy with an ensemble learner tuned by an MWS optimiser. Carried because short-horizon forecasting is the step most networks attempt after calibration, and ensemble skill claims usually hinge on whether validation is time-blocked. To be assessed when the Elsevier abstract deposits: horizon, held-out period, skill against persistence, and whether sensor or reference data feed it. Relevance: possible, unassessed.

doi: 10.1016/j.atmosenv.2026.122424

Wu Heyuan attention-LSTM SHAP hourly $PM_{2.5}$ O_3 forecasting persistence 2019-2023 Research Square

Baseline-relative predictability of hourly $PM_{2.5}$ and O_3 in a low-emission South Chinese city: an attention-LSTM with feature attribution

Preprint. Attention-LSTM on hourly observations from two Heyuan stations, 2019-2023, interpreted with SHAP. Test R2 0.9796-0.9877 for $PM_{2.5}$ and 0.9921-0.9975 for O_3, RMSE 1.28-2.02 and 1.77-2.83 $\mu g\,m^{-3}$; the most recent pollutant observations dominated attributions, humidity and temperature contributed for O_3. The authors themselves frame the result against persistence: at one-hour horizon a lag-1 baseline already achieves R2 near 0.95, and the abstract does not report the skill gain over it. Two stations in a low-emission city limit generality. Relevance: indirect - a useful cautionary example that high R2 in short-horizon PM forecasting mostly measures autocorrelation, relevant to edge-inference claims on sensor nodes.

doi: 10.21203/rs.3.rs-11616075/v1

Yadav CUPI-G low-cost sensor assimilation WRF-Chem Delhi stubble burning 2026 Research Square (preprint)

Can Low-Cost Sensor Networks Improve Analysis of Extreme Pollution Episodes in Delhi During Stubble Burning?

November 2024 post-monsoon burning season. Assimilating 33 rural source-region low-cost $PM_{2.5}$ nodes cut the mean fractional bias over Haryana from -116.7% to -56.8%, and the Severe AQI category was recovered in about 95% of Delhi cases and 100% of Punjab and Haryana cases, where the control run predicted no Severe case at all. A second experiment using a 1.5 $\mu g\,m^{-3}$ observation error, the reference-monitor value, improved further, so the assumed observation error, not the sensor count, sets the gain. Limitation: one month, one burning season, a preprint, and no co-location statistics for the CUPI-G nodes, so the 10% error is asserted rather than demonstrated. Relevance: direct - the strongest case to date that uncalibrated-grade source-region nodes carry forecast-relevant information.

doi: 10.21203/rs.3.rs-11388680/v1

Exposure assessment & modelling — 40 records

Carried from the daily issues — full summaries (40 of 40 records in this subtopic)

Alruqimi Alpine region multi-source machine learning daily NO_2 PM_{10} $PM_{2.5}$ estimation 2026 Atmos Environ

Multi-source machine learning for daily surface NO_2, PM_{10}, and $PM_{2.5}$ estimation in Alpine region

metadata only. Metadata only: Crossref deposit without abstract, Atmospheric Environment. The title describes a multi-source machine-learning model producing daily surface NO_2, PM_{10} and $PM_{2.5}$ estimates across the Alpine region. No inputs, resolution, validation statistics, station counts or period can be recovered, and no finding is asserted here. Complex terrain with inversions and valley pooling is where satellite-ML PM products usually degrade, so the cross-validation design (spatial versus random) will decide how much the reported skill means. Relevance: potentially direct - an Alpine daily PM surface is a candidate fusion target for sensor networks in mountain valleys; recheck for the abstract in the weekly sweep.

doi: 10.1016/j.atmosenv.2026.122454

Andino intra-urban $PM_{2.5}$ PM_{10} meteorological GAM arid San Juan Argentina 2026 Environ Monit Assess

Intra-urban variability and meteorological drivers of PM(2.5) and PM(10) concentrations in an arid Argentine city.

Cold-season campaign across three land-use zones in an arid metropolitan area, with generalized additive models over temperature, humidity, pressure, wind speed and direction, and radiation. Industrial and commercial zones ran highest and residential lowest, with morning and evening peaks. Pressure and wind speed were the dominant predictors; $PM_{2.5}$ responded to atmospheric stability and low wind, PM_{10} to wind-driven resuspension. No effect sizes, CIs or deviance explained in the abstract, and the campaign is short-term, so the zone contrast confounds season with siting. Relevance: moderate - the $PM_{2.5}/PM_{10}$ split in meteorological response is the physical reason a single correction factor fitted on one fraction transfers badly to the other in dust-dominated cities.

doi: 10.1007/s10661-026-16103-5

Ayittey Accra FRM gravimetric $PM_{2.5}$ three sites e-waste 2021-2022 2026 Environ Monit Assess

Persistent urban $PM_{2.5}$ pollution in Accra, Ghana: spatiotemporal patterns, meteorological and anthropogenic drivers, and associated health risks

Two years (2021-2022) of FRM gravimetric $PM_{2.5}$ at Adabraka (next to the Agbogbloshie e-waste burning area), Dansoman and Legon. All sites ran about 3-4 times the WHO 24-h guideline; Adabraka was highest at 60.76 (2021) and 60.54 $\mu g\,m^{-3}$ (2022), with negligible interannual change but strong spatial contrast. PCA ties variability positively to temperature and negatively to RH. Hazard quotients exceeded 1 for adults and children. Sampling was intermittent (~300 samples over 3 sites and 2 years), and HQ applied to PM mass is a screening device, not an epidemiological risk. Relevance: direct - one of very few reference-grade West African series, usable to anchor low-cost and satellite calibration. backfill, entry 3 March 2026.

doi: 10.1007/s10661-026-15117-3

Bonilla California CMAQ wildfire $PM_{2.5}$ evaluation 2008-2018 random forest diagnostic Res Sq

Evaluating CMAQ Representation of Wildfire $PM_{2.5}$ in California (2008-2018): Regime-Dependent Performance and Nonlinear Error Structure.

Preprint. 187,051 paired site-days across California fire seasons 2008-2018 (18 CMAQ values above 557 $\mu g\,m^{-3}$ quarantined). Observed and modelled means 10.69 and 9.29 $\mu g\,m^{-3}$; r 0.502, mean bias -1.40, RMSE 9.96 $\mu g\,m^{-3}$. CMAQ detected 45.4% of observed site-days above 12 and 40.8% above 35 $\mu g\,m^{-3}$, and at 35 it produced more events overall yet missed 1,273 of 2,151 observed ones because events did not coincide by site-day. A random forest using fire-only CMAQ $PM_{2.5}$ among its predictors reached R2 0.711, RMSE 5.82 on a held-out site-year partition. The diagnostic does not apportion error to emissions, plume rise or meteorology. Relevance: direct - quantifies why model-only smoke exposure misclassifies acute days; sensor-network fusion is the fix.

doi: 10.21203/rs.3.rs-11167409/v1

Bu Anmyeon-do GAW November 2025 Asian dust haze wildfire episodes aerosol optics 2026 Preprints.org

Episode-Dependent Variations in Atmospheric Composition at the Anmyeon-Do Global Atmosphere Watch Station in November 2025

Anmyeon-do GAW station, November 2025: Asian dust, haze, high-PM and wildfire-influenced episodes vs non-event periods. PM_{10} in all events exceeded twice the non-event mean (24.8 +/- 9.9 $\mu g\,m^{-3}$), peaking in Asian dust (66.1 +/- 19.2). Scattering Angstrom exponent was lowest in dust (1.1 +/- 0.5). Secondary ions made up 93.4% (haze) and 75.1% (high-PM) of measured composition; NOx was 1.5-1.7 times higher in haze/high-PM and CO rose most in the wildfire period. One month, one site, preprint. Relevance: direct - episode-dependent Angstrom and composition shifts are the conditions under which optical PM sensors' mass calibration breaks.

doi: 10.20944/preprints202611.0490.v1

Cheng pedestrian-level $PM_{2.5}$ crowd mobility 3D urban form CFD historic neighbourhood 2026 Environ Pollut

Dynamic Crowd Mobility, Socio-Economic, and Pedestrian-Level PM(2.5) in a High-Density Historic Neighbourhood: Transient Effects and Synergistic Mechanisms of 3D Urban Form.

Real-time $PM_{2.5}$ at 2-second resolution across 90 cells of a dense historic district was combined with CFD at breathing height and a socio-economic and physical 3D form indicator set, then analysed by stepwise regression and variance decomposition. Short-term population mobility, economic activity intensity, and building and road pattern were all associated with $PM_{2.5}$, forming an interacting chain with seasonally varying synergies; physical form dominated both internal and external areas, while social and economic synergy was stronger externally. The authors describe the crowd-behaviour link as a transient disturbance effect and keep it statistical. No effect sizes, no CI, and the instrument class behind 2-second logging is not named in the abstract. Limitation: one neighbourhood. Relevance: direct - this is the spatial scale at which a dense network is justified, measured rather than assumed.

doi: 10.1016/j.envpol.2026.129433

Gantina remote sensing CO HC NOx soot heavy-duty trucks buses Indonesia 2026 Aerosol Air Qual Res

Emission of CO, HC, NOx, and Soot from HDV Trucks and Buses using Remote Sensing in Indonesia

metadata only. Metadata only; no abstract in Crossref at issue time, so no finding is asserted. The title reports on-road remote sensing of CO, HC, NOx and soot from heavy-duty trucks and buses in Indonesia. Carried because measured real-world fleet soot factors are scarce for Southeast Asia, and because they anchor inventory inputs where reference networks are thin and low-cost networks are the main expansion route. To be assessed when an abstract deposits: fleet size sampled, soot retrieval method and any emission-standard contrast. Relevance: possible, unassessed.

doi: 10.1007/s44408-026-00174-7

Gonzalez organosulfates CMAQ hygroscopicity kappa US summers 2013-2022 2026 Environ Sci Technol

Modeling Decadal Organosulfates in Particulate Matter to Improve Estimations of Aerosol Hygroscopicity in the United States

CMAQ with updated isoprene organosulfate chemistry, contiguous US, summers 2013-2022, evaluated on IMPROVE organosulfur and sulfate plus speciated filters, with an interpretable ML surrogate. Summer mean organosulfate up to 0.51 $\mu g\,m^{-3}$ in the east. Net change in hygroscopicity kappa -5.6% (direct +2.8%, indirect -8.4% through sulfate reduction). No CI on these. Limitation: summer only, isoprene-derived only, and validation rests on sparse speciated measurements; evaluation metrics are not in the abstract. Relevance: moderate - humidity corrections for optical sensors commonly assume a kappa; a chemistry-driven shift of this size, spatially structured, is one physical reason a single continental correction does not transfer.

doi: 10.1021/acs.est.6c08752

Hansen spatial mediation GWR SVC $PM_{2.5}$ income cardiovascular simulation 2026 medRxiv (preprint)

Spatially varying effect estimates of mediation in the presence of spatial auto-correlation

An extensive simulation compares approaches to mediation when direct and indirect effects vary over space, concluding that geographically weighted regression and other spatially varying coefficient models combined with the product method recover spatially heterogeneous estimates accurately and handle spatial confounding in a grid setting. The applied case treats $PM_{2.5}$ as the mediator between income and cardiovascular disease. Limitation: a preprint, the performance claim rests on simulations the authors designed, the grid setting is not an irregular administrative geography, and no effect size is reported for the case study. Relevance: moderate - if the mediated fraction of a socioeconomic gradient is itself spatial, the exposure surface needs to resolve that scale, which is the methodological case for dense networks.

doi: 10.64898/2026.10.30.26366937

Hussain Bergin Pakistan smog episodes data-driven source attribution 2026 Atmos Environ

Attribution of Air Pollution Sources During Smog Episodes Across Major Urban Centers of Pakistan: A Data-Driven Approach

METADATA ONLY. Metadata only; no abstract in Crossref at issue time, so no finding is asserted. The title promises data-driven source attribution for smog episodes across several Pakistani cities; the author list includes Mike Bergin, whose group has built low-cost and satellite $PM_{2.5}$ work in South Asia, and Muhammad Fahim Khokhar. Carried because Punjab smog is among the most extreme PM regimes, and attribution there decides between crop burning, brick kilns and traffic. To be assessed when the abstract deposits: data sources (sensor, reference, satellite) and method. Relevance: possible, unassessed.

doi: 10.1016/j.atmosenv.2026.122473

Jaelani Surabaya Benowo landfill 15-min $PM_{2.5}$ 2020-2024 2026 Environ Sci: Atmos

The spatial impact assessment of landfill sites on air quality in Surabaya: a diurnal and seasonal perspective

Only a one-sentence abstract is available. Five years of 15-minute monitoring at the Benowo landfill show $PM_{2.5}$ rising from 37.17 to 50.32 $\mu g\,m^{-3}$ between 2020 and 2024, 7-10 times the WHO annual guideline, with exceedance peaks at 07:00 and overnight. The authors attribute the timing to boundary-layer mixing rather than landfill operations. Instrument class, completeness and any upwind comparison are not stated, so landfill attribution cannot be judged. Relevance: indirect - the diurnal peak set by mixing depth is a reminder that a single fixed node cannot separate source from meteorology without a paired background site.

doi: 10.1039/d6ea00048g

Jin personal monitoring inhaled dose mobility physical activity $PM_{2.5}$ 2026 Environ Monit Assess

Leveraging real-time personal monitoring of air pollution and physical activity levels, and location tracking for mobility- and inhaled dose-based assessments of exposure and health risk

Personal $PM_{2.5}$ monitoring was combined with real-time location, physical-activity intensity and physiological characteristics to compute mobility- and activity-weighted average daily inhaled dose, then compared against conventional residence- and concentration-based metrics. Gender differences in exposure appeared only in the inhaled-dose metric, with men higher; lower educational attainment showed higher exposure and risk in all mobility-based metrics but not in the residence-based one. A 3D geovisualization showed dose changes departing from concentration changes. No effect sizes, CIs or sample size in the abstract, so the disparity claims are directional only. Relevance: direct - it quantifies how much of a measured disparity is an artifact of the exposure metric rather than of the exposure. BACKFILL, entry 23 February 2026.

doi: 10.1007/s10661-026-15090-x

Jiratampradab daily $PM_{2.5}$ probability distribution upper northern Thailand log-normal 2026 Atmosphere

Probability Distribution Analysis of Daily $PM_{2.5}$ Concentration at Five Stations in Upper Northern Thailand

1,096 daily values at each of Chiang Rai, Mae Hong Son, Chiang Mai, Nan and Lampang, June 2023-May 2026. Log-normal, gamma and Weibull fitted by maximum likelihood and compared by Kolmogorov-Smirnov, Anderson-Darling, AIC and BIC. Log-normal won at all stations. Exceedance of the 24-h 37.5 $\mu g\,m^{-3}$ standard: Nan 25.12%, Lampang 21.70%, Chiang Rai 20.87%. No CI. Limitation: treating strongly autocorrelated, bimodal-season data as independent draws invalidates the goodness-of-fit p-values and understates tail uncertainty, and three years do not constrain extremes. Relevance: low - a descriptive baseline for a haze region where low-cost networks are already dense; useful for setting sensor range and alert thresholds.

doi: 10.3390/atmos17111081

Khan Pakistan winter smog $PM_{2.5}$-associated antimicrobial resistance genes qPCR 2026 Research Square

$PM_{2.5}$-Associated Variation in Airborne Antimicrobial Resistance Gene Abundance During Winter Smog Episodes in Urban Pakistan (2022–2023)

Winter $PM_{2.5}$ sampled in three Pakistani cities; eDNA qPCR for sul1, blaTEM, blaCTX-M-32, blaNDM-1, vanA, vanRA, tnpA, intI1. Lahore $PM_{2.5}$ reached 276.6 +/- 27.4 $\mu g\,m^{-3}$ (AQI 289). blaNDM-1 had the highest relative abundance (0.077 Lahore, 0.056 Islamabad); vanA was higher in Okara. $PM_{2.5}$ correlated with AQI (r=0.92) and humidity (r=0.91), which is near-tautological. No ARG-$PM_{2.5}$ regression, sample count or blank controls are reported, so the PM link is spatial co-variation across three sites. Preprint. Relevance: minimal for sensing - a bioaerosol dimension that mass sensors cannot resolve, at concentrations where optical sensors also saturate.

doi: 10.21203/rs.3.rs-11398870/v1

Kim CMAQ-UNIPAR secondary organic aerosol South Korea ASIA-AQ 2026 Atmos Environ

Prediction of Secondary Organic Aerosol Formation in South Korea using CMAQ-UNIPAR during the ASIA-AQ Campaign

METADATA ONLY. Metadata only; no abstract in Crossref at issue time, so no finding is asserted. The title places a CMAQ run with the UNIPAR SOA module against the ASIA-AQ campaign over South Korea. Carried because a mechanistic SOA module tested against an airborne and ground campaign bears on the organic fraction of $PM_{2.5}$, the component that most changes refractive index and hygroscopicity across a sensor's calibration range, and on the CTM surfaces health cohorts inherit. To be assessed when the Elsevier abstract deposits: bias against campaign OA and any change in simulated $PM_{2.5}$. Relevance: possible, unassessed.

doi: 10.1016/j.atmosenv.2026.122465

Kim Korea urban forest $PM_{2.5}$ stable isotopes receptor model seasonal 2026 Environ Pollut

Seasonal variation in urban PM(2.5) sources and the modulating role of urban forests.

Two urban-forest regions (Gomae, expressway-to-forest transect; Sihwa, industrial complex to residential with a forest buffer) sampled across seasons with $PM_{2.5}$ mass, stable carbon and nitrogen isotopes, meteorology and a vegetation index, plus receptor modelling. d15N pointed to vehicle emissions in both regions, rising in Sihwa in winter consistent with coal. Forest-interior $PM_{2.5}$ was significantly lower in the growing season, but the difference vanished in the non-growing season, even under evergreen canopy, when concentrations are highest. No effect sizes, site counts or sampling durations are given; two regions cannot separate canopy from distance-to-source decay. Relevance: indirect - transect designs like this are cheap to densify with paired low-cost nodes and would resolve the season-by-canopy interaction properly.

doi: 10.1016/j.envpol.2026.129432

Ku North China Plain new particle formation GEOS-Chem-APM spring campaign 2026 Environ Sci Technol

Investigations of New Particle Formation Mechanisms in North China Plain during the Spring Campaign Based on Observations and Simulations

One-month 2016 spring campaign in an industrial North China Plain city: 10 NPF days, each after nighttime northwesterly ventilation under dry daytime conditions. GEOS-Chem-APM reproduced 6 of 10 events; NH_3 was in excess; tripling H2SO4 raised nucleation but did not reproduce particle evolution, and simulated growth rates were 51% and 75% too low on well- and poorly-simulated days. Ten events and one season limit generality, and the campaign is a decade old. No mass or health metric. Relevance: indirect - NPF bursts raise ultrafine number with little $PM_{2.5}$ mass, the regime mass-calibrated optical nodes cannot see.

doi: 10.1021/acs.est.6c07540

Kutarna SWAPIT Toronto winter NO_2 O_3 $PM_{2.5}$ AQHI representativeness 2026 J Air Waste Manag Assoc

Insights into urban concentrations and variability of NO(2), O(3), and PM(2.5) from the study of winter air pollution in Toronto (SWAPIT).

Eight continuous sites across Toronto, six weeks January-March 2024, plus NAPS records from 1997. The published community AQHI, built from four NAPS stations, deviated by more than 2 AQHI units above and below values at other sites. NO_2 dominated short-term AQHI risk and rose under calm winds; O_3 rose with wind speed; $PM_{2.5}$ showed local and regional features. Median AQHI fell almost twice as fast in summer as in winter, leaving winter higher since 2013. No CI. Limitation: one six-week winter at eight sites, and the abstract names neither instrument class nor site typology, so the deviation cannot be assigned to siting or to method. Relevance: high - a quantified gap between a four-station index and the intra-urban field, the case for denser sensing, though NO_2 rather than $PM_{2.5}$ drives it.

doi: 10.1080/10962247.2026.2743096

Li Wuhan impervious surface morphology land surface temperature $PM_{2.5}$ 2015-2024 2026 Atmos Pollut Res

Combined effects of impervious surface morphology on land surface temperature and $PM_{2.5}$ pollution: Evidence from long-term observations (2015–2024) in Wuhan, China

METADATA ONLY. Metadata only; no abstract in Crossref at issue time, so no finding is asserted. The title promises a decade (2015-2024) of observations in Wuhan relating impervious-surface morphology to both land surface temperature and $PM_{2.5}$. Carried because urban-form predictors are standard LUR inputs and a joint heat-PM treatment matters for co-exposure studies, since heat and $PM_{2.5}$ share urban-form drivers and confound each other in health models. To be assessed when the abstract deposits: whether $PM_{2.5}$ comes from monitors or satellite retrievals, spatial unit, and whether morphology effects survive meteorological adjustment and wind. Relevance: possible, unassessed; a candidate LUR-predictor source.

doi: 10.1016/j.apr.2026.103261

Liu refined SOA hygroscopicity aerosol liquid water CCN uncertainty 2026 npj Clim Atmos Sci

Refined SOA hygroscopicity reduces climate-relevant uncertainties in aerosol liquid water and CCN predictions

METADATA ONLY. Metadata only; no abstract in Crossref at issue time. The title claims that a refined treatment of secondary organic aerosol hygroscopicity reduces uncertainty in predicted aerosol liquid water and cloud condensation nuclei. The watch carries it because hygroscopicity is the quantity that links composition to the humidity response of every optical PM sensor: the 1 November issue logged a CMAQ result moving kappa by about -5.6% through isoprene organosulfates alone. A refined kappa for SOA is the same term approached from the climate side, and if it is spatially structured it is another reason a single relative-humidity correction cannot be transferred between regions. Flagged for the weekly re-sweep.

doi: 10.1038/s41612-026-01587-2

Lv MERRA-2 CAMS aerosol components AERONET GRASP intercomparison trends 2026 Atmospheric Environment

Global Aerosol Component Distributions and Long-Term Trends in MERRA-2 and CAMS: Intercomparison and Evaluation with AERONET GRASP/Component Retrievals

METADATA ONLY. Metadata only; no abstract in Crossref at issue time. The title describes an intercomparison of aerosol component distributions and long-term trends between MERRA-2 and CAMS, evaluated against AERONET GRASP component retrievals. This is the sort of record the watch most wants and can say least about on metadata: reanalysis speciation is what feeds the composition term in satellite-surface $PM_{2.5}$ fusion, and a documented divergence between the two reanalyses would propagate straight into any exposure surface built on either. Flagged for the weekly Elsevier abstract re-sweep.

doi: 10.1016/j.atmosenv.2026.122466

Mohsin California disaggregated AA NHPI wildfire-specific $PM_{2.5}$ disparities 2026 J Clim Change Health

Race disparities in total PM(2.5) and wildfire-specific PM(2.5) exposure among disaggregated Asian American, Native Hawaiian, and Pacific Islander groups in California.

California census tracts, 2008-2012, total and wildfire-specific $PM_{2.5}$ weighted by 2010 Census counts across five broad and 29 disaggregated AA and NHPI groups. Hmong populations had the highest average total and wildfire-specific $PM_{2.5}$; Fijian, Laotian, Bhutanese, Marshallese and other Micronesian groups also ranked high for wildfire $PM_{2.5}$. The abstract gives no concentrations or contrasts, the exposure period predates the 2017-2021 fire seasons, and the method for isolating wildfire $PM_{2.5}$ from AQS data is unstated. Relevance: indirect - Central Valley Hmong communities are exactly where community sensor deployments could update a 15-year-old picture.

doi: 10.1016/j.joclim.2026.100755

Park dual-metric duration-aware rainfall regime surface particulate matter 2026 Atmospheric Research

A dual-metric, duration-aware assessment of how rainfall regimes control surface particulate matter

METADATA ONLY. Metadata only; no abstract in Crossref at issue time. The title promises a duration-aware treatment of how rainfall regimes control surface PM, using two metrics. Wet deposition is one of the larger unmodelled terms in short-term PM variability, and duration rather than intensity being the operative variable would matter for both forecast evaluation and for the humidity regime in which optical sensors are read: rain-period data are routinely the worst-behaved segment of a low-cost record. Nothing is asserted here. Flagged for the weekly Elsevier abstract re-sweep.

doi: 10.1016/j.atmosres.2026.109429

Qu karst city $PM_{2.5}$ rare earth elements heavy REE enrichment diurnal 2026 Atmospheric Environment

Heavy rare earth enrichment and diurnal source invariance of $PM_{2.5}$-bound rare earth elements in a karst city, southwestern China

METADATA ONLY. Metadata only; no abstract in Crossref at issue time and no Europe PMC record, so no finding is asserted. The title claims heavy rare earth enrichment and diurnal source invariance in $PM_{2.5}$ in a karst city. Were that supported, the value here is a composition tracer whose source signature does not move between day and night, which is unusual and would make it a useful constraint on apportionment in a terrain where boundary-layer behaviour dominates. Flagged for the weekly Elsevier abstract re-sweep.

doi: 10.1016/j.atmosenv.2026.122467

Rehman Lahore GTWR satellite $PM_{2.5}$ urban expansion climate 2015-2024 Research Square

Urban expansion, seasonal climate and fine particulate matter in Lahore, Pakistan: a spatiotemporal analysis of regional environmental change (2015-2024)

Preprint. Satellite $PM_{2.5}$ across Lahore 2015-2024 modelled with two geographically and temporally weighted regressions: climatic (1,400 point-seasons, R2 0.894) and socioeconomic (5,710 point-years, R2 0.562). Annual mean $PM_{2.5}$ 62.9-74.5 $\mu g\,m^{-3}$, rising about 4.7% over the decade and spreading to the periphery, peaking in winter. Humidity was positive and rainfall negative; wind speed was dispersive in the north-east but positive on the western fringe, consistent with crop-burning smoke transport; built-up area, road density and population were strongest on the south-western periphery. The satellite product's own error in Lahore smog is not reported and GTWR coefficients are descriptive, not causal. Relevance: indirect - maps where a Lahore sensor network should be densified.

doi: 10.21203/rs.3.rs-11602212/v1

Sathish Kanpur Mumbai nine-stage size-resolved aerosol chemistry ICRP deposited dose 2026 Environ Res

Size-resolved measurements matter for chemistry and health: paired inland-coastal size-resolved aerosol chemistry and the inadequacy of bulk PM(2.5) dose estimates.

Paired Kanpur (inland IGP) and Mumbai (coastal) impactor sampling over three seasons. Fine mode (PM2.1) is up to 62% of mass in Kanpur, not Mumbai; Kanpur fine aerosol is more acidic (pH 3.5-4.5 vs 4.5-5.0). ICRP 66 dosimetry: fine mode delivers 53-75% of alveolar dose from 40-62% of mass, and Kanpur alveolar doses are 1.3-2 times Mumbai's. The authors flag that SOC-water associations lose significance at one value per sample day, and dose depends on assumed deposition fractions. Relevance: direct - a calibration fitted at a coastal site will misweight the inland fine mode that carries the dose.

doi: 10.1016/j.envres.2026.126094

Schwarz Helmholtz letter ultrafine particle health studies forest for the trees 2026 Environ Res

Letter to the Editor: Health effect studies of ultrafine air pollution-Are we missing the forest for the trees?

Metadata only; a letter with no abstract, so no argument is asserted beyond the title. From the Helmholtz Munich epidemiology group (Schwarz, Pickford, Cyrys, Breitner-Busch), which runs long-term UFP number and size measurement in Augsburg. The title signals a critique of how UFP health studies are framed - plausibly metric choice (number vs surface vs size bins) or the pursuit of UFP-specific effects independent of correlated traffic pollutants. To be read when the full text is accessible. Relevance: possible - any UFP exposure standard decides what number-concentration sensors in a network should report.

doi: 10.1016/j.envres.2026.126091

Shankar meteorological drivers urban air quality weather-pollution interactions 2026 Atmos Environ

Understanding the Meteorological Drivers of Urban Air Quality: Weather-Air Pollution Interactions

METADATA ONLY. Metadata only; no abstract in Crossref at issue time, so no finding is asserted. The title points to an analysis, or possibly a review, of how weather variables drive urban air quality. Carried provisionally: meteorological normalisation is a prerequisite for interpreting network trends and for separating calibration drift from real change. Geography, pollutants and design are unknown, and it may be gas-dominated, in which case it will be noted as such in the weekly. Relevance: possible, unassessed.

doi: 10.1016/j.atmosenv.2026.122468

Sharma $PM_{2.5}$-bound water soluble organic carbon urban sources metadata only 2026 Atmos Pollut Res

Variability in sources and processes of $PM_{2.5}$-bound Water Soluble Organic Carbon over an urban location in the North-Western Himalayas

METADATA ONLY. Elsevier day-0 deposit, no abstract in Crossref or Semantic Scholar at issue time; no finding is asserted. The title places it in $PM_{2.5}$ water-soluble organic carbon source and process variability at an urban location. Limitation: content unread. Relevance: unknown, retry the abstract at the weekly sweep.

doi: 10.1016/j.apr.2026.103262

Tan Shenzhen PMF-ML firework burning $PM_{2.5}$ source apportionment 2026 Environ Res

Apportioning Firework Burning and Routine Source Contributions to PM(2.5) in a Chinese Megacity Using Two PMF-Machine Learning Frameworks.

Three Spring Festival seasons of hourly speciation in Shenzhen; PMF resolved 11 routine sources plus firework burning. Fireworks contributed 4.7% of $PM_{2.5}$ over the full period, 33% during display episodes and over 50% at peak hours. Two ML surrogates (direct and indirect) reproduced PMF contributions and flagged fireworks as dominant during episodes. Agreement with PMF is circular validation - the ML learns PMF output, so it inherits PMF's factor ambiguity and no independent tracer test is reported. Relevance: direct for networks - episodic sources at >50% of hourly mass are exactly where a fixed humidity/composition calibration on optical sensors breaks.

doi: 10.1016/j.envres.2026.126121

Wang Kwan Shenzhen 110,000 mobile signalling year-long temporal exposure averaging problem 2026 Environ Sci Ecotechnol

The temporal exposure averaging problem (TEAP).

Year-long mobile-signalling trajectories for 110,000 Shenzhen residents were combined with pollutant and greenspace surfaces; months were resampled to test how observation length shapes chronic-exposure estimates. Short windows produced large intra-individual deviations from annual exposure and inflated between-person variance; longer windows converged steadily, consistently across age, gender and indicators, and seasonally stratified vs random four-month samples agreed. No numerical deviation or convergence threshold is in the abstract, and the surfaces themselves carry model error that this framing does not address. Relevance: direct - a formal warning for wearable-sensor campaigns of days to weeks that report exposure inequality.

doi: 10.1016/j.ese.2026.100782

Wayland discussion data fusion air pollution exposure modeling metadata only 2026 J Air Waste Manag Assoc

Discussion of "Data fusion for air pollution exposure modeling".

METADATA ONLY. No abstract available from PubMed or Crossref at issue time, and no finding is asserted. PubMed lists this as a discussion at J Air Waste Manag Assoc 76(11):908-916, with authors from the retired EPA, Resources for the Future, Emory and Tennessee among others. Limitation: content unread. Relevance: on title alone high - data fusion is where sensor, monitor, satellite and model fields are combined; retrieve the full text and the paper under discussion.

doi: 10.1080/10962247.2026.2737061

Wei dust storm events Northeast China GEOS-Chem PM measurements metadata only 2026 Atmos Pollut Res

Dust storm events in Northeast China by GEOS-Chem modeling and PM measurements

METADATA ONLY. Elsevier day-0 deposit, no abstract in Crossref or Semantic Scholar at issue time; no finding is asserted. The title places it in dust storm events over Northeast China using GEOS-Chem and PM measurements. Limitation: content unread. Relevance: unknown on content; dust regimes are where low-cost optical sensors undercount, so retry the abstract at the weekly sweep.

doi: 10.1016/j.apr.2026.103259

Wu airborne plastics plastic-associated chemicals $PM_{2.5}$ Singapore PMF I168O 2026 Environ Sci Technol

Fingerprints of Airborne Plastics in Urban $PM_{2.5}$: Applications and Implications

240 daily $PM_{2.5}$ samples, Singapore 2018-2020, nine plastic-associated chemicals, seven apportioned sources. I168O made up 90% of quantified chemicals and identified at least 40% more $PM_{2.5}$ linked to respirable airborne plastic chemicals. A resolved plastic production, use and burning factor held 37% of quantified chemicals; traffic and road dust held over 20% of near-range chemicals, with DNBP/DEHP above 1 proposed as a non-exhaust marker. Over 90% of DEP sat with transported peat smoke. No uncertainty on factor shares in the abstract. Limitation: one site is implied, and I168O is an additive oxidation product with other potential sources, so the factor measures additives, not plastic mass. Relevance: low for mass sensing - a speciation layer invisible to optical nodes.

doi: 10.1021/acs.est.6c09617

Wu penguin colony chlorophyll-a mediation PM_1 $PM_{2.5}$ PM_{10} Ardley Island 2026 Mar Pollut Bull

Phytoplankton-mediated coupling between penguin colonies and atmospheric particulate matter in Antarctic coastal waters.

Synchronous ocean-atmosphere transects around Ardley Island. After controlling for spatial autocorrelation, chlorophyll-a remained associated with $PM_{2.5}$ (standardized coefficient 0.12). Across PM_1, $PM_{2.5}$ and PM_{10} indirect NH4-Chl-a-PM effects exceeded direct ones, up to eightfold, but only at alpha 0.10, with 95% CIs including zero. The authors nonetheless propose Chl-a as a proxy for the colony atmospheric footprint. Limitation: the mediation result does not exclude zero, the transects are one spatial sample, and particle composition is not reported, so marine biogenic origin is inferred. Relevance: low - a very clean background where optical PM readings are near any low-cost sensor's noise floor.

doi: 10.1016/j.marpolbul.2026.120512

Yao Harbin early-spring heavy pollution episodes natural anthropogenic drivers 2026 Atmos Pollut Res

Natural and anthropogenic drivers of heavy pollution episodes in early spring in Harbin, Northeast China

METADATA ONLY. Metadata only; no abstract in Crossref at issue time, so no finding is asserted. The title sets natural against anthropogenic drivers of early-spring heavy pollution in Harbin, a period where heating-season combustion, agricultural residue burning and stagnant boundary layers overlap. Carried because cold, humid episodes are the regime where optical low-cost sensors overestimate most and where humidity correction matters. To be assessed when the abstract deposits: whether it separates open burning from heating and uses PM composition. Relevance: possible, unassessed.

doi: 10.1016/j.apr.2026.103263

Yin brake-wear particle emissions thermal history ultrafine number optical PHEV 2026 J Hazard Mater

Mechanical and thermal controls on brake-wear particle emissions and physicochemical properties in a plug-in hybrid vehicle: A comparison of two brake-pad formulations.

A plug-in hybrid was tested with low-metallic and non-asbestos organic pads using wheel-end sampling with event-resolved hydraulic pressure. Decelerations with effective or high mechanical input were 36% of events but 80.3-92.7% of the PM_{10} mass response. Sustained heating raised sub-100 nm number concentrations 48.3 to 98.1-fold, and emissions stayed elevated through cooling while the micrometre mass mode persisted: number and mass decouple, and both depend on thermal history. Single-scattering albedo ran 0.606-0.766 and mass absorption efficiency 0.364-0.747 m2/g, evolving differently by pad material, with no increase in mass-normalized absorption from high-temperature carbonaceous enrichment. Limitation: one vehicle, two formulations, no CI. Relevance: direct - a hundredfold ultrafine excursion invisible to a mass sensor, with pad-dependent 630 nm optics that sit inside the wavelength band low-cost nephelometers actually use.

doi: 10.1016/j.jhazmat.2026.144101

Young CSU San Joaquin Valley agricultural community personal indoor $PM_{2.5}$ PM_{10} 2026 Environments

Particulate Matter Pollution in an Agricultural Setting: A Community-Engaged Research Study.

Community-engaged study, San Joaquin Valley: 35 Hispanic/Latino(a) participants from 18 households living a median 633 m from processing silos, sampled in December 2023, May 2024 and the September 2024 harvest. Median personal $PM_{2.5}$ 11.1 and PM_{10} 45.5 $\mu g\,m^{-3}$; indoor 12.9 and 24.3 $\mu g\,m^{-3}$. 29% of $PM_{2.5}$ and 33% of PM_{10} samples exceeded WHO 24-h guidelines. Agricultural work in the household and regional ambient PM were the strongest predictors. Small n, three visits, and no coarse-fraction composition to separate soil from processing dust. Relevance: direct - personal PM_{10} four times indoor PM_{10} shows a coarse take-home or activity component that ambient $PM_{2.5}$ sensors miss entirely.

doi: 10.3390/environments12100348

Yu Jeju Lunar New Year 2026 fireworks long-range transport K+ aerosol 2026 Preprints.org

Fireworks-Influenced Aerosols on Jeju Island, Korea, During the 2026 Lunar New Year

Seogwipo, Jeju, 14-23 February 2026: two K+-enriched episodes with mean K+ 2.59 +/- 1.44 and 1.50 +/- 0.66 $\mu g\,m^{-3}$, about 99% non-sea-salt. K+/total carbon was 5.5 and 3.9 times the pre-episode value, while absorption Angstrom exponents of 1.37 and 1.29 gave no optical sign of biomass burning; median transit from 120 E was 23 h, crossing on Lunar New Year's Day. Attribution to transported Chinese fireworks rests on K+ ratios and timing, not tracers such as Sr, Ba or Mg; $PM_{2.5}$ mass is not reported in the abstract. Preprint. Relevance: indirect - a transported episode with fresh-composition particles.

doi: 10.20944/preprints202611.0554.v1

Zhang source-resolved $PM_{2.5}$ metals Bayesian inversion California hazard index 2026 Research Square (preprint)

Source-Resolved $PM_{2.5}$ Metals Emissions, Exposure, and Health Risks in California Using an Observation-Constrained Hybrid Model

Source-oriented CTM output was reconciled with observations by Bayesian inversion, lowering the mean fractional error from 0.835 to 0.511. Miscellaneous sources supplied 9.1% of emissions but 24.1% of population-weighted exposure and 57.7% of the non-carcinogenic hazard index, driven by cadmium; industrial and miscellaneous sectors each carried roughly 35% of excess lifetime cancer risk across six carcinogenic metals. Census-tract clusters with joint contributions from both sectors were concentrated in metropolitan Los Angeles. No CI on any figure. Limitation: a preprint, and the inversion is only as good as the sparse speciation network that constrains it, which is the same sparsity the paper is trying to overcome. Relevance: moderate - mass-only sensing is blind to exactly the composition that drives these risk gradients.

doi: 10.21203/rs.3.rs-11579816/v1

Cardiovascular & metabolic — 9 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (9 OF 9 RECORDS IN THIS SUBTOPIC)

Alli ambient air pollution blood pressure schoolchildren Accra LUR 2026 Sci Rep

Urban ambient air pollution and blood pressure in schoolchildren in Accra, Ghana

BACKFILL (entry date 4 Jun 2026; Amass). 919 children aged 7-14 in 90 schools; one-third with elevated BP. Per 10 units home exposure: SBP $PM_{2.5}$ -0.70 mmHg (-3.34 to 1.94), BC -4.61 (-9.59 to 0.37); DBP $PM_{2.5}$ -0.83 (-2.93 to 1.28), BC -4.35 (-8.19 to -0.50); elevated BP $PM_{2.5}$ OR 0.75 (0.39-1.41). Exposures ran 5-8 times the WHO annual $PM_{2.5}$ guideline. Limitation: cross-sectional, and inverse associations in a high-exposure city suggest residual confounding by socioeconomic or adiposity patterning; the upper bound does not exclude a moderate SBP rise. Relevance: high - the LUR rests on a 146-site campaign in sub-Saharan Africa, the design a low-cost network could sustain.

doi: 10.1038/s41598-026-54135-6

He Beijing panel 152 young adults PNC5-560 gut microbiome pro-atherosclerotic biomarkers 2026 Environ Health Perspect

Exposure to Size-Fractioned Ambient Particles Associated with Alterations in Gut Microbiome and Pro-Atherosclerotic Biomarkers in Young Adults.

Beijing panel of 152 young adults, four monthly visits (Sep 2019-Jan 2020). PNC 5-560 nm sources: diesel 37.9%, gasoline 39.4%, nucleation 9.2%, secondary 11.9%. Per IQR of 7-day PNC, inflammatory and pro-atherosclerotic biomarkers rose by 2.1% (0.4-3.8) to 198.6% (29.3-365.8); Shannon, Simpson and Chao1 diversity fell. Microbial mediation reached 9.8% (0.8-38.3) and 4.8% (0.09-22.5). The 2-to-199% span across many biomarkers signals heavy multiplicity, mediation shares are small with wide CIs, and a single fixed-site PNC stands in for personal number exposure. Relevance: indirect - number concentration, not mass, carries the signal, and mass-calibrated optical sensors cannot see most of the 5-560 nm range.

doi: 10.1021/ehp.6c00374

Kulkayeva Almaty time-series $PM_{2.5}$ cardiovascular mortality null 2026 Front Public Health

Short-term effects of ambient $PM_{2.5}$ exposure on cardiovascular mortality in Almaty, Kazakhstan.

Almaty 2017-2025, 3,287 days, 43,095 cardiovascular deaths, mean $PM_{2.5}$ 36.9 $\mu g\,m^{-3}$. Cumulative lag 0-7 RR 1.011 (0.837-1.222), null; age >=65 RR 1.174 (0.926-1.489), also null; lag-specific risk peaked at 2-4 days. The comparison increment for the cumulative RR is not given in the abstract, the CI spans a 16% deficit to a 22% excess, and the $PM_{2.5}$ series source and station count are unstated. Relevance: indirect - Almaty's monitor record is thin and its PurpleAir network is one of the denser in Central Asia; a sensor-fused exposure series could narrow exactly this interval. Included as a reported null from an under-studied region.

doi: 10.3389/fpubh.2026.1950463

Lin prospective cohort plasma proteomics $PM_{2.5}$ type 2 diabetes mediation 2026 Environ Pollut

Plasma Proteomic Profiling Reveals Mediating Proteins and Preventive Targets Linking Ambient Air Pollution to Type 2 Diabetes Incidence: A Prospective Cohort Study

44,917 participants with 2,911 plasma proteins, 13.64 years of follow-up, 3,193 incident T2D. Per IQR of $PM_{2.5}$, NO_2, NOx and SO_2, T2D risk was 5-14% higher (pollutant-specific HRs not separated in the abstract). 187 proteins mediated the $PM_{2.5}$-T2D association; a protein score mediated 31.69% for $PM_{2.5}$. Hubs sat in cytokine, chemokine and leukocyte-migration pathways; 13 targets have approved drugs. The cohort is not named in the abstract. Proteins measured once, at baseline alongside the exposure window, make the mediation temporally ambiguous, and NO_2, NOx and SO_2 are correlated with $PM_{2.5}$. Relevance: indirect - mechanism-oriented, with interpolated residential exposure and no new measurement contribution.

doi: 10.1016/j.envpol.2026.129476

Saz-Lara CarVasCLiMa Castilla-La Mancha cohort protocol air pollution satellite arterial stiffness BMJ Open

Impact of environmental factors on cardiovascular health in Castilla-La Mancha: the CarVasCLiMa cohort study protocol.

Protocol, no results. A representative urban, semi-urban and rural cohort in Castilla-La Mancha followed for 2 years, with air pollution and climate assigned from high-resolution satellite products supplemented by local stations, greenness by NDVI, and outcomes of blood pressure, arterial stiffness, C-reactive protein, quality of life and lifestyle. Sample size, pollutant list, spatial resolution and the exposure averaging windows are not given; two years is short for arterial-stiffness change, and rural Spain has few stations to anchor the satellite surface. Relevance: indirect - a rural cohort where co-located low-cost sensors could validate the satellite assignment before outcomes accrue.

doi: 10.1136/bmjopen-2026-123803

Shi CFPS compound dry-hot extremes $PM_{2.5}$ joint incident CVD China 2026 BMC Public Health

Joint effects of air pollution and compound dry-hot extremes on cardiovascular disease risk across adults: a cohort study in China.

China Family Panel Studies, 9,265 adults with 69,297 repeated observations, 2010-2018, mixed-effects Cox. Per IQR, $PM_{2.5}$ HR 1.54 (1.42-1.67) for CVD and 1.75 (1.46-2.11) for coronary disease; dry-hot cumulative days HR 1.30 (1.20-1.41). Concurrent high exposure to both gave a joint HR of 1.92 (1.67-2.21), called synergistic, and repeated across PM_1, PM_{10}, NO_2 and O_3. Limitation: an IQR HR of 1.54 is far above the cohort literature and neither the IQR nor the exposure source is stated; the abstract does not specify how CVD was ascertained; joint exposure and synergy are different quantities, and no interaction scale is given. Relevance: low - no measurement content, but a compound-exposure design that a network could sharpen.

doi: 10.1186/s12889-026-28927-4

Siddiqui narrative review AQI ASCVD South Asians 2026 Curr Atheroscler Rep

Impact of Air Quality Index on Atherosclerotic Cardiovascular Disease Risk in South Asians: A Narrative Review.

Narrative review arguing that worse AQI and its constituent pollutants raise ASCVD risk in South Asians, who already carry high genetic and metabolic predisposition, and that pollution may compound traditional risk factors. Reports no search strategy, no pooled estimate and no South Asia-specific effect sizes; it concedes that region-specific intervention evidence is thin. Using AQI as the exposure is the core weakness: it is a max-of-pollutants communication index, not a dose metric, and conflates PM with gases. Relevance: indirect - the region with the highest burden still lacks cohort-grade exposure, which is where dense sensing would add most.

doi: 10.1007/s11883-026-01486-0

Singh German Diabetes Study long-term $PM_{2.5}$ clamp insulin sensitivity albuminuria T1D T2D 2026 Diabetologia

Associations of long-term exposure to air pollution with glucose homeostasis biomarkers and diabetes-related comorbidities in people with recent-onset type 1 and type 2 diabetes.

Cross-sectional analysis of 1,120 German Diabetes Study participants with recent-onset diabetes (417 T1D, 703 T2D), gold-standard clamp phenotyping. Per IQR $PM_{2.5}$: T1D HbA1c exp(b) 1.07 (1.03-1.11), albuminuria 1.72 (1.34-2.20), VO2max 0.94 (0.89-0.99); T2D HbA1c 1.04 (1.01-1.06), insulin sensitivity b -0.42 (-0.82 to -0.03), insulin clearance 0.92 (0.87-0.98), albuminuria 1.71 (1.41-2.08), DSPN OR 1.53 (1.02-2.32). Many outcomes are tested without stated multiplicity control, and a single-time-point design cannot order exposure and disease progression. The IQR value is not in the abstract. Relevance: indirect - deep phenotyping makes this a strong mechanistic anchor; exposure is modelled, not measured.

doi: 10.1007/s00125-026-06908-x

Xu CHARLS PM_1 $PM_{2.5}$ PM_{10} cardiometabolic multimorbidity cohort 2026 J Investig Med

EXPRESS: Air pollution increases cardiometabolic diseases in China: A nationwide longitudinal cohort study.

CHARLS, 8,725 adults aged >=45 free of cardiometabolic multimorbidity (two or more of hypertension, diabetes, CVD, stroke) at 2011 baseline, followed to 2020. All five pollutants - PM_1, $PM_{2.5}$, PM_{10}, O_3, NO_2 - were associated with incident multimorbidity, more strongly with lower education and short sleep. The abstract reports no hazard ratios, increments, case counts or exposure model, so magnitude cannot be judged; participants with one condition at baseline were retained, which mixes progression with onset. Relevance: indirect - one more CHARLS reuse with a satellite-type exposure inferred; no new measurement contribution.

doi: 10.1177/10815589261507081

Neuro / mental health — 7 records

Carried from the daily issues — full summaries (7 of 7 records in this subtopic)

Calderon-Garciduenas ferrimagnetic nanoparticles olfactory bulb cortex Mexico City children 2026 Gac Med Mex

The olfactory bulb, prefrontal and anterior cingulate cortex are early pediatric targets for anthropogenic magnetic nanoparticles and their motion behavior under electromagnetic fields.

48 brain samples (olfactory bulb, anterior cingulate, prefrontal cortex) from 12 children (8.5 +/- 7.6 y) and 16 adults (39.2 +/- 19.6 y), Metropolitan Mexico City. Magnetophoresis at 25-1,000 mT AC showed small displacements and significant vector-direction variation; TEM and EDX located ferrimagnetic nanoparticles. No effect size. Limitation: autopsy convenience series with no comparison population or exposure measure in the abstract, so 'early pediatric targets' and the neural-damage risk are inferred, not tested; particle attribution to anthropogenic sources is not apportioned here. Relevance: low for sensing - mass-calibrated optical nodes are blind to this metallic ultrafine fraction, which is the actual argument for co-located number or magnetic measurements.

doi: 10.24875/gmm.m26001158

Cardenas-Iniguez ABCD linked external data environment review first decade 2026 Dev Cogn Neurosci

Environmental influences using linked external data in ABCD: A review of the first decade.

Ten years of Linked External Data work in the Adolescent Brain Cognitive Development Study (~11,800 US youth), covering neighbourhood conditions, built environment, air pollution, greenspace and state policy. The authors conclude that development reflects cumulative influences and that timing and study design bound what can be concluded, and they call for refined geocoding and exposure methods. No effect sizes in the abstract. Limitation: an internal working-group review of its own dataset, and residential geocoding ignores school and transit time, which matters at this age. Relevance: moderate - the largest US neurodevelopment cohort names exposure resolution as a limiting method.

doi: 10.1016/j.dcn.2026.101822

Cho exercise under particulate matter exposure BDNF neurotrophic narrative review 2026 Atmosphere

Exercise Under Particulate Matter Exposure: Neurotrophic Responses and Functional Brain Outcomes

Focused narrative review, PubMed and Web of Science, 37 reports (33 main, four indirect). Circulating BDNF responses to exercise under PM were heterogeneous; VEGF and IGF-1 gave complementary peripheral evidence. Human studies mixed attenuation in selected cognitive, dementia and imaging contrasts with retained activity benefit and possible buffering. Central molecular evidence from inhalation was limited; some animal improvements came from intravenous or instillation models. The authors find support for neither a uniform loss of exercise benefit nor a general reversal threshold. No pooling. Limitation: non-systematic, overlapping cohorts possible. Relevance: low-moderate - the stated gap is activity-specific exposure, which is what wearable PM sensing alongside exercise intensity could supply.

doi: 10.3390/atmos17111077

Defrancesco lung-brain axis review $PM_{2.5}$ Alzheimer disease mechanisms 2026 Clin Sci (Lond)

Mechanistic understanding of the role of the lung-brain axis in the pathogenesis of $PM_{2.5}$-induced Alzheimer's disease.

Narrative review positioning pulmonary injury, rather than direct translocation, as the main route from inhaled $PM_{2.5}$ to AD pathology: systemic inflammation, oxidative stress, blood-brain barrier disruption and extracellular-vesicle signalling converge on amyloid, tau, microglial and autophagy-lysosomal dysfunction. Modifiers named are composition, co-pollutants, sex, APOE and epigenetic reprogramming. No search protocol, no quantitative synthesis; the framework is a hypothesis map and the human causal link remains epidemiological association. Relevance: indirect - if the pulmonary route dominates, deposited dose in the alveolar region, not ambient mass, is the exposure metric a sensor should be calibrated to serve.

doi: 10.1042/cs20262109

Huang UK Biobank CKD dementia multidomain risk factors air pollution null 2026 BMC Neurol

Disease susceptibility and risk of dementia in patients with chronic kidney disease: a prospective cohort study based on UK Biobank.

16,083 UK Biobank participants with CKD followed 13.8 years; 806 incident dementias. Cox models with inverse probability weighting, competing-risk and FDR correction screened lifestyle, clinical, social, device-use, environmental and sensory factors. Hypertension, diabetes, depression, hearing loss and low social activity raised risk; moderate activity and computer use lowered it. Air pollution showed no nominally significant association. With 806 events spread over dozens of exposures and UK Biobank's 2010 single-year LUR surfaces, power for a PM effect of the expected size (HR ~1.05-1.10 per few $\mu g\,m^{-3}$) in a subgroup is low, so the null is uninformative rather than reassuring. Relevance: weak - a null by underpowered screening.

doi: 10.1186/s12883-026-05212-w

Zareba systematic review PM neurological health older adults 21 studies 2026 Curr Environ Health Rep

Impact of Particulate Matter Exposure on Neurological Health in Older Adults: A Systematic Review.

PRISMA review of PubMed, Scopus, Web of Science and ScienceDirect to January 2026; 21 observational studies in older adults. Cognitive performance, decline and MCI were the most studied outcomes, with several adverse associations for long-term $PM_{2.5}$; dementia and Alzheimer evidence came from large cohorts but varied with exposure metric and endpoint; Parkinson hospitalisation, amyloid burden, retinal neurodegeneration and constituent-specific findings rested on one or a few studies. No pooled estimate, no risk-of-bias grading in the abstract, and 21 studies is a narrow catch for a field with dozens of large dementia cohorts, so the search or eligibility was restrictive. Relevance: indirect - restates that exposure-assessment heterogeneity, not absence of signal, limits the neuro evidence.

doi: 10.1007/s40572-026-00572-4

Zundel Detroit adolescents $PM_{2.5}$ above 9 $\mu g\,m^{-3}$ fear conditioning amygdala 2026 Biol Psychiatry Glob Open Sci

Outdoor Air Pollution Exposure Alters Fear Regulation and Fear-Related Neural Circuitry in Adolescence.

102 adolescents aged 10-17 (68 with fMRI) completed Pavlovian conditioning, extinction and recall. Past-month $PM_{2.5}$ above vs below 9 $\mu g\,m^{-3}$: greater skin conductance and subjective fear to threat cues, greater return of fear at recall, higher overall SCR, and greater right amygdala (conditioning, extinction) and left insula (recall) activation, adjusted for age, sex and neighbourhood disadvantage. No effect sizes are given in the abstract. Dichotomising a modelled one-month mean at a regulatory cut-point in 102 people invites residual neighbourhood confounding. Relevance: indirect - short-window exposure contrasts like this need residence-level resolution that dense networks can supply.

doi: 10.1016/j.bpsgos.2026.100831

Respiratory & allergic — 16 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (16 OF 16 RECORDS IN THIS SUBTOPIC)

Alimu Kashi Xinjiang ExWAS elastic-net air pollution meteorology pulmonary tuberculosis 2026 BMC Pulm Med

Associations between air pollution, meteorology, and pulmonary tuberculosis risk in Kashi, China.

TB cases from a surveillance platform compared with hospital non-TB controls in Kashi, Xinjiang, using ExWAS, adaptive elastic-net and XGBoost. Retained predictors: PM_{10} OR 1.013, $PM_{2.5}$ OR 1.008, NO_2 OR 1.018 (all $p<0.0001$), wind speed 0.983 and humidity 0.944; XGBoost ranked PM_{10} most important. No confidence intervals, increments, exposure windows or sample size are given in the abstract, and controls drawn from a hospital record system against cases from a separate platform build in selection bias; Kashi PM_{10} is dominated by Taklamakan dust, so PM_{10} also marks season and desert-dust episodes. Relevance: weak - a dust-dominated setting where PM_{10} sensors would be informative, but the design cannot carry a causal reading.

doi: 10.1186/s12890-026-04696-z

Drummond mini-review climate exposures paediatric lung-function trajectories 2026 Eur Respir Rev

Climate change as a modifier of paediatric lung-function trajectories.

Mini-review using lung-function trajectories as its frame. Climate-sensitive exposures shift trajectories downward by three routes: early developmental disruption (gestational pollution and temperature, via growth restriction and preterm birth), sustained exposure impairing postnatal growth and peak, and exacerbations with incomplete recovery. Evidence that cleaner air accelerates lung-function growth is taken as proof the trajectory can be protected. No effect sizes or search method in the abstract. Limitation: co-occurring heat, smoke, allergen and damp are not separated, so attribution to PM is not possible from this frame. Relevance: low - frames why childhood exposure windows matter, without a measurement requirement.

doi: 10.1183/16000617.0222-2026

Espinoza-Guillen Lima case-crossover $PM_{2.5}$ asthma ED road proximity 2026 Environ Res

Ambient fine particulate matter and asthma emergency visits in Lima, Peru: Does Residential Proximity to Major Roads Matter?

Time-stratified case-crossover of 75,602 asthma ED visits (mean age 16.2 y) at 10 Lima public hospitals, 2017-2023, with DLNM. At lag 0, $PM_{2.5}$ rising from 5 to 15 $\mu g\,m^{-3}$ gave OR 1.09 (1.01-1.16); the exposure-response curve was steeper at low concentrations. Cumulative lag 0-15 OR was 4.02 (1.97-8.19) within 400 m of a major road versus 1.66 (0.81-3.38) beyond it (interaction $p=0.04$). The cumulative ORs are for a contrast the abstract does not state and are implausibly large for a 15-day DLNM sum; the road modifier probably carries unmodelled traffic NO_2 and ultrafine particles that a 1-km $PM_{2.5}$ surface cannot resolve. Relevance: direct - a near-road gradient finer than the exposure grid is exactly what curbside low-cost nodes would test.

doi: 10.1016/j.envres.2026.126120

Hao Shanghai case-crossover lab-confirmed influenza six pollutants 2026 Int J Biometeorol

Association between short-term air pollutant exposure and influenza incidence in Shanghai, China: A case-crossover study

Laboratory-confirmed influenza cases, Shanghai 2013-2017, time-stratified case-crossover with IDW exposure at the individual level. Per 10 $\mu g\,m^{-3}$ PM_{10}, OR 1.019 (1.004-1.034); O_3 was inverse (OR 0.959, 0.941-0.977). $PM_{2.5}$, SO_2, NO_2 and CO were positive across lags but their point estimates are not in the abstract. Laboratory confirmation is the design's strength over syndromic counts; IDW from a sparse regulatory grid and selection into testing are the weaknesses, and the negative O_3 term likely absorbs season. Read alongside the same-day Shanxi series: two independent Chinese designs give PM effects near 2% per 10 $\mu g\,m^{-3}$. Relevance: indirect. BACKFILL, entry 20 March 2026.

doi: 10.1007/s00484-026-03163-y

Huang All of Us case-crossover wildfire smoke $PM_{2.5}$ adult asthma exacerbation 2026 GeoHealth

Asthma Exacerbations Following Wildfire Smoke $PM_{2.5}$ Exposure Among U.S. Adults

NIH All of Us: 18,935 adult asthma exacerbations, four matched control periods each, warm months 2006-2019. Wildfire-smoke $PM_{2.5}$ was associated with exacerbation at every lag 0-7 d, adjusted OR 1.27 (1.22-1.33) to 1.38 (1.32-1.44); higher in older adults, non-Hispanic White, Midwestern residents and in May. The abstract does not state the exposure increment or the smoke-PM product, so the OR cannot be scaled, and an OR near 1.3 is large for smoke PM per usual increments - likely a binary smoke-day or large contrast. All of Us EHR linkage selects insured, engaged participants. Relevance: indirect - the Midwestern excess points at long-range transported smoke, where PurpleAir density is thinnest.

doi: 10.1029/2026gh001972

Jean-Luc Kinshasa PurpleAir PA-II-FLEX asthma cross-sectional ISAAC 2026 Environ Geochem Health

Abiotic and socioenvironmental factors of $PM_{2.5}$ pollution associated with asthma in a dense urban road network environment in Kinshasa, D.R. Congo: a cross-sectional study

424 households at 36 sites in six health areas, August-September 2024, asthma prevalence 15.8%. High $PM_{2.5}$ was associated with asthma (adjusted OR 1.210, p<0.001), as were more than 12 h daily outdoor time (OR 5.111) and low-status occupations; proximity to a major road was not (OR 0.939, p=0.069), which the authors read as near-homogeneous traffic pollution across a dense road network. $PM_{2.5}$ reached 291.20 $\mu g\,m^{-3}$ early in the campaign, falling with the rains. No CI is reported for any OR, and the exposure contrast is cross-sectional and ecological at site level. Relevance: direct - a locally corrected PurpleAir network carrying a health inference in a city with no reference network. BACKFILL, entry 12 March 2026.

doi: 10.1007/s10653-026-03125-8

Kutzinski CAPNETZ air pollution sputum pathogen spectrum community-acquired pneumonia 2026 Pneumonia (Nathan)

Air pollution is associated with a distinct pathogen spectrum in community-acquired pneumonia: a prospective multicentre cohort study.

CAPNETZ, 4,446 CAP patients with sputum; pathogen found in 1,581, cell counts in 4,014. Mixed-effects logistic models: S. pneumoniae with SO_2 (OR 1.75, p=0.008) and CO (1.57); E. coli with NO_2 (2.75); higher $PM_{2.5}$ with lower detection of the pathogen examined (OR 0.37, p=0.001). No CIs or increments in the abstract, so magnitudes are ungradable. City-level exposure over a month is coarse, many pollutant-pathogen pairs were tested, and an inverse $PM_{2.5}$ association invites a sampling or antibiotic-pretreatment explanation the abstract does not exclude. Relevance: indirect - outcome definition, not exposure metrology, limits this.

doi: 10.1186/s41479-026-00212-3

Li Shandong ICU cardiac surgery $PM_{2.5}$ hospital-acquired pneumonia DLNM 2026 Front Public Health

Association between short-term fine particulate matter exposure and hospital-acquired pneumonia risk in critically ill post-cardiac surgery patients: a 10-year cohort study using distributed lag non-linear models.

6,196 post-cardiac-surgery ICU patients at one Shandong hospital, 2014-2023; 1,938 (31.3%) developed HAP. With DLNM adjusted for co-pollutants, trend and weather, $PM_{2.5}$ at 30 versus 10 $\mu g\,m^{-3}$ gave 14.2% higher odds of HAP at lag 1 (95% CI 2.6-27%), stronger at age >=60, BMI >23.9 and with diabetes, and robust to ventilation duration. The exposure is ambient city-wide $PM_{2.5}$ assigned to patients inside an ICU, whose indoor air is filtered and decoupled from outdoors; the association may instead reflect seasonal case-mix, staffing or infection-control variation tracking pollution. Relevance: indirect - an indoor clinical endpoint fitted to outdoor monitors; an ICU indoor sensor would test the pathway directly.

doi: 10.3389/fpubh.2026.1867760

Pulczinski EPA controlled human exposure 3 consecutive days 35 $\mu g\,m^{-3}$ $PM_{2.5}$ FEV1 exercise 2026 Environ Health Perspect

Repeated Exposure to Near-Ambient PM(2.5) Decreases Pulmonary Function: Controlled Human Exposure Study.

First multi-day controlled human exposure: 17 healthy volunteers, 3 consecutive days at 35 $\mu g\,m^{-3}$ $PM_{2.5}$ (the US 24-h standard) vs clean air, crossover 2 weeks apart. Day 1 FEV1 unchanged (-0.9%, -2.1 to 0.3). Day 2 FEV1 -1.9% (-3.1 to -0.7) immediately and -1.7% (-3.3 to -0.2) at +20 h; Day 3 -1.8% (-3.5 to -0.1) and -1.7% (-3.2 to -0.1). Exercise output fell 2.0 W (Day 2) and 2.4 W (Day 3). n=17 healthy young adults, so sex-modification claims are underpowered, and the decrements are small against spirometric reproducibility. Relevance: direct - cumulative-day effects at a concentration low-cost networks routinely report argue for multi-day running means, not single-day exceedance flags.

doi: 10.1021/ehp.6c00055

Ruvuna ambient $PM_{2.5}$ NO_2 plasma proteomics SPIROMICS SomaScan congress 2026 Eur Respir J (congress abstract)

Associations of ambient air pollution and protein expression among individuals from the Subpopulations and intermediate outcome measures in COPD study (SPIROMICS)

BACKFILL (Crossref deposit 29 Oct 2026; Amass; no PMID). ERS congress abstract. SPIROMICS tobacco-exposed cohort, n=1,627 for $PM_{2.5}$ and n=1,618 for NO_2; SomaScan 7k against modelled prior-year exposure, adjusted for site, COPD status, smoking, BMI, occupation, income and deprivation. 860 proteins associated with $PM_{2.5}$ at FDR<0.05, none with NO_2; enrichment in vascular remodelling, insulin signalling and inflammation. Limitation: abstract only, no effect sizes or replication, cross-sectional, and a one-pollutant-sided result may partly reflect exposure-model quality or spatial confounding. Relevance: low - no sensing element, but the proteomic contrast depends on exposure accuracy.

doi: 10.1183/13993003.congress-2026.oa5535

Salimbene Turin mild COPD lung function $PM_{2.5}$ microenvironment MANCOVA 2026 Atmosphere

Ambient, Indoor Air Pollution Exposures and Mild Chronic Obstructive Pulmonary Disease Trajectories in Non-Hospitalized Volunteers in Turin, Italy

50 non-hospitalized men with mild COPD, resident in Turin at least five years, spirometry monthly for a year. MANCOVA gave an overall multivariate effect (Wilks lambda 0.125, p<0.001); monthly $PM_{2.5}$ (p=0.001), proximity to primary roads (p=0.004) and gas cooking (p=0.002) were independently associated with spirometric decline, while PM_{10} and internal boilers were not. Mixed models gave per-interval decline rates and clustering separated patients by indoor source profile. Only p-values are reported: no coefficient, no CI, no decline rate in ml/year, which for a 50-man single-sex sample is the whole inference. Relevance: moderate - $PM_{2.5}$ assigned by residence while gas cooking carries an independent effect is the standard argument for indoor nodes. Crossref deposits a Ljubljana affiliation for a Turin study; recorded as deposited.

doi: 10.3390/atmos17111082

Tsogtbayar Ulaanbaatar time-series PM_1 $PM_{2.5}$ under-5 pneumonia outpatient 2026 J Thorac Dis

Associations between air pollution and pediatric pneumonia incidence in Ulaanbaatar, Mongolia: age and sex susceptibility.

Ulaanbaatar 2014-2019, 48,488 under-5 outpatient pneumonia visits, adjusted for season, weather, holidays and the raw-coal ban. At lag 2, per IQR: $PM_{2.5}$ RR 1.057 (1.015-1.100), PM_1 1.065 (1.025-1.106), SO_2 1.136 (1.023-1.261), NO_2 1.383 (1.108-1.726); PM_{10} and O_3 non-significant. Infants <1 y showed PM_1 RR 1.098 (1.046-1.152) at lag 0. $PM_{2.5}$ and NO_2 survived co-pollutant adjustment. IQR values are not in the abstract, outpatient visits depend on care-seeking, and NO_2 dwarfs PM. Relevance: indirect - PM_1 outperforming PM_{10} in a coal-heating city favours sub-micron-capable nodes in ger districts, where the regulatory monitors that fed this series are sparse.

doi: 10.21037/jtd-2026-1940

Ubuane Lagos protocol paediatric asthma control household $PM_{2.5}$ CO cotinine 2026 Research Square

Association of asthma control with household particulate matter less than 2.5 microns, carbon monoxide and second-hand-smoking among children with asthma in Lagos, Nigeria: protocol of an ongoing observational cross-sectional study

Protocol, no results. Children and adolescents up to 18 years attending LASUTH asthma clinics in Lagos are being recruited consecutively; asthma control, lung function (peak flow, FEV1, home PEF variability), adherence, inhaler technique, quality of life, urinary cotinine and household $PM_{2.5}$ and CO will be related by hierarchical logistic regression. The rationale is that questionnaire-based household air pollution studies found no association. The $PM_{2.5}$ instrument, sampling duration and sample size are not stated. Relevance: direct if it uses portable optical monitors - the protocol should specify instrument and calibration before data collection ends.

doi: 10.21203/rs.3.rs-9634309/v1

Yang UK Biobank life-course $PM_{2.5}$ constituents COPD asthma LRTI lung cancer 2026 Lung

Life-Course Exposure to PM(2.5) Constituents and Major Respiratory Diseases: Constituent-Specific Associations and Candidate Immune-Metabolic Signatures in a Large Prospective Cohort.

47,737 UK Biobank participants with birth-to-baseline residential histories, 13-14 years of follow-up. Per IQR of life-course $PM_{2.5}$: COPD HR 1.138 (1.069-1.211), asthma 1.171 (1.105-1.241), lower respiratory infection 1.046 (1.002-1.091), lung cancer 1.193 (1.064-1.337). SO4 and EC were positive for all four outcomes; mixture associations held for COPD and asthma. The IQR in $\mu g\,m^{-3}$ is not given in the abstract. Limitation: decades of exposure rest on one retrospective CTM with no available monitor validation for most of the period, and the 47,737 with histories are a selected fraction of the Biobank. Relevance: moderate - composition, not mass, is the signal, which a mass-calibrated optical sensor cannot resolve.

doi: 10.1007/s00408-026-00948-1

Zhang Shanxi 11-city time-series pollutant mixtures influenza vaccination 2026 Environ Res

Influenza Incidence in Relation to Air Pollution Mixtures and the Modifying Role of Vaccination in Shanxi Province, China.

11 Shanxi cities, 2014-2022, DLNM with zero-inflated negative binomial and meta-analysis. Per 10 $\mu g\,m^{-3}$ $PM_{2.5}$, RR 1.024 (1.015-1.034); PM_{10} 1.018 (1.010-1.025); NO_2 1.115 (1.086-1.146). WQS (1.078, 1.005-1.157) and qgcomp (1.122, 1.064-1.182) mixture effects were driven by SO_2 and NO_2, not PM. Vaccination modification is city-level coverage, therefore ecological, and labelled exploratory. Surveillance incidence is care-seeking and testing dependent; 2020-2022 includes COVID-era behavioural shifts the abstract does not say were handled. Relevance: indirect - PM signal is small next to gases, which argues for multi-pollutant nodes rather than PM-only networks for infectious endpoints.

doi: 10.1016/j.envres.2026.126109

Zhu meta-analysis air pollutants allergic rhinitis children $PM_{2.5}$ TRAP mold 2026 Int J Pediatr Otorhinolaryngol

Association between exposure to gaseous air pollutants and the risk of allergic rhinitis in children and adolescents: A meta-analysis.

Meta-analysis of 21 studies (13 cross-sectional, 6 cohort, 2 case-control). Pooled ORs: mold 1.45 (1.16-1.80), traffic-related pollution 1.39 (1.21-1.59), $PM_{2.5}$ 1.23 (1.05-1.43), O_3 1.15 (1.05-1.26), NO_2 1.09 (1.00-1.17); PM_{10}, SO_2 and CO not significant. Estimates for PM_{10}, NO_2 and O_3 were higher in Chinese studies. Limitation: the abstract gives no exposure increment, so the pooled ratios cannot be compared across pollutants or to any concentration, and 13 of 21 studies are cross-sectional. Mold is not an ambient pollutant, and a 'gaseous' title over a $PM_{2.5}$ estimate shows loose scoping. Relevance: low - a pooled OR with no increment cannot feed a sensor-based exposure-response.

doi: 10.1016/j.ijporl.2026.113068

Reproductive & developmental — 6 records

Carried from the daily issues — full summaries (6 of 6 records in this subtopic)

Karimi PMOS polycystic ovary metabolic syndrome Arak Iran $PM_{2.5}$ PR IDW 2026 Chemosphere

Air pollution and polycystic ovary metabolic syndrome (PMOS) in a heavily industrialized Middle Eastern city: A cross-sectional study.

301 women at gynecology clinics in Arak (153 PMOS, 148 non-PMOS), April 2024-March 2025. Modified Poisson, per 1-SD: $PM_{2.5}$ PR 1.22 (1.09-1.35), PM_{10} PR 1.23 (1.09-1.36), NO_2 1.21 (1.07-1.33), SO_2 1.15 (1.01-1.28). Cases had 3.00 $\mu g\,m^{-3}$ more $PM_{2.5}$. Sensitivity windows and propensity matching agreed. SD widths are not in the abstract. Limitation: exposure from three stations by IDW in an industrial city carries little real spatial contrast, so residence and socioeconomic position are confounded with it; clinic controls and prevalent cases rule out temporal inference. Relevance: moderate - a study whose result hangs on three monitors is where a small low-cost network would directly test the contrast.

doi: 10.1016/j.chemosphere.2026.145136

Kim prenatal $PM_{2.5}$ pediatric inguinal hernia hydrocele kriging cohort 2026 J Clin Med

Prenatal Fine Particulate Matter ($PM_{2.5}$) Exposure and the Risk of Pediatric Inguinal Hernia or Hydrocele: A Retrospective Cohort Study

BACKFILL (entry 4 May 2026). 1,093 mother-offspring pairs, one tertiary centre, deliveries July 2016-June 2019, follow-up to March 2024; 53 cases (4.85%). Second-trimester $PM_{2.5}$ OR 1.07 (1.01-1.14) per 1 $\mu g\,m^{-3}$; an IQR increase gave OR 1.64 (no CI, IQR width not stated). Male sex OR 24.71 (5.95-102.54). AUC 0.804. Limitation: 53 events against several trimester windows with no stated multiplicity correction, referral-centre cohort, hernia and hydrocele pooled, and a mechanism inferred rather than tested. Treat as hypothesis. Relevance: low - kriging from fixed stations, with sparse-network error in a trimester-specific contrast, the setting where denser residential estimates would matter.

doi: 10.3390/jcm15083089

Marques Generation R multi-cohort LUR $PM_{2.5}$ NO_2 epigenetic age birth to adolescence 2026 Environ Res

Outdoor Air pollution Exposure and Epigenetic Age from birth to adolescence - a multi-cohort study.

European birth cohorts: 4,166 at birth, 1,531 at 4-7 y, 1,748 at 8-10 y, 2,546 at 16-18 y; harmonised LUR exposures in three windows, fixed-effects meta-analysis. No pollutant, $PM_{2.5}$ included, was associated with epigenetic age at birth or in childhood. In adolescence, per 10 $\mu g\,m^{-3}$ lifetime NO_2, epigenetic age was 0.23 years higher (0.06-0.39), about 84 days, surviving adapted Bonferroni. PM associations were null in the main analysis. LUR at address omits school and transit time, which dominates adolescent exposure. Relevance: indirect - a well-powered PM null; the one signal is traffic NO_2, not particle mass.

doi: 10.1016/j.envres.2026.126058

Qiu stage-specific models pregnancy loss recurrent abortion $PM_{2.5}$ O_3 random forest 2026 Int J Womens Health

Stage-Specific Models for Predicting Subsequent Pregnancy Loss in Women with Recurrent Spontaneous Abortion.

Single-centre prospective cohort of 470 women with recurrent spontaneous abortion, 86 (18.3%) with another loss. Three staged models by logistic regression, random forest and XGBoost, repeated 10-fold cross-validation. Third-stage random forest ROC-AUC 0.935 (0.874-0.996); XGBoost 0.870 (0.801-0.938), after adding gestational age, hormones and $PM_{2.5}$ and O_3 metrics. No PM effect estimate. Limitation: internal validation only, 86 events or fewer against many predictors, and gestational age at measurement risks leaking the outcome; the PM contribution cannot be separated from the bundle, and the exposure source is not in the abstract. Relevance: low - PM is a feature in a classifier, and this is no evidence that exposure drives discrimination.

doi: 10.2147/ijwh.s637498

Tao Peking ART women 1-km $PM_{2.5}$ follicular fluid metals 2026 J Hazard Mater

Associations of ambient fine particulate matter exposure with metal and metalloid concentrations in follicular fluid.

198 women undergoing assisted reproduction; 12-month residential $PM_{2.5}$ at 1 km. Per IQR, follicular-fluid chromium +22.10% (9.59-36.04), lead +23.15% (1.27-49.76), cadmium +22.81% (8.95-38.44), arsenic +29.25% (8.13-54.49), germanium +15.72% (4.42-28.25), lithium +25.84% (9.65-44.43); five survived FDR, lead did not. The IQR value is not reported, n is small, diet and water - major Cd/As sources - are not mentioned as adjusted, and the $PM_{2.5}$ surface has no composition, so the link to particle-borne metals is inferred. Relevance: indirect - a biomarker endpoint where metal-resolved sampling, not optical mass, would be needed to close the pathway.

doi: 10.1016/j.jhazmat.2026.144126

Zhang paternal $PM_{2.5}$ concentrated ambient particles multigenerational lipid DNA methylation mice 2026 Environ Pollut

Paternal PM(2.5) exposure is associated with lipid metabolic dysfunction across generations and altered sperm and hepatic DNA methylation in mice.

F0 males were exposed by inhalation for 22 weeks, then F1-F3 were bred through unexposed partners. Concentrated ambient $PM_{2.5}$ was associated with dyslipidemia and hepatic steatosis in F1 and F2 males and no significant serum lipid change in F3; unfiltered ambient air gave only minor alterations. Hepatic fatty-acid oxidation, lipid degradation and cholesterol transport genes were disrupted in F1 and F2, Apoe_1 hypermethylation survived multiple-testing correction in F1 liver, and F0 sperm carried abundant DMRs enriched for lipid metabolism, with Apoe and Apoa2 concordant between sperm and offspring liver. No CI or exposure concentration in the abstract. Limitation: concentrated-particle exposure, one male per litter, and an association framing the authors keep throughout. Relevance: low for sensing; it is the cleanest multigenerational dose this watch has logged.

doi: 10.1016/j.envpol.2026.129477

Mechanistic toxicology — 14 records

Carried from the daily issues — full summaries (14 of 14 records in this subtopic)

Ba rat intratracheal $PM_{2.5}$ IL-17A TXNIP NLRP3 cardiac fibrosis 2026 Kaohsiung J Med Sci

Neutralization of IL-17A Ameliorates $PM_{2.5}$-Induced Cardiac Inflammation and Fibrosis in Rats by Suppressing the TXNIP/NLRP3 Signaling Pathway.

Rat cardiac-injury model by intratracheal $PM_{2.5}$ instillation with an IL-17A-neutralising antibody arm, plus neonatal rat cardiac fibroblasts in vitro. $PM_{2.5}$ raised splenic Th17 fraction and cardiac IL-17A/TRAF6, impaired cardiac function, and increased collagen I/III, alpha-SMA, NF-kB activation, IL-6, IL-1b, TXNIP and NLRP3-pyroptosis proteins (ASC, cleaved caspase-1, GSDMD-N, IL-18); antibody neutralisation reversed these. TXNIP knockdown abolished IL-17A effects in fibroblasts. No dose, particle source, group size or effect magnitude is reported in the abstract, and bolus instillation is far from inhaled ambient dose rates, so the human-relevant dose is unknown. Relevance: indirect - mechanism for cardiac endpoints; adds nothing to measurement.

doi: 10.1002/kjm2.70316

Guardado lipocalin-2 knockout organic dust extract lung inflammation recovery 2026 PLoS One

The role of lipocalin-2 in lung homeostasis, inflammation response, and recovery following organic dust exposure.

Wild-type and lipocalin-2 knockout C57Bl/6 mice were given repetitive intranasal swine-dust extract for fifteen days, with and without a three-day recovery window. Knockouts failed to resolve histopathological inflammation and had lower IL-10 after exposure, and exogenous IL-10 partially rescued them; T cell and macrophage responses diverged between lavage fluid and lung tissue. No fold-changes, doses or group sizes in the abstract. Limitation: dust extract rather than inhaled particles, so deposition and dose are not comparable to an occupational aerosol, and a single mouse strain. Relevance: low for sensing - but recovery capacity as the endpoint, rather than peak response, is the structure that intermittent high-concentration exposures would act through, and intermittency is what a dense network measures and a 24-hour average destroys.

doi: 10.1371/journal.pone.0359338

Jain air pollution redox proteomics oxidative modification review 2026 Redox Biol

Air pollution leaves a redox-proteomic imprint: From shared inflammation to disease-specific vulnerability.

Metadata only. Metadata only: PubMed record without abstract, Redox Biology. The title frames a synthesis of air-pollution-induced redox-proteomic modifications, moving from shared inflammatory signatures to disease-specific vulnerability. Whether it is a narrative review or presents primary proteomic data, which pollutants or particle fractions are covered, and which tissues or diseases are addressed cannot be recovered, and no finding is asserted here. Relevance: indirect - such oxidative protein modifications are candidate biomarkers that would link oxidative-potential measurements on sensor or filter samples to biological effect in exposed people; recheck once the abstract is deposited.

doi: 10.1016/j.redox.2026.104471

Jeon rat $PM_{2.5}$ inhalation sperm abnormality ACE2 vs renal insufficiency 2026 Transl Androl Urol

Particulate matter inhalation induces male reproductive dysfunction comparable to renal failure: insights from a rat model.

Male rats: control, $PM_{2.5}$ (30 mg/kg/day, 3 h/day, 10 days) and renal insufficiency by unilateral ureteral obstruction. $PM_{2.5}$ increased sperm morphological abnormalities; count and motility changes were not significant. Testicular ACE2 staining and inflammatory/endothelial gene expression shifted in both exposed and RI groups. Dose expressed per kg body weight for an inhalation exposure is unusual and cannot be converted to an air concentration; group sizes and effect magnitudes are absent, and the PM source is unstated. Relevance: minimal for sensing - mechanistic, at a dose far above ambient; logged for the reproductive-endpoint map.

doi: 10.21037/tau-2026-0425

Li poultry-house $PM_{2.5}$-derived material APEC sequential exposure chick cell 2026 Poult Sci

Inflammatory and redox responses associated with sequential exposure to poultry-house $PM_{2.5}$-derived material and avian pathogenic Escherichia coli in chicken models.

Chicken type II-like lung cells and 40 SPF chicks (n=10/group, 14 days) received control, single or sequential $PM_{2.5}$-derived material then avian pathogenic E. coli. The sequential group showed the largest changes in several inflammatory and redox endpoints and lower CFTR; N-acetyl-L-cysteine and CBIQ shifted cell responses toward control. No effect sizes in the abstract. Limitation, stated by the authors: the eluted, lyophilized, autoclaved material was not characterized for size, composition or endotoxin, and schedules differed in timing, so no interaction, synergy or particle-specific effect is established. Relevance: low - a sterilized eluate is not an airborne particle.

doi: 10.1016/j.psj.2026.108064

Liu BALF fibroblast $PM_{2.5}$ house dust mite FAM45A allergic asthma 2026 Environ Health Perspect

FAM45A-Mediated Persistent Memory-Like Fibroblast Reprogramming Drives $PM_{2.5}$-Exacerbated Allergic Asthma.

BALF fibroblasts from 8 HDM-sensitised asthmatics and 8 sensitised non-asthmatic controls: only combined $PM_{2.5}$ + HDM stimulation induced a persistent antigen-presenting, IL-33-secreting phenotype, with FAM45A the top dual-exposure transcript; FAM45A slows lysosomal allergen degradation via V-ATPase, is required by CRISPR knockout, and its neutralisation reduced $PM_{2.5}$/HDM airway inflammation in humanised mice. In an expanded cohort fibroblast FAM45A correlated with residential $PM_{2.5}$ exposure duration, HDM IgE and GINA severity. Primary-cell n=8 per group; $PM_{2.5}$ dose and source are not given, and the clinical correlation is unquantified in the abstract. Relevance: indirect - an interaction mechanism that argues for allergen-season stratification of sensor-based asthma alerts.

doi: 10.1021/ehp.6c00353

Mali Rutgers wildfire PM0.1 mouse ovarian hyperandrogenism AhR 2026 Environ Health Perspect

Wildfire-Emitted Particulate Matter Induces Ovarian Hyperandrogenism through Aryl Hydrocarbon Receptor Activation

Mouse intratracheal exposure to lab-synthesised wildfire PM0.1 at a dose described as environmentally relevant disrupted estrous cycles and raised serum testosterone; follicles showed AhR activation and altered steroidogenic genes, reproduced by real Canadian 2023 wildfire PM0.1, BaP and TCDD, and reversed by AhR inhibition. Single-follicle and single-oocyte RNA-seq flagged inflammatory and mitochondrial genes. The abstract gives no dose in ug or lung-deposited equivalent, no group sizes and no effect magnitudes, so translation to human wildfire episodes cannot be judged. Relevance: indirect - the toxicity is attributed to the nanosized fraction, which optical sensors used for smoke alerts barely register.

doi: 10.1021/ehp.6c00189

Mishra ultrafine PM black carbon polystyrene nanoplastic mitochondrial stress PBMC 2026 Front Immunol

Conserved and particle-specific mitochondrial stress responses drive immunotoxicity of ultrafine particulate matter, black carbon, and polystyrene nanoplastics.

Three particle classes were dosed on human peripheral blood mononuclear cells under matched conditions to ask whether the mitochondrial stress response is generic or particle-specific. All three raised oxidative and mitochondrial stress markers but with distinct kinetics: ultrafine PM gave an acute antioxidant and adaptive response, black carbon sustained dysfunction with OMA1 and DELE1 induction and inflammation, nanoplastics a low but sustained adaptation with biogenesis. Oxidative stress, adaptation, the integrated stress response and inflammation were correlated independently of physicochemistry. No doses or fold-changes in the abstract, and PBMCs are a surrogate for an airway compartment these particles reach only after deposition. Relevance: moderate - black carbon separating from bulk ultrafine mass is an argument for composition-resolved rather than mass-only nodes.

doi: 10.3389/fimmu.2026.1993305

Pitchai pulmonary nanomaterial exposure inflammatory lipid mediators sex disease 2026 Part Fibre Toxicol

Pulmonary exposure to nanomaterials dysregulates inflammatory lipid mediators in a sex- and disease-dependent manner

METADATA ONLY. Metadata only. Crossref carried no abstract at issue time and there is no Europe PMC record yet, so no finding is asserted. The title claims dysregulation of inflammatory lipid mediators after pulmonary nanomaterial exposure, modified by both sex and pre-existing disease. If the full text supports it, the interest for this watch is the effect modification rather than the mediator panel: a sex- and comorbidity-dependent response means a population-average concentration-response function understates risk in some strata, which bears on who a monitoring network should be sited to protect. Flagged for the weekly re-sweep.

doi: 10.1186/s12989-026-00713-5

Qureshi narrative review bioaccessibility particle-bound metals PAHs tiered framework 2026 Research Square (preprint)

Bioaccessibility and Toxicity of Particle-Bound Metals and Polycyclic Aromatic Hydrocarbons: Sources, Respiratory Fate, Mechanisms, and Risk Assessment

Single-author critical narrative review, not peer-reviewed. Metal release varies with element, source, size, extraction conditions and solid-to-liquid ratio, and acidic lysosomal fluid usually dissolves more than near-neutral extracellular fluid; PAH release depends on hydrophobicity, soot binding, surfactants and proteins. Poorly soluble particles can stay active with little released. Proposes a tiered framework joining mass, source, size, speciation, compartment-specific bioaccessibility, oxidative potential and delivered dose. No search method or quantitative synthesis in the abstract, and differences in simulated lung fluids limit comparison. Relevance: low - explains why equal mass need not mean equal hazard, beyond what a mass sensor reports.

doi: 10.21203/rs.3.rs-11258579/v1

Stegmann atmospheric aging biomass burning particle toxicity BEAS-2B PHOTO-LAC 2026 Environ Sci Technol

Atmospheric Aging Modifies the Toxicological Profile of Biomass Burning Particles in Epithelial Lung Cells.

Biomass burning particles from controlled flaming and smoldering combustion were aged in the PHOTO-LAC chamber under alternating photochemical and dark conditions, then extracts dosed on BEAS-2B bronchial cells. Aging enhanced intracellular ROS and oxidative DNA damage, most for smoldering emissions, with cytotoxicity largely unchanged. Inflammatory and matrix-remodelling responses were attenuated with aging in smoldering particles and stable in flaming particles. No fold-changes or doses in the abstract. Limitation: extracts on a submerged immortalised line, not air-liquid interface deposition, and one chamber aging protocol, so dose relevance to ambient plumes is open. Relevance: moderate - mass from a sensor is blind to aging state; a plume reading the same $\mu g\,m^{-3}$ near source and downwind need not be the same exposure.

doi: 10.1021/acs.est.6c02821

Sun Waite review molecular mechanisms respirable silica coal dust asbestos lung disease 2026 Part Fibre Toxicol

Molecular mechanisms of dust-induced lung disease: exploring the role of physicochemical composition and cellular pathways.

Review of respirable occupational dust toxicity: oxidative stress, mitochondrial and lysosomal damage, inflammasome activation, ferroptosis, necrosis and pyroptosis, macrophage/neutrophil activation and NK dysfunction driving fibrosis and tumorigenesis, with emphasis on silica and on coal dust bearing reduced iron. Covers antifibrotic and immunotherapy approaches. Narrative, no stated search or appraisal. The physicochemical angle (surface iron, crystallinity) is the useful part: it is why mass concentration alone mis-ranks hazard across dust types. Relevance: indirect - optical low-cost sensors are blind to mineralogy, so occupational dust monitoring needs composition or a source-specific mass-to-hazard factor.

doi: 10.1186/s12989-026-00696-3

Wang naphthalene SOA aerosol-organoid exposure platform glioma ALDH1A3 ferroptosis 2026 J Hazard Mater

Secondary organic aerosol from polycyclic aromatic hydrocarbon modulates glioma proliferation.

Naphthalene SOA, generated photochemically, was delivered at the air-liquid interface through a new aerosol-organoid exposure platform to patient-derived glioblastoma organoids and U87/U251 cells. Highly oxygenated SOA increased tumour-cell growth by upregulating ALDH1A3, which suppressed ferroptosis; high tumour ALDH1A3 tracked poor patient survival in public data. Dose, deposited mass, oxidation state metrics and effect magnitudes are not in the abstract, and SOA reaching glioma tissue in vivo is unshown, so this is a tumour-progression hypothesis rather than an exposure effect. Relevance: indirect - the exposure platform itself, delivering characterised aerosol to organoids, is the transferable contribution.

doi: 10.1016/j.jhazmat.2026.144002

Zhang SMR multi-omics $PM_{2.5}$-annotated genes diabetic nephropathy TNF NOS2 methylation 2026 Clin Epigenetics

A multi-omics Mendelian randomization framework identifies $PM_{2.5}$-annotated candidate regulatory genes and methylation-linked pathways in diabetic nephropathy.

Summary-data MR integrating eQTL, pQTL and mQTL with diabetic nephropathy GWAS, restricted to a curated set of $PM_{2.5}$-annotated genes. Ten genes (TNF, NOS2, DUSP1 among them) showed genetically predicted effects; colocalisation supported a shared variant only at EIF2AK3 (PP.H4 0.90). 48 CpGs in 28 genes associated with DN risk, TNF methylation strongest; cell $PM_{2.5}$ exposure reproduced TNF/NOS2 up and DUSP1 down; a two-gene nomogram gave AUC 0.766 (training) and 0.726 (external). The MR tests gene expression on DN, not $PM_{2.5}$ on DN: the PM link is a database annotation, so the causal claim about pollution is not identified. Relevance: weak - no exposure measurement of any kind.

doi: 10.1186/s13148-026-02215-9

Occupational & indoor — 20 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (20 OF 20 RECORDS IN THIS SUBTOPIC)

Apak office-to-education studio zone CO_2 $PM_{2.5}$ PM_{10} TVOC perception Istanbul 2026 Indoor Air

Indoor Air Quality and Occupant Perception in an Office Building Adapted for Educational Use: A Zone-Based Case Study

One mechanically ventilated design studio, three zones, questionnaire from 48 users. CO_2 was highest and most variable in the central zone (527 +/- 114 ppm, 95th percentile 760 ppm), and upper-end particulate indicators were greater in the circulation zone, but no zonal difference in CO_2, PM or TVOC survived multiplicity adjustment. Perceived freshness tracked overall satisfaction (Spearman rho 0.475, 0.219-0.682). Limitation, stated by the authors: respondents cannot be linked to measurement times or locations, so this is no exposure-perception relationship; sequential monitoring and few points restrict inference. Relevance: low - sequential single-instrument sampling cannot resolve within-room PM differences, which simultaneous nodes could.

doi: 10.1155/ina/4355214

Arts letter health-based nanomaterial guidance value 4 $\mu g\,m^{-3}$ occupational critique 2026 NanoImpact

Letter to the editor regarding the paper entitled "Health-based nanomaterial guidance value (HNGV) for occupational exposure to spheroidal biodurable engineered nanomaterials of relatively low substance-specific toxicity".

Nine authors contest a generic health-based guidance value of 4 $\mu g\,m^{-3}$ for low-toxicity biodurable nanomaterials. Their case: a small transient rise in rat pulmonary neutrophils without biochemical or histopathological corroboration is an adaptive rather than adverse response and does not match OECD or ECHA practice; assessment factors designed for systemic toxicity were applied to a local, concentration-driven effect; existing occupational limits for benchmark low-toxicity dusts are far higher with long human experience behind them. They note 4 $\mu g\,m^{-3}$ sits at or below typical background $PM_{2.5}$ and current measurement capability. Limitation: a letter, no new data, and the authors are largely industry-affiliated, which the record does not state. Relevance: direct on metrology - a limit below ambient background cannot be enforced by any instrument class in routine occupational use.

doi: 10.1016/j.impact.2026.100649

Bittner Grieshop rural Malawi domestic biofuel emissions characterisation 2026 Environ Sci Atmos

Characterizing emissions from diverse domestic biofuel uses in rural Malawi

METADATA ONLY. Metadata only: the Crossref deposit carries only the opening of the abstract, stating that household and small-scale industry energy in Malawi and similar sub-Saharan settings is almost entirely biofuel and that household air pollution is a primary concern. The title indicates emission characterisation across diverse domestic biofuel uses in rural Malawi. Emission factors, pollutants, instruments, sample size and any PM size information are not recoverable, and no finding is asserted. Relevance: potentially direct - field emission factors for non-cooking biofuel uses (brewing, brick and fish drying) are the missing source terms for household and community PM models in rural Africa; recheck for full abstract.

doi: 10.1039/d6ea00123h

Borgini e-cigarette real-life indoor PM AirBeam3 bar traffic walk pilot 2026 Multidiscip Respir Med

Investigating the Real-Life Indoor Air Quality Impact of E-Cigarette Vaping.

Pilot walk with an AirBeam3 along low- and heavy-traffic roads, then inside a bar with three people vaping. Bar $PM_{2.5}$ averaged 28.8 $\mu g\,m^{-3}$ with a peak of 84.7 $\mu g\,m^{-3}$, 6.5 times the low-traffic and 3.1 times the heavy-traffic street value. No CI; a single session is implied, with no replicate or vape-free control in the same room. Limitation: e-cigarette aerosol is volatile liquid droplets, outside an ambient-dust factory calibration, and no gravimetric or reference co-location is reported, so the mass values are nominal; bar sources other than vaping are uncontrolled. Relevance: direct - a low-cost optical sensor reporting a mass ratio in an aerosol regime it was never characterised for. Read it as a relative signal.

doi: 10.5826/mrm.2026.1117

Gupta Maharashtra opencast coal mine $PM_{2.5}$ composition oxidative potential 2026 Environ Geochem Health

Comprehensive assessment of $PM_{2.5}$ exposures around a regional mining site in India: integrating chemical composition, oxidative potential and health risk

Six sites around an active opencast coal mine in eastern Maharashtra: two roadside (~400 $\mu g\,m^{-3}$), residential at 2 and 5 km (122 +/- 43 and 116 +/- 36 $\mu g\,m^{-3}$), and background. Carbonaceous species and metals were ~40-50% of mass; Cr, Ni, Fe, Mg and Zn ran 5-10-fold higher near the mine. OPvAA and OPvDTT peaked roadside and correlated only weakly with metals and organics (r=0.3-0.5); regression explained 25-40% of OP. EC cancer risk ~1e-3 rests on a risk-assessment slope, not outcomes. Season and sample counts are not in the abstract. Relevance: direct - at this composition, optical-sensor mass calibrations from urban sites will not transfer. BACKFILL, entry 25 June 2026.

doi: 10.1007/s10653-026-03301-w

Hu Shanghai catering cooking fume emission multi-criteria assessment 2026 Atmospheric Environment

Comprehensive Assessment of Cooking Fumes Pollution Emission Levels in Selected Shanghai Catering Businesses: An Exploratory Multi-Criteria Assessment

METADATA ONLY. Metadata only. Crossref carried no abstract at issue time and the record is absent from Europe PMC, so no finding is asserted here. The title places it in commercial-kitchen cooking-fume emissions in Shanghai under a multi-criteria assessment framing. Relevance, if the full text supports it, would be to the occupational and near-source indoor band rather than to ambient sensing: commercial kitchen exhaust is a dominant and highly intermittent neighbourhood PM source, and the emission levels such an assessment reports are the inputs a near-source siting study needs. Flagged for the weekly Elsevier abstract re-sweep.

doi: 10.1016/j.atmosenv.2026.122460

Khoshakhlagh occupational airborne pollutants hematological outcomes systematic review 2026 J Appl Toxicol

Occupational Airborne Pollutant Exposure and Hematological Outcomes in Workplace Settings: A Systematic Review.

PRISMA review of 1,855 records over five decades, 55 studies, predominantly developing and industrialising countries, JBI appraisal, no pooled estimate. Workplace VOCs, heavy metals such as lead, and PM were linked to subclinical haematological change: lower hemoglobin, altered MCH and MCHC. Linear correlations with pollutant levels were often weak despite high concentrations. BMI associated positively with hemoglobin; age weakly, read as a healthy-worker effect. Limitation: no effect size, PM not separated from co-exposures, and a protective BMI 'buffer' is confounding by nutrition until shown otherwise. Relevance: low - weak exposure-response at high concentrations is also what uncalibrated or area-based workplace exposure metrics would produce.

doi: 10.1002/jat.70474

Kim New Mexico Wyoming coal miners self-reported dust SARS-CoV-2 seroconversion 2026 JMIR Form Res

Novel Occupational and Environmental Risk Factors for SARS-CoV-2 Infection Among Coal Miners: Longitudinal Cohort Study.

230 surface coal miners (New Mexico 169, Wyoming 61), February 2021-March 2022, nucleocapsid IgG at 0, 3, 6 and 12 months. Higher self-reported mining-dust exposure was associated with lower odds of incident seroconversion, adjusted OR 0.42 (0.17-1.07), non-significant; wood burning and shared transport showed no association, and no frequent wood burner seroconverted (0/35). The hypothesis that particulate exposure raises infection risk via ACE2 upregulation was not supported. Exposure is a single questionnaire answer with no dust measurement, Firth-penalised models on few events, and vaccination plus prevalent seropositivity confound strongly. Relevance: weak - a negative occupational result resting entirely on perceived dust, a case for personal samplers in mining cohorts.

doi: 10.2196/89412

Krevanko UW cannabis processing local exhaust ventilation inhalable PM terpenes 2026 Front Public Health

Pilot study of a local exhaust ventilation system to control particle and terpene concentrations during cannabis processing.

Partially enclosed local exhaust ventilation hood with MERV-13 particle and activated-charcoal terpene filters, tested in three paired trials during cannabis flower grinding and automated pre-roll filling. Inhalable PM fell on average 95% during grinding and 83% during pre-roll filling; terpenes fell below the limit of detection during grinding. Three paired trials, no absolute concentrations in the abstract, and area rather than personal sampling limit inference about worker dose. Relevance: indirect - inhalable organic dust is a coarse fraction that low-cost $PM_{2.5}$ sensors under-report, so sensor-based workplace screening would miss the hazard this hood controls.

doi: 10.3389/fpubh.2026.1964267

Kumar firefighter occupational exposure neurotoxicity narrative review 2026 Toxicol Mech Methods

Occupational exposure and neurotoxic health effects in firefighters: toxicological mechanisms and health risks.

PRISMA-informed narrative review of literature to June 2026, without meta-analysis or risk-of-bias assessment. Proposed pathways: CO and HCN hypoxia with ATP depletion acutely; fine PM and PAHs reaching the brain, NLRP3 glial activation and ferroptosis chronically; heat, fatigue and sleep loss as amplifiers. The authors state the firefighter literature is mostly cross-sectional and the mechanisms come from non-firefighter models, so causality is uncertain. Relevance: indirect - fireground PM exposure is essentially unmeasured at the individual level; wearable PM sensing on turnout gear is the obvious gap.

doi: 10.1080/15376516.2026.2748390

Laferriere VA burn pits registry headache phenotypes mixed methods 2026 Front Neurol

Veteran deployed burn pits exposure: headache phenotypes, trajectories, and outcomes: a mixed-methods evaluation.

Airborne Hazards and Open Burn Pit Registry: 247,583 linked participants; 1,980 surveyed, 772 responded (39.0%), 335 completed headache phenotyping, 49 interviewed. Migraine and post-traumatic headache carried more disability and insomnia than other headache; 68.0% of SDOH responders reported social isolation; many dated onset to deployment. There is no unexposed comparator, burn-pit exposure is self-reported, and 335 of 247,583 is a highly selected fraction. No PM metric is measured, and headache is not compared with other registry outcomes. Relevance: indirect - deployment smoke exposure remains unquantified, a gap personal samplers in future deployments could close.

doi: 10.3389/fneur.2026.1899580

Lin Taiwan metro track maintenance respirable dust metals noise WBGT task-based 2026 Ann Work Expo Health

Exposure to dust, metals, noise, and heat stress among metro track maintenance workers: task-based onsite measurements in two urban metro systems.

Seven personal sampling events across an underground high-capacity and a half-open medium-capacity Taiwanese metro, covering track cutting, aluminothermic welding and grinding. Task-duration respirable dust ranged <0.10-1.72 mg/m3, with short grinding samples to 5.71 mg/m3; cutting noise reached 106.7 dBA LAeq and peak 122.7 dB; welding WBGT peaked at 27.4 C. Results are descriptive at task-episode level because real-time records are not independent; one system per environment and day-versus-night sampling confound site contrasts, and seven events cannot support exceedance statistics. Metal speciation values are not in the abstract. Relevance: indirect - task-tagged real-time photometry against gravimetric respirable dust is the occupational analogue of sensor co-location, but no photometer-to-filter agreement is reported.

doi: 10.1093/annweh/wxag091

Molina Rueda AirPen facility-wide personal PM noise formaldehyde one shift 2026 J Expo Sci Environ Epidemiol

Facility-wide assessment of occupational exposure to dust, noise, and formaldehyde

Five people deployed 83 AirPen compact personal monitors across one full shift at a furniture plant of about 100 workers. Valid after screening: 67 PM, 67 noise, 22 formaldehyde (72, 67, 31 counting samples below the LOD). On-board sensors identified anomalies and confirmed wear. Distribution parameters were estimated with high certainty, and confidence intervals grew several-fold when fewer samples were analysed. Limitation: one facility on one shift says nothing about day-to-day variance, usually the main occupational term; 16 of 83 PM samples were lost or below LOD, and no comparison values are in the abstract. Relevance: direct - sample count, not instrument class, sets the uncertainty. BACKFILL, entry 28 May 2026.

doi: 10.1038/s41370-026-00883-6

Myers NIOSH CWHSP pneumoconiosis dusty job type underground coal 1991-2024 Front Public Health

Assessment of pneumoconiosis by dusty job type among underground coal miners participating in the Coal Workers' Health Surveillance Program,1991-2024.

30,319 underground miners in the national surveillance programme, 1991-2024, tenure-adjusted. Continuous miner operators had CWP PR 1.8 (1.6-2.0) and PMF PR 3.0 (2.3-4.0) against those never in the job; roof bolters 1.4 (1.3-1.6) and 1.4 (1.1-1.9). CWP prevalence was 8.4% in Central Appalachia versus 1.5% elsewhere, adjusted PR 3.9 (3.4-4.5); PMF PR 10 (5.7-17.5). Limitation: job title stands in for dust, and programme participants are self-selected, so healthy-worker and participation bias run in unknown directions. Prevalence pools 33 years of changing limits. Relevance: moderate - a tenfold regional contrast in a monitored regulated industry is the strongest argument today for personal rather than compliance-point dust data.

doi: 10.3389/fpubh.2026.1970741

Nie personal air purification device multi-scale modelling fan speed occupant exposure 2026 Build Environ

Multi-scale modeling and fan speed optimization of a personal air-purification device for occupant exposure control in polluted indoor environments

METADATA ONLY. Metadata only: Elsevier day-0 deposit, no abstract in Crossref or Semantic Scholar; no finding asserted. The title describes multi-scale modelling and fan-speed optimisation of a personal air-purification device. Limitation: the title does not confirm particulate matter as the pollutant, and no performance figures, validation or occupant data are available. Relevance: to be assessed - breathing-zone control is where stationary sensors misstate exposure; retain only if the abstract confirms a PM target and measured validation.

doi: 10.1016/j.buildenv.2026.115523

Olsen LC-APCI-MS resin acids personal softwood sawmill dust Norway 2026 J Sep Sci

A Versatile Method for Determination of Resin Acids in Occupational Softwood Sawmill Dust Samples by Liquid Chromatography-Atmospheric Pressure Chemical Ionization-Mass Spectrometry.

Six reversed-phase columns and multiple mobile phases were screened; a C18-amide column with acetonitrile and 30 mM ammonium acetate at pH 6.30 and negative APCI gave a 15-minute isocratic run. Applied to six personal sawmill dust extracts, it determined abietic, dehydroabietic and levopimaric acids, coeluting isopimaric and pimaric acids, and 7-oxodehydroabietic acid. Limitation: six extracts demonstrate applicability, not exposure levels, and two isomers still coelute despite that being the problem the method targets. LODs are not in the abstract. Relevance: low - constituent chemistry of an occupational dust that no optical sensor could separate, useful only as a reference analysis.

doi: 10.1002/jssc.70568

Rana Bhopal indoor DustTrak PM_1 $PM_{2.5}$ PM_{10} TVOC 30 households seasonal 2026 Research Square

Estimation of Emerging Indoor Air Pollutants and Associated Health Effects on Humans in the Capital City of Central India

Preprint. 30 randomly selected Bhopal households across residential, commercial and industrial localities, monitored in winter, monsoon and summer with a DustTrak aerosol monitor (270 samples). Residential homes carried higher PM_1-PM_{10} in most seasons; industrial areas peaked in summer; 258 of 270 samples were classed unhealthy, and morbidity prevalence was highest in industrial areas. No concentrations, effect estimates or photometer correction are given in the abstract; DustTrak readings are uncorrected optical mass and overestimate in humid, biomass-smoke-rich Indian homes. Morbidity is self-reported and not linked to measured exposure. Relevance: indirect - indoor optical photometry without gravimetric correction, the same error budget as low-cost indoor sensors.

doi: 10.21203/rs.3.rs-11610717/v1

Subramanian chamber size-resolved emission rates PM oxidative potential six indoor sources 2026 Environ Sci Technol

Size- and Time-Resolved Emission Rates of Particulate Matter Mass, Chemical Constituents, and Oxidative Potential from Indoor Sources

Chamber tests of two incenses, candle, toaster, humidifier and cigarette. Emission was mostly unimodal and concentrated below 0.25 um, where mass-normalised DTT OP was highest; composition and OPm varied with size but little with time. 37 +/- 3% of emitted PM_{10} mass and OP is deposition-capable, 45 +/- 2% of it alveolar. For equal PM_{10} mass and OP, these indoor sources give an estimated 76 +/- 10% higher alveolar dose than ambient PM. Six sources in one chamber; the dose comparison is modelled, not measured. Relevance: direct - optical sensors undercount below ~0.3 um, so indoor nodes systematically miss the fraction carrying most OP.

doi: 10.1021/acs.est.6c08882

Wang campus indoor air $PM_{2.5}$ phthalate esters gas-particle partitioning 2026 J Hazard Mater

Cross-media partitioning, sources, and health risks of phthalate esters in campus indoor air and $PM_{2.5}$.

Paired gas-phase and $PM_{2.5}$ samples from classrooms, dormitories, canteens and courier stations, spring and summer 2025, 15 phthalates quantified. Totals ran 1.11e2 to 1.53e3 ng/m3 in air and 3.14e1 to 2.94e2 ng/m3 in $PM_{2.5}$, with lower-molecular-weight species shifting to the gas phase in summer. Partitioning tracked KOA, and a dynamic partitioning framework beat the instantaneous-equilibrium model. Ingestion of $PM_{2.5}$-bound phthalates dominated exposure, DEHP drove carcinogenic risk, and DBP and DiBP ranked first for control. No CI. Limitation: one campus, two seasons, modelled rather than measured intake. Relevance: moderate - the same microgram of $PM_{2.5}$ carries a different phthalate burden by room, which mass-only monitoring cannot see.

doi: 10.1016/j.jhazmat.2026.144031

Zulian urinary chromium nickel welders TIG metalworkers Friuli biomonitoring 2026 Occup Med (Lond)

Biological chromium and nickel monitoring in metalworkers: welders versus non-welders.

245 male metalworkers, Friuli Venezia Giulia (179 welders, 66 non-welders), end-of-week urine by ICP-MS. Welders had higher median Cr (0.75 vs 0.56 ug/l, P=0.020) and Ni (1.70 vs 1.10 ug/l, P=0.0001). 50.0% exceeded the Cr limit of 0.7 ug/l. Welding vs non-welding OR 1.71 (0.96-3.06) for exceeding the Cr limit. Age and smoking were unrelated. Limitation: no airborne fume measurement, so job title stands in for exposure, non-welders share the shop air, and the group contrast is correspondingly compressed; Cr speciation not reported. Relevance: low for ambient sensing - a biomarker reference for a workplace aerosol that task-level PM sensors could stratify.

doi: 10.1093/occmed/kqag126

Burden, policy & mitigation 12 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (12 OF 12 RECORDS IN THIS SUBTOPIC)

Ali Somalia ecological time-series $PM_{2.5}$ urbanization under-five mortality 1990-2023 Research Square

Assessing Temporal Trends and Ecological Associations Between $PM_{2.5}$ Exposure, Urbanization, and Under-Five Mortality in Somalia, 1990-2023: An Ecological Time-Series Analysis

Preprint. 34 annual national observations, Somalia 1990-2023, Theil-Sen trends and first-difference regression with Newey-West errors. $PM_{2.5}$ rose 0.090 $\mu g\,m^{-3}$ per year (0.042-0.143), urbanisation 0.661 points per year, and under-five mortality fell 2.30 per 1,000 per year. Year-on-year changes in $PM_{2.5}$ were unrelated to under-five mortality (b 1.28, -8.81 to 11.36). With n=34 national points, conflict, drought and aid dominate mortality, and annual national $PM_{2.5}$ for Somalia is itself a satellite-model product with little ground truth; the analysis has no power for an effect of the expected size. Relevance: weak - illustrates that national ecological series cannot detect PM effects, the opposite of what dense local monitoring provides.

doi: 10.21203/rs.3.rs-11210072/v1

Aretz sick leave long-term fine particulate German health claims 13-year 2026 Environ Int

Assessing the health and economic burden of sick leave associated with long-term fine particulate matter exposure: a 13-year analysis using German health claims data

METADATA ONLY. Metadata only: Elsevier day-0 deposit, no abstract in Crossref, Europe PMC or Semantic Scholar; no finding asserted. The title describes a 13-year analysis of German health claims linking long-term fine PM to sick leave and its economic burden. Limitation: exposure source, increment, cohort size and estimate are all unknown until the abstract deposits. Relevance: to be assessed - claims-linked sick leave is a working-age endpoint rarely paired with PM, and this is the first record to retry on the next weekly abstract sweep.

doi: 10.1016/j.envint.2026.110586

Dyer urban burden of disease 825 European cities joint $PM_{2.5}$ NO_2 green space 2026 Eur J Public Health

Urban burden of disease estimation for policy making: health impacts of environmental stressors

BACKFILL (Crossref deposit 26 October 2026; congress abstract). Comparative risk assessment across 825 European cities, three reporting periods 2015-2024. Exceeding WHO guidelines gave 213,889 deaths (95% CI 89,224-344,223) jointly attributable to $PM_{2.5}$ and NO_2 in 2018, reported as 76,833 fewer than summing single-pollutant estimates ($PM_{2.5}$ 145,747; NO_2 143,646; the printed figures actually differ by 75,504). Limitation: abstract only; the joint method is not specified, the interval is wide, and attribution inherits the concentration-response function and the exposure model with no measured validation stated. Relevance: moderate - summing pollutant burdens overstates by about a quarter here, and city-scale burden accounting is where dense sensor fields could test the modelled surface.

doi: 10.1093/eurpub/ckag169.841

Hauck USC plasma-enhanced electrostatic precipitator self-cleaning electrode diesel PM 2026 ACS Omega

Plasma-Enhanced Electrostatic Precipitator (ESP) with Self-Cleaning Electrode for Scalable Remediation of Particulate Matter in Large Diesel Engines.

Nanosecond-pulsed (20 kV, 20 ns) plasma-enhanced ESP, with ion densities about six orders above DC corona, removed >90% of diesel PM at 1% of engine power; a porous stainless-steel collector regenerated by Joule heating kept >89% removal over four cycles and >200 h. Target is megawatt-class marine and locomotive engines without acceptable after-treatment. Efficiency is mass-based in the abstract; size-resolved removal for ultrafine particles - the hardest for ESPs - and regeneration emissions are not given. Relevance: minimal for sensing - port and rail-yard networks would be the field test of whether such controls change fenceline PM.

doi: 10.1021/acsomega.6c06682

Hess low-level $PM_{2.5}$ mortality Native American Medicare geographic attenuation 2026 Int J Environ Res Public Health

Untangling Impacts of Socioeconomic Position, Chronic Disease, and Low-Level $PM_{2.5}$ Exposure on Mortality Among Native American Medicare Beneficiaries

BACKFILL (entry date 4 May 2026; Amass). Case-control within 1,713,399 traditional Medicare beneficiaries in the lowest $PM_{2.5}$ decile, 2015-2016, with PCA summaries of area SEP and chronic disease. Fully adjusted OR per $\mu g\,m^{-3}$: Native American 1.12 (1.06-1.18), non-Native American 1.01 (1.001-1.02). Sequential adjustment attributed most attenuation in Native American beneficiaries to state-level clustering, less to SEP and disease. Limitation: the abstract gives no pre-adjustment coefficient, zip-level exposure over a narrow range makes a per-$\mu g\,m^{-3}$ slope fragile, and state effects may absorb real exposure contrast. Relevance: moderate - low-level gradients are where sparse monitoring error is largest.

doi: 10.3390/ijerph23040464

Laban scoping review household backup generators emissions health Africa 2026 BMC Public Health

Health effects associated with emissions from household generators used for backup power in Africa: a scoping review.

PubMed and Google Scholar scoping review; nine studies met inclusion, and only ~2% of screened studies addressed health outcomes directly. Generator emissions carried $PM_{2.5}$, PM_{10}, NOx and SO_2 at high levels; reported links to respiratory symptoms, CO poisoning, and possibly cancer and hypertension rest on self-report, modelling or indirect evidence. No study established a causal link. Two databases, English only. Relevance: direct - generator plumes are intermittent, near-field and household-specific, a source that only distributed indoor/outdoor low-cost sensing can capture at the needed resolution in Nigeria, Ghana or South Africa.

doi: 10.1186/s12889-026-28946-1

Madej Poland 379 counties spatial Mundlak GAM mortality inequality $PM_{2.5}$ 2010-2025 medRxiv

Place-based structural disadvantage and mortality inequality across 379 Polish counties, 2010-2025: a spatial Mundlak analysis

Preprint. 379 harmonised Polish counties, 2010-2025, spatial Mundlak GAMs separating between-county from within-county associations. Pre-COVID median between-county slope per SD: +30.2 deaths per 100,000 for $PM_{2.5}$, against -26.1 for higher education and +29.4 for low education; these gradients persisted across windows, while within-county terms were unstable and sometimes reversed in pandemic-inclusive models. Leave-one-year-out validation showed weak temporal transportability, and the $PM_{2.5}$ contribution contrast shrank after COVID-19. The between-county PM term is cross-sectional and collinear with Silesian coal-mining deprivation; no uncertainty on the slope is given in the abstract. Relevance: indirect - $PM_{2.5}$ matches education in explaining Poland's mortality geography at ecological scale.

doi: 10.64898/2026.11.02.26367044

Ni GBD 2021 otitis media macroeconomic burden value of lost welfare PM secondhand smoke J Int Med Res

The global macroeconomic burden of otitis media and its modifiable risk factors.

GBD 2021 DALYs for otitis media in 183 countries converted to value of lost welfare against PPP-adjusted GDP. Total burden was US$13.82 billion (0.009% of world GDP), highest in sub-Saharan Africa (0.016%) and Somalia (>0.042%). Of three modifiable risks (secondhand smoke, low birth weight/short gestation, particulate matter), secondhand smoke contributed most, US$0.15 billion; the PM-attributable share is not stated in the abstract. The analysis inherits GBD exposure surfaces and risk curves without independent uncertainty, and value-of-welfare metrics scale with income. Relevance: weak - PM is a minor attributed risk for this endpoint; no new exposure information.

doi: 10.1177/03000605261504275

Sangkham Upper Northern Thailand $PM_{2.5}$ hazard quotient biomass burning 2021-2023 Toxicol Rep

Spatial assessment of PM(2.5) exposure and health risks in Upper Northern Thailand: Implications for public health.

Eight Upper Northern Thai provinces, 2021-2023: regional mean $PM_{2.5}$ 27.57 +/- 23.40 $\mu g\,m^{-3}$ (8.40-78.64). Hazard quotients rose from wet to dry (biomass-burning) season: 1.19 to 4.31 against WHO, 0.50 to 1.80 against US NAAQS, 0.24 to 0.86 against Thai NEB. A hazard quotient is the concentration divided by a guideline, so these numbers restate exceedance rather than estimate risk, and the choice of benchmark alone moves the conclusion across 1. Station coverage in mountain valleys, where burning smoke pools, is not described. Relevance: indirect - northern Thailand is a region where sparse reference stations and strong valley gradients make a dense low-cost network the obvious complement.

doi: 10.1016/j.toxrep.2026.102360

Shao greening Chinese cities air pollution deaths 25-year scenario health impact 2026 Environ Int

Greening Chinese cities could prevent air pollution—related deaths: A 25-year analysis and scenario-based health impact assessment

METADATA ONLY. Metadata only; no abstract in Crossref at issue time, so no finding is asserted. The title describes a 25-year analysis of urban greening in Chinese cities and scenario-based estimates of air-pollution-related deaths avoidable by further greening. Read against the 6 November Korean urban-forest isotope study, which found vegetation lowered $PM_{2.5}$ only in the growing season. To be assessed when the abstract deposits: pollutant(s), the greening-to-concentration function and whether it is causal or cross-sectional, and the concentration-response function used, since greening scenarios usually borrow risk functions rather than observe deaths. Relevance: possible, unassessed.

doi: 10.1016/j.envint.2026.110595

Sternbach Beijing Clean Heating Policy difference-in-differences blood pressure 2026 Environ Health Perspect

China's Clean Heating Policy and Effects on Blood Pressure in Rural Beijing Adults

1,003 adults >40 y in 50 rural Beijing villages enrolled before treatment; 20 villages switched from coal to electric or gas heating during follow-up. Two-way fixed-effects difference-in-differences: brachial systolic -1.4 mmHg (-3.3 to 0.5), central systolic -1.5 (-3.3 to 0.4), brachial diastolic -1.6 (-2.9 to -0.3), central diastolic -1.6 (-2.9 to -0.3). Effects varied by treatment year, larger in early-treated villages. The exposure is policy assignment, not measured PM, and indoor temperature rises alongside; the abstract cannot separate the two pathways. Relevance: indirect - the clearest causal design in the household-energy literature; personal PM monitoring in the parent study is what would assign the mechanism.

doi: 10.1021/ehp.6c00053

US coal-fired power plant proximity Area Deprivation Index cross-sectional 2026 JAMA Netw Open

Distribution of Coal Power, Pollution, and Poverty.

2023 Neighborhood Atlas and census block groups: 2,783,004 people (0.83% of the US) live within 3 miles of a coal plant, disproportionately in the most deprived ADI decile and more often American Indian/Alaska Native. The most advantaged decile had OR 0.04 (0.00-0.81) of living within 3 miles versus the most deprived - a CI spanning near-zero that signals sparse cells. Proximity stands in for exposure; no PM, metal or stack-emission data, and plant size and wind direction are ignored. Relevance: indirect - a siting map for fenceline nodes, not an exposure estimate.

doi: 10.1001/jamanetworkopen.2026.42335

Other clinical endpoints — 6 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (6 OF 6 RECORDS IN THIS SUBTOPIC)

Ammons NIH-AARP $PM_{2.5}$ NO_2 ovarian endometrial cancer cohort 2026 J Expo Sci Environ Epidemiol

Long-term outdoor air pollution and risk of ovarian and endometrial cancers in a large prospective cohort

140,457 women in NIH-AARP, enrolled 1995-96 and followed to 2018, with 3,088 endometrial and 1,582 ovarian cancers. No association with either cancer overall. $PM_{2.5}$ and endometrial cancer appeared in Atlanta (HR 1.70, 1.16-2.48) and Pennsylvania (HR 1.15, 1.02-1.29) per 5 $\mu g\,m^{-3}$, with a weaker NO_2 signal in the same areas; ovarian associations in Detroit were positive but imprecise (HR 1.42, 0.92-2.18). Limitation: these are state- and city-stratified subgroups of a null overall result, so the multiplicity is severe and the authors frame them as suggestive. Relevance: moderate - the geographic heterogeneity is attributed to composition, which a mass-only exposure surface cannot represent. BACKFILL, entry 15 June 2026.

doi: 10.1038/s41370-026-00901-7

Kim district carcinogen emissions thyroid cancer Korea Bayesian PM_{10} null 2026 J Prev Med Public Health

Association Between District-level Aggregate Carcinogen Emissions and Thyroid Cancer Incidence in Korea: An Ecological Study Using Bayesian Spatial Models.

229 Korean districts, thyroid cancer 2014-2018, Bayesian hierarchical Poisson with mutual adjustment. Aggregate IARC Group 1/2A/2B release indicator RR 1.04 (95% CrI 1.01-1.06) per SD overall, 1.04 (1.02-1.06) in women, 1.02 (0.99-1.06) in men. PM_{10}, NO_2, SO_2, radon and nuclear-plant proximity showed no association after urban-rural adjustment; CO was inverse (0.96, 0.93-0.99); population density 1.13 (1.06-1.21). Limitation: ecological, and thyroid cancer incidence in Korea is strongly shaped by screening and detection intensity; authors call it hypothesis-generating. PM_{10} estimate not reported in the abstract. Relevance: low - a coarse-metric null at district scale, informative only in showing that urbanicity adjustment absorbs the PM signal.

doi: 10.3961/jpmph.26.726

Madsen systematic review environmental risk factors acute myeloid leukemia GRADE 2026 Blood Rev

Environmental risk factors for acute myeloid leukemia in adults, a systematic review.

Systematic review with risk-of-bias and GRADE assessment, 6,685 records screened, 15 studies from 11 populations included; no meta-analysis. Seven studies were high risk of bias. Of those not at high risk, associations appeared in four, for air pollution ($PM_{2.5}$, black carbon, secondary inorganic aerosol, ammonium, NOx, nitrate) and farm residency. Certainty was low for air pollution and very low for residential benzene, chlorinated water, radiation, industry proximity and power lines. Four studies supporting the PM signal is thin, and AML is rare, so exposure misclassification biases toward the null. Relevance: indirect - PM remains the best-supported environmental AML exposure, at low certainty.

doi: 10.1016/j.blre.2026.101443

Smith sickle cell disease outdoor air pollution scoping review ED visits 2026 Curr Environ Health Rep

The Effects of Outdoor Air Pollution on Health Among People Living with Sickle Cell Disease: A Scoping Review.

2,246 records screened, 1,454 abstracts, 30 full texts, 12 studies included, all with children in the population and half set in the United States. Associations with all-cause ED visits were consistently positive for CO, NO_2, $PM_{2.5}$ and PM_{10}, but only three reached statistical significance. The authors stop short of pooling, citing marked methodological heterogeneity across designs, exposure windows and outcome definitions. Limitation: a 12-study base, a literature search closing at the end of 2024, and no quantitative synthesis, so the direction is all that is on offer. Relevance: low for sensing, but it names a susceptible group whose crises are plausibly triggered on hours-to-days timescales, which is the timescale a dense network resolves and a regulatory network does not.

doi: 10.1007/s40572-026-00557-3

Xiao air pollution kidney disease mortality subtypes China case-crossover 2026 Chin Med J (Engl)

Air pollution exposure and death from different subtypes of kidney diseases: A nationwide, individual-level, case-crossover study in China.

466,543 kidney deaths, China 2013-2019, time-stratified case-crossover with conditional logistic regression. Per IQR: $PM_{2.5}$ lag 0-2 OR 1.0195 (1.0122-1.0269); coarse $PM_{2.5}$-10 lag 0-3 OR 1.0240 (1.0158-1.0323); O_3 +2.05% (1.05-3.07); NO_2 lag 0-3 +3.62% (2.47-4.78). Renal failure and glomerular deaths responded to all four pollutants; tubulointerstitial deaths to O_3 only; urolithiasis null. Limitation: the abstract does not give the IQR widths, so 'NO_2 strongest' is not a per-mass comparison, the lag differs by pollutant, and renal failure on a death certificate is often a proximate cause. Relevance: moderate - coarse-mode signal comparable to $PM_{2.5}$ per IQR, in a size fraction low-cost optical sensors undercount.

doi: 10.1097/cm9.0000000000004318

Yang US claims case-control $PM_{2.5}$ sinonasal malignant neoplasm 2026 JAMA Otolaryngol Head Neck Surg

Fine Particulate Matter Exposure and Sinonasal Malignant Neoplasm.

National commercial-claims case-control, 2007-2020: 3,600 SNM cases, 17,695 controls (~1:5). Per SD $PM_{2.5}$ (about 2.5 $\mu g\,m^{-3}$), adjusted OR 1.25 (1.05-1.49) for the 6-month mean, 1.31 (1.08-1.59) for 1 year and 1.36 (1.11-1.66) for 5 years, rising with averaging window. Exposure is assigned at metropolitan-area level, so a 1-km product is collapsed to an ecological contrast of ~0.2-0.3 $\mu g\,m^{-3}$ between groups; smoking and deprivation are regional, not individual, and occupational wood/leather dust - the dominant SNM cause - is unmeasured. Cases are 10 years older than controls. Relevance: indirect - a rare endpoint where residence-level exposure from dense sensing would remove the MSA misclassification.

doi: 10.1001/jamaoto.2026.3453

Pooled corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Calderon-Garciduenas ferrimagnetic nanoparticles olfactory bulb cortex Mexico City children	The olfactory bulb, prefrontal and anterior cingulate cortex are early pediatric targets for anthropogenic magnetic nanoparticles and their motion behavior under electromagnetic fields.	*Gac Med Mex*	Autopsy tissue analysis	Ferrimagnetic nanoparticles in brain tissue by magnetophoresis, TEM and EDX	Mexico (Mexico City)
Cardenas-Iniguez ABCD linked external data environment review first decade	Environmental influences using linked external data in ABCD: A review of the first decade.	*Dev Cogn Neurosci*	Narrative review	Geocoded residential air pollution among linked neighbourhood and policy variables; metric not specified	North America
Cho exercise under particulate matter exposure BDNF neurotrophic narrative review	Exercise Under Particulate Matter Exposure: Neurotrophic Responses and Functional Brain Outcomes	*Atmosphere*	Narrative review	Particulate or traffic-related pollution during exercise, heterogeneous metrics	Global / multi-region
Defrancesco lung-brain axis review $PM_{2.5}$ Alzheimer disease mechanisms	Mechanistic understanding of the role of the lung-brain axis in the pathogenesis of $PM_{2.5}$-induced Alzheimer's disease.	*Clin Sci (Lond)*	Mechanistic review	$PM_{2.5}$ (general; composition discussed as modifier)	Global
Huang UK Biobank CKD dementia multidomain risk factors air pollution null	Disease susceptibility and risk of dementia in patients with chronic kidney disease: a prospective cohort study based on UK Biobank.	*BMC Neurol*	Cohort	UK Biobank residential air pollution estimates (LUR), one of many candidate factors	UK
Zareba systematic review PM neurological health older adults 21 studies	Impact of Particulate Matter Exposure on Neurological Health in Older Adults: A Systematic Review.	*Curr Environ Health Rep*	Systematic review	Long- and short-term $PM_{2.5}$, PM_{10} and constituents as reported by included studies	Global / multi-region
Zundel Detroit adolescents $PM_{2.5}$ above 9 $\mu g/m^3$ fear conditioning amygdala	Outdoor Air Pollution Exposure Alters Fear Regulation and Fear-Related Neural Circuitry in Adolescence.	*Biol Psychiatry Glob Open Sci*	Cross-sectional experimental (fear conditioning + fMRI)	Past-month $PM_{2.5}$ from a spatiotemporal model, dichotomised at 9 $\mu g/m^3$	USA

First author	Focus	Journal	Design	Metric	Region
Cardiovascular & metabolic					
Alli ambient air pollution blood pressure schoolchildren Accra LUR	Urban ambient air pollution and blood pressure in schoolchildren in Accra, Ghana	*Sci Rep*	School-based cross-sectional	Home and school $PM_{2.5}$, BC, NO2 from spatiotemporal LUR (146 measurement sites)	Ghana (Accra)
He Beijing panel 152 young adults PNC5-560 gut microbiome pro-atherosclerotic biomarkers	Exposure to Size-Fractioned Ambient Particles Associated with Alterations in Gut Microbiome and Pro-Atherosclerotic Biomarkers in Young Adults.	*Environ Health Perspect*	Panel study (repeated measures)	Particle number concentration 5-560 nm from a fixed site, 7-day mean; PMF source split	China
Kulkayeva Almaty time-series $PM_{2.5}$ cardiovascular mortality null	Short-term effects of ambient $PM_{2.5}$ exposure on cardiovascular mortality in Almaty, Kazakhstan.	*Front Public Health*	Time-series (DLNM, quasi-Poisson)	Daily city $PM_{2.5}$, mean 36.9 +/- 33.3 $\mu g/m^3$, lags 0-7 d	Kazakhstan
Lin prospective cohort plasma proteomics $PM_{2.5}$ type 2 diabetes mediation	Plasma Proteomic Profiling Reveals Mediating Proteins and Preventive Targets Linking Ambient Air Pollution to Type 2 Diabetes Incidence: A Prospective Cohort Study	*Environ Pollut*	Prospective cohort + proteomic mediation	Long-term $PM_{2.5}$, NO2, NOx, SO2 by bilinear interpolation of gridded estimates, per IQR	Not stated
Saz-Lara CarVasCLiMa Castilla-La Mancha cohort protocol air pollution satellite arterial stiffness	Impact of environmental factors on cardiovascular health in Castilla-La Mancha: the CarVasCLiMa cohort study protocol.	*BMJ Open*	Study protocol	Air pollution and climate from high-resolution satellite data with local monitoring stations	Spain
Shi CFPS compound dry-hot extremes $PM_{2.5}$ joint incident CVD China	Joint effects of air pollution and compound dry-hot extremes on cardiovascular disease risk across adults: a cohort study in China.	*BMC Public Health*	Prospective cohort	$PM_{2.5}$ with PM1, PM_{10}, NO2 and O3; exposure source not stated in the abstract	China
Siddiqui narrative review AQI ASCVD South Asians	Impact of Air Quality Index on Atherosclerotic Cardiovascular Disease Risk in South Asians: A Narrative Review.	*Curr Atheroscler Rep*	Narrative review	AQI and constituent pollutants ($PM_{2.5}$, PM_{10}, NO2, CO, O3, NO)	South Asia
Singh German Diabetes Study long-term $PM_{2.5}$ clamp insulin sensitivity albuminuria T1D T2D	Associations of long-term exposure to air pollution with glucose homeostasis biomarkers and diabetes-related comorbidities in people with recent-onset type 1 and type 2 diabetes.	*Diabetologia*	Cross-sectional (cohort baseline)	Five-year mean $PM_{2.5}$, PM_{10} and NO2 at residential address	Germany
Xu CHARLS PM1 $PM_{2.5}$ PM_{10} cardiometabolic multimorbidity cohort	EXPRESS: Air pollution increases cardiometabolic diseases in China: A nationwide longitudinal cohort study.	*J Investig Med*	Prospective cohort	Long-term PM1, $PM_{2.5}$, PM_{10}, O3, NO2 (exposure model not stated in the abstract)	China
Reproductive & developmental					
Karimi PMOS polycystic ovary metabolic syndrome Arak Iran $PM_{2.5}$ PR IDW	Air pollution and polycystic ovary metabolic syndrome (PMOS) in a heavily industrialized Middle Eastern city: A cross-sectional study.	*Chemosphere*	Cross-sectional study	365-day residential $PM_{2.5}$ and PM_{10} by inverse-distance weighting from three stations	Iran (Arak)
Kim prenatal $PM_{2.5}$ pediatric inguinal hernia hydrocele kriging cohort	Prenatal Fine Particulate Matter ($PM_{2.5}$) Exposure and the Risk of Pediatric Inguinal Hernia or Hydrocele: A Retrospective Cohort Study	*J Clin Med*	Retrospective cohort	Monthly residential $PM_{2.5}$ by kriging from fixed monitoring stations	South Korea
Marques Generation R multi-cohort LUR $PM_{2.5}$ NO2 epigenetic age birth to adolescence	Outdoor Air pollution Exposure and Epigenetic Age from birth to adolescence - a multi-cohort study.	*Environ Res*	Multi-cohort meta-analysis	Harmonised LUR NO2, NOx, PM_{10}, $PM_{2.5}$, black carbon at home address; pregnancy, recent and lifetime windows	Europe
Qiu stage-specific models pregnancy loss recurrent abortion $PM_{2.5}$ O3 random forest	Stage-Specific Models for Predicting Subsequent Pregnancy Loss in Women with Recurrent Spontaneous Abortion.	*Int J Womens Health*	Prediction model development (prospective cohort)	Pregnancy-course ambient $PM_{2.5}$ and O3 metrics as predictors	China
Tao Peking ART women 1-km $PM_{2.5}$ follicular fluid metals	Associations of ambient fine particulate matter exposure with metal and metalloid concentrations in follicular fluid.	*J Hazard Mater*	Cross-sectional (clinical cohort)	12-month residential $PM_{2.5}$ at 1-km resolution, per IQR	China

First author	Focus	Journal	Design	Metric	Region
Zhang paternal $PM_{2.5}$ concentrated ambient particles multigenerational lipid DNA methylation mice	Paternal PM(2.5) exposure is associated with lipid metabolic dysfunction across generations and altered sperm and hepatic DNA methylation in mice.	*Environ Pollut*	Animal multigenerational inhalation experiment	Filtered air, unfiltered ambient air, or concentrated ambient $PM_{2.5}$, 6 h/day, 5 days/week, 22 weeks	East Asia (laboratory, China)
Respiratory & allergic					
Alimu Kashi Xinjiang ExWAS elastic-net air pollution meteorology pulmonary tuberculosis	Associations between air pollution, meteorology, and pulmonary tuberculosis risk in Kashi, China.	*BMC Pulm Med*	Exposome-wide association	Ambient PM_{10}, $PM_{2.5}$, NO2 and meteorology; source and averaging window not stated	China
Drummond mini-review climate exposures paediatric lung-function trajectories	Climate change as a modifier of paediatric lung-function trajectories.	*Eur Respir Rev*	Narrative review	Air pollution and wildfire smoke among climate-sensitive exposures; no PM metric specified	Global
Espinoza-Guillen Lima case-crossover $PM_{2.5}$ asthma ED road proximity	Ambient fine particulate matter and asthma emergency visits in Lima, Peru: Does Residential Proximity to Major Roads Matter?	*Environ Res*	Case-crossover	Daily $PM_{2.5}$ from a 1x1 km multi-source model at geocoded residence	Peru
Hao Shanghai case-crossover lab-confirmed influenza six pollutants	Association between short-term air pollutant exposure and influenza incidence in Shanghai, China: A case-crossover study	*Int J Biometeorol*	Time-stratified case-crossover	$PM_{2.5}$, PM_{10}, SO2, NO2, O3, CO interpolated by inverse distance weighting to individual address	China
Huang All of Us case-crossover wildfire smoke $PM_{2.5}$ adult asthma exacerbation	Asthma Exacerbations Following Wildfire Smoke $PM_{2.5}$ Exposure Among U.S. Adults	*GeoHealth*	Time-stratified case-crossover	Wildfire-smoke $PM_{2.5}$ at residence, lags 0-7 d, April-August 2006-2019	USA
Jean-Luc Kinshasa PurpleAir PA-II-FLEX asthma cross-sectional ISAAC	Abiotic and socioenvironmental factors of $PM_{2.5}$ pollution associated with asthma in a dense urban road network environment in Kinshasa, D.R. Congo: a cross-sectional study	*Environ Geochem Health*	Cross-sectional study	PurpleAir PA-II-FLEX sensors with a locally derived correction for Kinshasa aerosol	Africa (Kinshasa, DR Congo)
Kutzinski CAPNETZ air pollution sputum pathogen spectrum community-acquired pneumonia	Air pollution is associated with a distinct pathogen spectrum in community-acquired pneumonia: a prospective multicentre cohort study.	*Pneumonia (Nathan)*	Prospective cohort	City-level $PM_{2.5}$, NO2, SO2, CO up to one month before enrolment	Germany
Li Shandong ICU cardiac surgery $PM_{2.5}$ hospital-acquired pneumonia DLNM	Association between short-term fine particulate matter exposure and hospital-acquired pneumonia risk in critically ill post-cardiac surgery patients: a 10-year cohort study using distributed lag non-linear models.	*Front Public Health*	Cohort	City-level daily $PM_{2.5}$ from public monitoring data (mean 59 $\mu g/m^3$)	China
Pulczinski EPA controlled human exposure 3 consecutive days 35 $\mu g/m^3$ $PM_{2.5}$ FEV1 exercise	Repeated Exposure to Near-Ambient PM(2.5) Decreases Pulmonary Function: Controlled Human Exposure Study.	*Environ Health Perspect*	Randomised double-blind crossover controlled exposure	Concentrated ambient $PM_{2.5}$ at 35 $\mu g/m^3$, 4 h/day with intermittent exercise, 3 consecutive days vs clean air	USA
Ruvuna ambient $PM_{2.5}$ NO2 plasma proteomics SPIROMICS SomaScan congress	Associations of ambient air pollution and protein expression among individuals from the Subpopulations and intermediate outcome measures in COPD study	*Eur Respir J (congress abstract)*	Cross-sectional (cohort baseline)	Modelled prior-year average $PM_{2.5}$ and NO2	United States (SPIROMICS)
Salimbene Turin mild COPD lung function $PM_{2.5}$ microenvironment MANCOVA	Ambient, Indoor Air Pollution Exposures and Mild Chronic Obstructive Pulmonary Disease Trajectories in Non-Hospitalized Volunteers in Turin, Italy	*Atmosphere*	Prospective cohort	Residential-location ambient $PM_{2.5}$ and PM_{10} with indoor microenvironmental profiling	Europe (Turin, Italy)
Tsogtbayar Ulaanbaatar time-series PM1 $PM_{2.5}$ under-5 pneumonia outpatient	Associations between air pollution and pediatric pneumonia incidence in Ulaanbaatar, Mongolia: age and sex susceptibility.	*J Thorac Dis*	Time-series (quasi-Poisson GAM)	Daily city PM1, $PM_{2.5}$, PM_{10}, SO2, NO2, O3 from monitors, per IQR	Mongolia

First author	Focus	Journal	Design	Metric	Region
Ubuane Lagos protocol paediatric asthma control household $PM_{2.5}$ CO cotinine	Association of asthma control with household particulate matter less than 2.5 microns, carbon monoxide and second-hand-smoking among children with asthma in Lagos, Nigeria: protocol of an ongoing observational cross-sectional study	*Research Square*	Protocol (cross-sectional)	Objectively measured household $PM_{2.5}$ and CO; urinary cotinine for second-hand smoke	Nigeria
Yang UK Biobank life-course $PM_{2.5}$ constituents COPD asthma LRTI lung cancer	Life-Course Exposure to PM(2.5) Constituents and Major Respiratory Diseases: Constituent-Specific Associations and Candidate Immune-Metabolic Signatures in a Large Prospective Cohort.	*Lung*	Prospective cohort	$PM_{2.5}$, SO4, NO3, NH4, OC and EC from CMAQ 1930-2010 linked to birth-to-baseline residential histories	UK
Zhang Shanxi 11-city time-series pollutant mixtures influenza vaccination	Influenza Incidence in Relation to Air Pollution Mixtures and the Modifying Role of Vaccination in Shanxi Province, China.	*Environ Res*	Multi-city time-series + random-effects meta-analysis	Daily city-mean $PM_{2.5}$, PM_{10}, NO2, SO2, CO, O3 from regulatory monitors	China
Zhu meta-analysis air pollutants allergic rhinitis children $PM_{2.5}$ TRAP mold	Association between exposure to gaseous air pollutants and the risk of allergic rhinitis in children and adolescents: A meta-analysis.	*Int J Pediatr Otorhinolaryngol*	Comprehensive review	$PM_{2.5}$, PM_{10}, traffic-related pollution, NO2, O3, SO2, CO and mold as reported by included studies	Global
Mechanistic toxicology					
Ba rat intratracheal $PM_{2.5}$ IL-17A TXNIP NLRP3 cardiac fibrosis	Neutralization of IL-17A Ameliorates $PM_{2.5}$-Induced Cardiac Inflammation and Fibrosis in Rats by Suppressing the TXNIP/NLRP3 Signaling Pathway.	*Kaohsiung J Med Sci*	Animal model	Collected $PM_{2.5}$ by intratracheal instillation; dose not stated in the abstract	China
Guardado lipocalin-2 knockout organic dust extract lung inflammation recovery	The role of lipocalin-2 in lung homeostasis, inflammation response, and recovery following organic dust exposure.	*PLoS One*	Animal knockout experiment	Aqueous extracts of swine-confinement organic dust, intranasal, 15 days	North America (laboratory)
Jain air pollution redox proteomics oxidative modification review	Air pollution leaves a redox-proteomic imprint: From shared inflammation to disease-specific vulnerability.	*Redox Biol*	Metadata only	Air pollution including PM (per title)	Global / multi-region
Jeon rat $PM_{2.5}$ inhalation sperm abnormality ACE2 vs renal insufficiency	Particulate matter inhalation induces male reproductive dysfunction comparable to renal failure: insights from a rat model.	*Transl Androl Urol*	In vivo rat exposure	$PM_{2.5}$ at 30 mg/kg/day, 3 h/day for 10 days (route described as inhalation)	South Korea
Li poultry-house $PM_{2.5}$-derived material APEC sequential exposure chick cell	Inflammatory and redox responses associated with sequential exposure to poultry-house $PM_{2.5}$-derived material and avian pathogenic Escherichia coli in chicken models.	*Poult Sci*	Animal + in vitro	Autoclaved eluate of broiler-house $PM_{2.5}$-fraction material, uncharacterized after sterilization	China
Liu BALF fibroblast $PM_{2.5}$ house dust mite FAM45A allergic asthma	FAM45A-Mediated Persistent Memory-Like Fibroblast Reprogramming Drives $PM_{2.5}$-Exacerbated Allergic Asthma.	*Environ Health Perspect*	Ex vivo primary cells + CRISPR + humanised mouse	$PM_{2.5}$ co-stimulation with house dust mite in vitro; residential $PM_{2.5}$ exposure duration in a clinical cohort	China
Mali Rutgers wildfire PM0.1 mouse ovarian hyperandrogenism AhR	Wildfire-Emitted Particulate Matter Induces Ovarian Hyperandrogenism through Aryl Hydrocarbon Receptor Activation	*Environ Health Perspect*	In vivo mouse + 3D follicle culture	Lab-synthesised and June-2023 Canadian wildfire PM0.1, intratracheal instillation and in vitro	USA
Mishra ultrafine PM black carbon polystyrene nanoplastic mitochondrial stress PBMC	Conserved and particle-specific mitochondrial stress responses drive immunotoxicity of ultrafine particulate matter, black carbon, and polystyrene nanoplastics.	*Front Immunol*	In vitro cell exposure	Ultrafine PM, black carbon and polystyrene nanoplastics on human PBMCs under matched conditions	South Asia (laboratory)
Pitchai pulmonary nanomaterial exposure inflammatory lipid mediators sex disease	Pulmonary exposure to nanomaterials dysregulates inflammatory lipid mediators in a sex- and disease-dependent manner	*Part Fibre Toxicol*	Metadata only: no abstract in Crossref at issue time	Pulmonary nanomaterial exposure, materials not recoverable	Not stated

First author	Focus	Journal	Design	Metric	Region
Qureshi narrative review bioaccessibility particle-bound metals PAHs tiered framework	Bioaccessibility and Toxicity of Particle-Bound Metals and Polycyclic Aromatic Hydrocarbons: Sources, Respiratory Fate, Mechanisms, and Risk Assessment	*Research Square (preprint)*	Narrative / critical review	Particle-bound metals and PAHs; PM mass, size, speciation and oxidative potential	Global
Stegmann atmospheric aging biomass burning particle toxicity BEAS-2B PHOTO-LAC	Atmospheric Aging Modifies the Toxicological Profile of Biomass Burning Particles in Epithelial Lung Cells.	*Environ Sci Technol*	In vitro cell exposure	Fresh vs chamber-aged biomass burning particle extracts, flaming and smoldering	Germany (laboratory)
Sun Waite review molecular mechanisms respirable silica coal dust asbestos lung disease	Molecular mechanisms of dust-induced lung disease: exploring the role of physicochemical composition and cellular pathways.	*Part Fibre Toxicol*	Narrative review	Respirable crystalline silica, coal mine dust, asbestos; reduced-iron minerals	Global
Wang naphthalene SOA aerosol-organoid exposure platform glioma ALDH1A3 ferroptosis	Secondary organic aerosol from polycyclic aromatic hydrocarbon modulates glioma proliferation.	*J Hazard Mater*	In vitro + organoid + murine	Photochemically generated naphthalene SOA delivered by an aerosol-organoid exposure platform	China
Zhang SMR multi-omics $PM_{2.5}$-annotated genes diabetic nephropathy TNF NOS2 methylation	A multi-omics Mendelian randomization framework identifies $PM_{2.5}$-annotated candidate regulatory genes and methylation-linked pathways in diabetic nephropathy.	*Clin Epigenetics*	Two-sample Mendelian randomisation	No exposure measurement; genes annotated as $PM_{2.5}$-responsive from curated databases, plus in vitro $PM_{2.5}$ exposure	Global / multi-region
Occupational & indoor					
Apak office-to-education studio zone CO2 $PM_{2.5}$ PM_{10} TVOC perception Istanbul	Indoor Air Quality and Occupant Perception in an Office Building Adapted for Educational Use: A Zone-Based Case Study	*Indoor Air*	Field measurement	Indoor $PM_{2.5}$ and PM_{10} by sequential point measurement; instrument not named in the abstract	Turkiye
Arts letter health-based nanomaterial guidance value 4 $\mu g/m^3$ occupational critique	Letter to the editor regarding the paper entitled "Health-based nanomaterial guidance value (HNGV) for occupational exposure to spheroidal biodurable engineered nanomaterials of relatively low substance-specific toxicity".	*NanoImpact*	Letter to the editor	Proposed 4 $\mu g/m^3$ 8-h TWA occupational limit for biodurable engineered nanomaterials	Global / multi-region
Bittner Grieshop rural Malawi domestic biofuel emissions characterisation	Characterizing emissions from diverse domestic biofuel uses in rural Malawi	*Environ Sci Atmos*	Metadata only	Emissions from household biofuel use (firewood and other biofuels) in field conditions (per title and abstract stub)	Malawi
Borgini e-cigarette real-life indoor PM AirBeam3 bar traffic walk pilot	Investigating the Real-Life Indoor Air Quality Impact of E-Cigarette Vaping.	*Multidiscip Respir Med*	Pilot field measurement	PM1, $PM_{2.5}$, PM_{10} by portable optical particle counter (AirBeam3)	Italy
Gupta Maharashtra opencast coal mine $PM_{2.5}$ composition oxidative potential	Comprehensive assessment of $PM_{2.5}$ exposures around a regional mining site in India: integrating chemical composition, oxidative potential and health risk	*Environ Geochem Health*	Filter sampling + composition	$PM_{2.5}$ mass, EC/OC, metals, PAHs, OP (AA and DTT) at six sites	India
Hu Shanghai catering cooking fume emission multi-criteria assessment	Comprehensive Assessment of Cooking Fumes Pollution Emission Levels in Selected Shanghai Catering Businesses: An Exploratory Multi-Criteria Assessment	*Atmospheric Environment*	Metadata only: no abstract in Crossref at issue time	Cooking-fume emissions from commercial catering premises, species not recoverable	East Asia (Shanghai, China)
Khoshakhlagh occupational airborne pollutants hematological outcomes systematic review	Occupational Airborne Pollutant Exposure and Hematological Outcomes in Workplace Settings: A Systematic Review.	*J Appl Toxicol*	Systematic review	Occupational PM, VOCs and heavy metals across 55 studies	Global / multi-region
Kim New Mexico Wyoming coal miners self-reported dust SARS-CoV-2 seroconversion	Novel Occupational and Environmental Risk Factors for SARS-CoV-2 Infection Among Coal Miners: Longitudinal Cohort Study.	*JMIR Form Res*	Cohort	Self-reported occupational dust, residential wood burning and shared transport frequency	USA

First author	Focus	Journal	Design	Metric	Region
Krevanko UW cannabis processing local exhaust ventilation inhalable PM terpenes	Pilot study of a local exhaust ventilation system to control particle and terpene concentrations during cannabis processing.	*Front Public Health*	Paired intervention pilot	Inhalable PM and total terpenes with vs without LEV (MERV-13 + activated charcoal)	USA
Kumar firefighter occupational exposure neurotoxicity narrative review	Occupational exposure and neurotoxic health effects in firefighters: toxicological mechanisms and health risks.	*Toxicol Mech Methods*	Narrative review	Fireground combustion products: fine PM, PAHs, CO, HCN	Global
Laferriere VA burn pits registry headache phenotypes mixed methods	Veteran deployed burn pits exposure: headache phenotypes, trajectories, and outcomes: a mixed-methods evaluation.	*Front Neurol*	Mixed methods (registry survey + interviews)	Self-reported open burn pit smoke exposure (AH&OBP Registry)	USA
Lin Taiwan metro track maintenance respirable dust metals noise WBGT task-based	Exposure to dust, metals, noise, and heat stress among metro track maintenance workers: task-based onsite measurements in two urban metro systems.	*Ann Work Expo Health*	Field measurement	Personal respirable dust and particulate metals (gravimetric) plus task-tagged real-time particle counts	Taiwan
Molina Rueda AirPen facility-wide personal PM noise formaldehyde one shift	Facility-wide assessment of occupational exposure to dust, noise, and formaldehyde	*J Expo Sci Environ Epidemiol*	Field measurement	Total particulate by personal AirPen sampler with on-board sensors, plus formaldehyde and A-weighted noise	North America
Myers NIOSH CWHSP pneumoconiosis dusty job type underground coal 1991-2024	Assessment of pneumoconiosis by dusty job type among underground coal miners participating in the Coal Workers' Health Surveillance Program,1991-2024.	*Front Public Health*	Cross-sectional (occupational)	Respirable coal mine dust, proxied by occupation and region; no dust measurement in the analysis	North America
Nie personal air purification device multi-scale modelling fan speed occupant exposure	Multi-scale modeling and fan speed optimization of a personal air-purification device for occupant exposure control in polluted indoor environments	*Build Environ*	Metadata only (no abstract)	Not stated (indoor pollutant class unavailable)	Not stated
Olsen LC-APCI-MS resin acids personal softwood sawmill dust Norway	A Versatile Method for Determination of Resin Acids in Occupational Softwood Sawmill Dust Samples by Liquid Chromatography-Atmospheric Pressure Chemical Ionization-Mass Spectrometry.	*J Sep Sci*	Analytical method validation	Resin acids in personal wood dust samples	Europe
Rana Bhopal indoor DustTrak PM1 $PM_{2.5}$ PM_{10} TVOC 30 households seasonal	Estimation of Emerging Indoor Air Pollutants and Associated Health Effects on Humans in the Capital City of Central India	*Research Square*	Cross-sectional environmental survey	Indoor PM1, $PM_{2.5}$, PM4, PM_{10}, total PM by DustTrak photometer; TVOC	India
Subramanian chamber size-resolved emission rates PM oxidative potential six indoor sources	Size- and Time-Resolved Emission Rates of Particulate Matter Mass, Chemical Constituents, and Oxidative Potential from Indoor Sources	*Environ Sci Technol*	Chamber emission characterisation	Size-resolved PM mass, composition and DTT oxidative potential, <0.25 to 10 um	USA
Wang campus indoor air $PM_{2.5}$ phthalate esters gas-particle partitioning	Cross-media partitioning, sources, and health risks of phthalate esters in campus indoor air and $PM_{2.5}$.	*J Hazard Mater*	Paired indoor sampling campaign	$PM_{2.5}$-bound and gas-phase phthalate esters in four campus microenvironments	East Asia (Hebei, China)
Zulian urinary chromium nickel welders TIG metalworkers Friuli biomonitoring	Biological chromium and nickel monitoring in metalworkers: welders versus non-welders.	*Occup Med (Lond)*	Cross-sectional study	Welding fume by job role; urinary Cr and Ni by ICP-MS (no airborne measure)	Italy
Sensing, forecasting & instrumentation					
Agrawal Bihar low-cost sensor network ML exposure inequity short-term mortality	Unveiling hidden air pollution exposure and impacts with low-cost sensor network-based frameworks.	*Nat Commun*	Network deployment + statistical model	$PM_{2.5}$ from a dense statewide low-cost sensor network, gap-filled by machine learning	India

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Ainslie ECCC calibrated ventilation index Lagrangian stochastic Richardson number	The scientific development of Environment and Climate Change Canada's new ventilation index.	*J Air Waste Manag Assoc*	Model development and operational evaluation	Ventilation defined as inversely proportional to modelled ground-level concentration from a surface-layer source	North America (Canada)
Angita Jhansi semi-arid campus automatic sensor $PM_{2.5}$ PM_{10} annual cycle	Year-Round Seasonal Variation of $PM_{2.5}$ and PM_{10} at a Semi-Arid Institutional Campus Site in Jhansi, Uttar Pradesh, India	*Research Square*	Field measurement	Continuous $PM_{2.5}$ and PM_{10} from an automatic sensor, one campus site, 2023	India
Chen TRACER Houston micropulse lidar aerosol vertical profile boundary layer parameterisation	Aerosol vertical distributions shaped by boundary layer dynamics in a coastal urban environment: insights from the TRACER campaign	*Atmos Chem Phys*	Remote sensing / lidar retrieval	Aerosol concentration profiles from micropulse lidar constrained by surface aerosol measurements	USA
Glienke HOLODROPS holographic droplet sensor event-based PIV characterization	Development and characterization of an advanced holographic instrument for atmospheric research	*Atmos Meas Tech*	Instrument development and laboratory validation	Digital holography of droplets and ice crystals above 8.8 um, with event-based particle image velocimetry	Not stated (laboratory and one outdoor test)
Goyal PurpleAir PA-II kappa-Kohler Lorentz-Lorenz random forest moisture artefact	A physics-aware digital twin for correcting moisture-induced material artifacts in $PM_{2.5}$ sensors	*Discover Electronics*	Calibration model comparison	$PM_{2.5}$ from a pair of PurpleAir PA-II units co-located with an EPA FEM monitor (site 220330009)	North America
Kloppenborg ARMIE SPS30 node laboratory DustTrak SMPS APS source-out calibration	Laboratory Evaluation of ARMIE, a Portable SPS30-Based Low-Cost Sensor Node for $PM_{2.5}$ Monitoring	*Sensors*	Instrument evaluation	$PM_{2.5}$ from Sensirion SPS30 in a portable node, against DustTrak and combined SMPS+APS	Europe
Li Nanjing jute cellulose filter $PM_{2.5}$ sampling recovery oxidative potential	An All-Natural Cellulose Fiber Filter Developed for High Recovery Sampling and In Vitro Toxicity Assessment of PM(2.5).	*Environ Res*	Laboratory sampler-media development	Filter-collected $PM_{2.5}$ for recovery, oxidative potential and cytotoxicity	China
Liu CNN-xLSTM attention SHAP $PM_{2.5}$ forecasting Beijing hourly	Hybrid Network Models Based on SHAP Interpretability for $PM_{2.5}$ Concentration Prediction: A Case Study of Beijing	*Atmosphere*	Machine-learning model development	Hourly $PM_{2.5}$ with co-pollutants and meteorology as predictors	East Asia (Beijing, China)
Mao wavelet attention LSTM indoor aerosol concentration prediction	A Wavelet-Based Attention LSTM Framework for Aerosol Concentration Prediction in Indoor Environments	*J Aerosol Sci*	Metadata only: no abstract in Crossref at issue time	Indoor aerosol concentration	Not stated
Park micro-3D-printed liquid-metal coil inductive ferrous aerosol impactor sensor	Micro-3D-printed integrated system with liquid-metal coils for simultaneous size-selective sampling and concentration sensing of respirable ferrous metal aerosols.	*Lab Chip*	Instrument development and laboratory validation	Respirable ferrous aerosol, 1-5 and >5 um fractions, inductive detection	South Korea (laboratory)
Peletidou fluorescence Raman depolarization lidar aerosol typing Thessaloniki	Vertical profiling and aerosol typing using fluorescence Raman and depolarization lidar measurements over Thessaloniki, Greece	*Atmos Chem Phys*	Multi-year lidar measurement campaign	Fluorescence backscatter, spectral fluorescence capacity and particle depolarization ratio, vertically resolved	Europe (Thessaloniki, Greece)
Rogers VUV-AMS siloxane radical mass calibration source aerosol mass spectrometry	Siloxane-Radical Species: Chemical Characteristics and Use for Mass Calibration in Aerosol Mass Spectrometry.	*J Phys Chem A*	Laboratory instrument development	Soot particles carrying vaporised PDMS siloxane oligomers	Chamber / laboratory
Sharon IoT vape detection school $PM_{2.5}$ PM_{10} 37 locations 18 months	Smart Vape Detection in Schools for Mitigating Student E-Cigarette Use	*Int J Environ Res Public Health*	Longitudinal case study of a sensor deployment	$PM_{2.5}$ and PM_{10} at one-minute resolution, threshold alerting	Australia
Soderblom Laplace-domain in situ network calibration Helsinki Antwerp SensEURCity	Laplace domain dynamic calibration in city-wide sensor networks	*Meas Sci Technol*	Calibration model comparison	PM from deployed urban low-cost sensor networks, alongside temperature, RH and pressure	Europe

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Sotiropoulou smart bench low-cost sensors Attica exposure AirQ+ FAIRCITY	Smart city developments: Pilot case study of smart benches usage for impact assessment due to exposure to air pollution in the urban environment.	*J Environ Health Sci Eng*	Pilot field measurement with health-impact modelling	$PM_{2.5}$, PM_{10}, NO2 and O3 from an IoT low-cost smart-bench station	Europe (Attica, Greece)
Suter LDSA lung-deposited surface area summer measurements spatiotemporal modelling	Measurements and exploratory spatiotemporal modelling of LDSA in summer	*Atmos Environ*	Metadata only: no abstract in Crossref at issue time	Lung-deposited surface area (LDSA)	Not stated
Tirpitz UV aerosol absorption retrieval actinic flux FIREX-AQ plume age	Retrieval of ultraviolet aerosol absorption from radiation measurements in young wildfire plumes	*Atmos Chem Phys*	Airborne retrieval algorithm development	Imaginary refractive index k(308-440 nm) of biomass-burning aerosol from airborne actinic flux	North America (western US wildfires)
Wang $PM_{2.5}$ PM_{10} forecasting dual-data MWS-optimised ensemble learning	$PM_{2.5}$ and PM_{10} Concentration Forecasting Using a Dual-Data Strategy and MWS-Optimized Ensemble Learning	*Atmos Environ*	Metadata only: no abstract in Crossref at issue time	$PM_{2.5}$ and PM_{10}	Not stated
Wu Heyuan attention-LSTM SHAP hourly $PM_{2.5}$ O3 forecasting persistence 2019-2023	Baseline-relative predictability of hourly $PM_{2.5}$ and O3 in a low-emission South Chinese city: an attention-LSTM with feature attribution	*Research Square*	Deep-learning forecasting	Hourly $PM_{2.5}$ and O3 from two urban monitoring stations, 2019-2023	China
Yadav CUPI-G low-cost sensor assimilation WRF-Chem Delhi stubble burning	Can Low-Cost Sensor Networks Improve Analysis of Extreme Pollution Episodes in Delhi During Stubble Burning?	*Research Square* (*preprint*)	Data assimilation experiment	Twice-daily $PM_{2.5}$ from 33 CUPI-G low-cost sensors assimilated into WRF-Chem by 3D-Var	India (Punjab, Haryana, Delhi)
Exposure assessment & modelling					
Alruqimi Alpine region multi-source machine learning daily NO2 PM_{10} $PM_{2.5}$ estimation	Multi-source machine learning for daily surface NO2, PM_{10}, and $PM_{2.5}$ estimation in Alpine region	*Atmos Environ*	Metadata only	Daily surface $PM_{2.5}$, PM_{10} and NO2 by multi-source machine learning (per title)	Europe
Andino intra-urban $PM_{2.5}$ PM_{10} meteorological GAM arid San Juan Argentina	Intra-urban variability and meteorological drivers of PM(2.5) and PM(10) concentrations in an arid Argentine city.	*Environ Monit Assess*	Multi-site monitoring campaign	$PM_{2.5}$ and PM_{10} with co-located meteorology across residential, industrial and commercial zones	South America (San Juan, Argentina)
Ayittey Accra FRM gravimetric $PM_{2.5}$ three sites e-waste 2021-2022	Persistent urban $PM_{2.5}$ pollution in Accra, Ghana: spatiotemporal patterns, meteorological and anthropogenic drivers, and associated health risks	*Environ Monit Assess*	Long-term field measurement campaign	24-h gravimetric $PM_{2.5}$ by USEPA FRM samplers, >300 samples at three sites	Ghana
Bonilla California CMAQ wildfire $PM_{2.5}$ evaluation 2008-2018 random forest diagnostic	Evaluating CMAQ Representation of Wildfire $PM_{2.5}$ in California (2008-2018): Regime-Dependent Performance and Nonlinear Error Structure.	*Res Sq*	Chemical transport model evaluation	Daily CMAQ total and fire-only $PM_{2.5}$ paired with monitor observations, June-October	USA
Bu Anmyeon-do GAW November 2025 Asian dust haze wildfire episodes aerosol optics	Episode-Dependent Variations in Atmospheric Composition at the Anmyeon-Do Global Atmosphere Watch Station in November 2025	*Preprints.org*	Measurement campaign (GAW station)	PM_{10}, scattering/absorption coefficients, Angstrom exponent, ion composition, reactive gases	South Korea
Cheng pedestrian-level $PM_{2.5}$ crowd mobility 3D urban form CFD historic neighbourhood	Dynamic Crowd Mobility, Socio-Economic, and Pedestrian-Level PM(2.5) in a High-Density Historic Neighbourhood: Transient Effects and Synergistic Mechanisms of 3D Urban Form.	*Environ Pollut*	Fine-grid monitoring campaign with CFD	Pedestrian-level $PM_{2.5}$ logged at 2-second intervals over 90 grid cells of 120 x 150 m, with CFD at 1.5 m	East Asia (China)
Gantina remote sensing CO HC NOx soot heavy-duty trucks buses Indonesia	Emission of CO, HC, NOx, and Soot from HDV Trucks and Buses using Remote Sensing in Indonesia	*Aerosol Air Qual Res*	Metadata only: no abstract in Crossref at issue time	Soot from on-road heavy-duty vehicles by remote sensing	Indonesia

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Gonzalez organosulfates CMAQ hygroscopicity kappa US summers 2013-2022	Modeling Decadal Organosulfates in Particulate Matter to Improve Estimations of Aerosol Hygroscopicity in the United States	*Environ Sci Technol*	Chemical-transport modelling	Isoprene-derived organosulfates and aerosol hygroscopicity from CMAQ, evaluated on IMPROVE	United States
Hansen spatial mediation GWR SVC $PM_{2.5}$ income cardiovascular simulation	Spatially varying effect estimates of mediation in the presence of spatial auto-correlation	*medRxiv (preprint)*	Methodological simulation with applied case study	$PM_{2.5}$ as a spatially varying mediator between income and cardiovascular disease	North America (United States)
Hussain Bergin Pakistan smog episodes data-driven source attribution	Attribution of Air Pollution Sources During Smog Episodes Across Major Urban Centers of Pakistan: A Data-Driven Approach	*Atmos Environ*	Metadata only: no abstract in Crossref at issue time	Smog-episode air pollution, likely $PM_{2.5}$-dominated	Pakistan
Jaelani Surabaya Benowo landfill 15-min $PM_{2.5}$ 2020-2024	The spatial impact assessment of landfill sites on air quality in Surabaya: a diurnal and seasonal perspective	*Environ Sci: Atmos*	Statistical analysis of monitoring data	15-minute $PM_{2.5}$ at the Benowo landfill, 2020-2024	Indonesia
Jin personal monitoring inhaled dose mobility physical activity $PM_{2.5}$	Leveraging real-time personal monitoring of air pollution and physical activity levels, and location tracking for mobility- and inhaled dose-based assessments of exposure and health risk	*Environ Monit Assess*	Personal monitoring campaign	Real-time personal $PM_{2.5}$ with GPS location and activity-dependent inhalation rates	North America (United States)
Jiratampradab daily $PM_{2.5}$ probability distribution upper northern Thailand log-normal	Probability Distribution Analysis of Daily $PM_{2.5}$ Concentration at Five Stations in Upper Northern Thailand	*Atmosphere*	Statistical analysis of monitoring data	Daily $PM_{2.5}$ at five monitoring stations, June 2023-May 2026	Thailand (upper north)
Khan Pakistan winter smog $PM_{2.5}$-associated antimicrobial resistance genes qPCR	$PM_{2.5}$-Associated Variation in Airborne Antimicrobial Resistance Gene Abundance During Winter Smog Episodes in Urban Pakistan	*Research Square*	Bioaerosol sampling campaign (preprint)	Filter $PM_{2.5}$ in Lahore, Islamabad, Okara, Nov-Dec 2022 and 2023; ARGs by qPCR	Pakistan
Kim CMAQ-UNIPAR secondary organic aerosol South Korea ASIA-AQ	Prediction of Secondary Organic Aerosol Formation in South Korea using CMAQ-UNIPAR during the ASIA-AQ Campaign	*Atmos Environ*	Metadata only: no abstract in Crossref at issue time	Secondary organic aerosol	South Korea
Kim Korea urban forest $PM_{2.5}$ stable isotopes receptor model seasonal	Seasonal variation in urban PM(2.5) sources and the modulating role of urban forests.	*Environ Pollut*	Field campaign	Filter $PM_{2.5}$ with d13C and d15N isotopes, transects from expressway and industrial edge into forest	South Korea
Ku North China Plain new particle formation GEOS-Chem-APM spring campaign	Investigations of New Particle Formation Mechanisms in North China Plain during the Spring Campaign Based on Observations and Simulations	*Environ Sci Technol*	Chemical transport model evaluation	Particle number size distributions; nucleation and growth rates	China
Kutarna SWAPIT Toronto winter NO2 O3 $PM_{2.5}$ AQHI representativeness	Insights into urban concentrations and variability of NO(2), O(3), and PM(2.5) from the study of winter air pollution in Toronto (SWAPIT).	*J Air Waste Manag Assoc*	Multi-site monitoring campaign	$PM_{2.5}$ with NO2 and O3 at eight continuous sites, plus NAPS history from 1997	Canada (Toronto)
Li Wuhan impervious surface morphology land surface temperature $PM_{2.5}$ 2015-2024	Combined effects of impervious surface morphology on land surface temperature and $PM_{2.5}$ pollution: Evidence from long-term observations (2015–2024) in Wuhan, China	*Atmos Pollut Res*	Metadata only: no abstract in Crossref at issue time	$PM_{2.5}$ (observation source not available)	China
Liu refined SOA hygroscopicity aerosol liquid water CCN uncertainty	Refined SOA hygroscopicity reduces climate-relevant uncertainties in aerosol liquid water and CCN predictions	*npj Clim Atmos Sci*	Metadata only: no abstract in Crossref at issue time	Secondary organic aerosol hygroscopicity, aerosol liquid water and CCN	Global / multi-region
Lv MERRA-2 CAMS aerosol components AERONET GRASP intercomparison trends	Global Aerosol Component Distributions and Long-Term Trends in MERRA-2 and CAMS: Intercomparison and Evaluation with AERONET GRASP/Component Retrievals	*Atmospheric Environment*	Metadata only: no abstract in Crossref at issue time	Speciated aerosol components in two global reanalyses, evaluated against AERONET GRASP retrievals	Global / multi-region

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Mohsin California disaggregated AA NHPI wildfire-specific $PM_{2.5}$ disparities	Race disparities in total PM(2.5) and wildfire-specific PM(2.5) exposure among disaggregated Asian American, Native Hawaiian, and Pacific Islander groups in California.	*J Clim Change Health*	Cross-sectional population-weighted exposure	Census-tract total and wildfire-specific $PM_{2.5}$, 2008-2012, from EPA AQS-derived estimates	USA
Park dual-metric duration-aware rainfall regime surface particulate matter	A dual-metric, duration-aware assessment of how rainfall regimes control surface particulate matter	*Atmospheric Research*	Metadata only: no abstract in Crossref at issue time	Surface particulate matter under rainfall regimes, fractions not recoverable	East Asia (Republic of Korea, by author affiliation)
Qu karst city $PM_{2.5}$ rare earth elements heavy REE enrichment diurnal	Heavy rare earth enrichment and diurnal source invariance of $PM_{2.5}$-bound rare earth elements in a karst city, southwestern China	*Atmospheric Environment*	Metadata only: no abstract in Crossref at issue time	$PM_{2.5}$-bound rare earth elements, diurnally resolved	East Asia (southwestern China)
Rehman Lahore GTWR satellite $PM_{2.5}$ urban expansion climate 2015-2024	Urban expansion, seasonal climate and fine particulate matter in Lahore, Pakistan: a spatiotemporal analysis of regional environmental change	*Research Square*	Ecological / spatial	Satellite-derived $PM_{2.5}$, 2015-2024	Pakistan
Sathish Kanpur Mumbai nine-stage size-resolved aerosol chemistry ICRP deposited dose	Size-resolved measurements matter for chemistry and health: paired inland-coastal size-resolved aerosol chemistry and the inadequacy of bulk PM(2.5) dose estimates.	*Environ Res*	Filter sampling + composition	Nine-stage impactor PM (<0.43 to >9 um), ions and OC/EC; ISORROPIA-II pH and water	India
Schwarz Helmholtz letter ultrafine particle health studies forest for the trees	Letter to the Editor: Health effect studies of ultrafine air pollution-Are we missing the forest for the trees?	*Environ Res*	Letter to the editor	Ultrafine particles	Global
Shankar meteorological drivers urban air quality weather-pollution interactions	Understanding the Meteorological Drivers of Urban Air Quality: Weather-Air Pollution Interactions	*Atmos Environ*	Metadata only: no abstract in Crossref at issue time	Urban air pollutants (unspecified)	Not stated
Sharma $PM_{2.5}$-bound water soluble organic carbon urban sources metadata only	Variability in sources and processes of $PM_{2.5}$-bound Water Soluble Organic Carbon over an urban location in the North-Western Himalayas	*Atmos Pollut Res*	Metadata only: no abstract in Crossref at issue time	$PM_{2.5}$-bound water-soluble organic carbon (from title)	Not stated
Tan Shenzhen PMF-ML firework burning $PM_{2.5}$ source apportionment	Apportioning Firework Burning and Routine Source Contributions to PM(2.5) in a Chinese Megacity Using Two PMF-Machine Learning Frameworks.	*Environ Res*	Source-apportionment method	Hourly $PM_{2.5}$ chemical constituents, Shenzhen, Jan-Mar 2023-2025	China
Wang Kwan Shenzhen 110,000 mobile signalling year-long temporal exposure averaging problem	The temporal exposure averaging problem (TEAP).	*Environ Sci Ecotechnol*	Methodological simulation on mobility data	Mobility-weighted exposure to multiple air pollutants and greenspace from year-long mobile signalling	China
Wayland discussion data fusion air pollution exposure modeling metadata only	Discussion of "Data fusion for air pollution exposure modeling".	*J Air Waste Manag Assoc*	Metadata only: invited discussion of a data-fusion paper	Not recoverable	Not stated
Wei dust storm events Northeast China GEOS-Chem PM measurements metadata only	Dust storm events in Northeast China by GEOS-Chem modeling and PM measurements	*Atmos Pollut Res*	Metadata only: no abstract in Crossref at issue time	PM from measurement and GEOS-Chem (from title)	China (Northeast)
Wu airborne plastics plastic-associated chemicals $PM_{2.5}$ Singapore PMF I168O	Fingerprints of Airborne Plastics in Urban $PM_{2.5}$: Applications and Implications	*Environ Sci Technol*	Source apportionment	Nine plastic-associated chemicals in 240 daily $PM_{2.5}$ filters, 2018-2020	Singapore
Wu penguin colony chlorophyll-a mediation PM1 $PM_{2.5}$ PM_{10} Ardley Island	Phytoplankton-mediated coupling between penguin colonies and atmospheric particulate matter in Antarctic coastal waters.	*Mar Pollut Bull*	Field campaign / shipborne	Along-track PM1, $PM_{2.5}$ and PM_{10} with seawater chlorophyll-a and nutrients; instrument not named in the abstract	Antarctica
Yao Harbin early-spring heavy pollution episodes natural anthropogenic drivers	Natural and anthropogenic drivers of heavy pollution episodes in early spring in Harbin, Northeast China	*Atmos Pollut Res*	Metadata only: no abstract in Crossref at issue time	Heavy-pollution episodes ($PM_{2.5}$-dominated in Harbin early spring)	China

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Yin brake-wear particle emissions thermal history ultrafine number optical PHEV	Mechanical and thermal controls on brake-wear particle emissions and physicochemical properties in a plug-in hybrid vehicle: A comparison of two brake-pad formulations.	*J Hazard Mater*	Source-isolated emission test	Brake-wear PM mass, number-size distribution, 630 nm optical properties, composition and morphology	East Asia (Tianjin, China)
Young CSU San Joaquin Valley agricultural community personal indoor $PM_{2.5}$ PM_{10}	Particulate Matter Pollution in an Agricultural Setting: A Community-Engaged Research Study.	*Environments*	Personal and indoor exposure campaign	24-h gravimetric personal and indoor $PM_{2.5}$ and PM_{10}, three seasons	USA
Yu Jeju Lunar New Year 2026 fireworks long-range transport K+ aerosol	Fireworks-Influenced Aerosols on Jeju Island, Korea, During the 2026 Lunar New Year	*Preprints.org*	Measurement campaign + back-trajectory	Filter $PM_{2.5}$ and PM_{10}, water-soluble ions, carbonaceous aerosol, size distributions	South Korea
Zhang source-resolved $PM_{2.5}$ metals Bayesian inversion California hazard index	Source-Resolved $PM_{2.5}$ Metals Emissions, Exposure, and Health Risks in California Using an Observation-Constrained Hybrid Model	*Research Square (preprint)*	Chemical-transport modelling with Bayesian inversion	23 $PM_{2.5}$-bound metals across 11 source sectors, source-oriented CTM constrained by monitor data	North America (California)
Burden, policy & mitigation					
Ali Somalia ecological time-series $PM_{2.5}$ urbanization under-five mortality 1990-2023	Assessing Temporal Trends and Ecological Associations Between $PM_{2.5}$ Exposure, Urbanization, and Under-Five Mortality in Somalia, 1990-2023: An Ecological Time-Series Analysis	*Research Square*	Ecological	National annual population-weighted $PM_{2.5}$ (modelled)	Somalia
Aretz sick leave long-term fine particulate German health claims 13-year	Assessing the health and economic burden of sick leave associated with long-term fine particulate matter exposure: a 13-year analysis using German health claims data	*Environ Int*	Metadata only (no abstract)	Long-term fine PM (exposure source not available)	Germany
Dyer urban burden of disease 825 European cities joint $PM_{2.5}$ NO2 green space	Urban burden of disease estimation for policy making: health impacts of environmental stressors	*Eur J Public Health*	Comparative risk assessment	Modelled $PM_{2.5}$ and NO2, joint exposure estimation, against WHO guidelines	Europe (825 cities)
Hauck USC plasma-enhanced electrostatic precipitator self-cleaning electrode diesel PM	Plasma-Enhanced Electrostatic Precipitator (ESP) with Self-Cleaning Electrode for Scalable Remediation of Particulate Matter in Large Diesel Engines.	*ACS Omega*	Engineering control device test	Diesel exhaust particulate matter	USA
Hess low-level $PM_{2.5}$ mortality Native American Medicare geographic attenuation	Untangling Impacts of Socioeconomic Position, Chronic Disease, and Low-Level $PM_{2.5}$ Exposure on Mortality Among Native American Medicare Beneficiaries	*Int J Environ Res Public Health*	Case-control	Long-term low-level $PM_{2.5}$ (lowest-decile zip codes; exposure source not stated)	United States (Medicare)
Laban scoping review household backup generators emissions health Africa	Health effects associated with emissions from household generators used for backup power in Africa: a scoping review.	*BMC Public Health*	Scoping review	Petrol and diesel generator emissions: $PM_{2.5}$, PM_{10}, NOx, SO2, CO	Sub-Saharan Africa
Madej Poland 379 counties spatial Mundlak GAM mortality inequality $PM_{2.5}$ 2010-2025	Place-based structural disadvantage and mortality inequality across 379 Polish counties, 2010-2025: a spatial Mundlak analysis	*medRxiv*	Ecological panel	County-level $PM_{2.5}$ (source not stated in the abstract)	Poland
Ni GBD 2021 otitis media macroeconomic burden value of lost welfare PM second-hand smoke	The global macroeconomic burden of otitis media and its modifiable risk factors.	*J Int Med Res*	Burden estimation (GBD)	GBD 2021 particulate matter risk factor (ambient and household)	Global / multi-region
Sangkham Upper Northern Thailand $PM_{2.5}$ hazard quotient biomass burning 2021-2023	Spatial assessment of PM(2.5) exposure and health risks in Upper Northern Thailand: Implications for public health.	*Toxicol Rep*	Community cross-sectional + USEPA HQ	Monitoring-station $PM_{2.5}$ across eight provinces, 2021-2023	Thailand
Shao greening Chinese cities air pollution deaths 25-year scenario health impact	Greening Chinese cities could prevent air pollution—related deaths: A 25-year analysis and scenario-based health impact assessment	*Environ Int*	Metadata only: no abstract in Crossref at issue time	Air pollution (pollutant set not available)	China
Sternbach Beijing Clean Heating Policy difference-in-differences blood pressure	China's Clean Heating Policy and Effects on Blood Pressure in Rural Beijing Adults	*Environ Health Perspect*	Quasi-experimental difference-in-differences	Policy-driven coal-to-clean heating transition (household air pollution and indoor temperature as pathways)	China

First author	Focus	Journal	Design	Metric	Region
US coal-fired power plant proximity Area Deprivation Index cross-sectional	Distribution of Coal Power, Pollution, and Poverty.	*JAMA Netw Open*	Population-wide administrative linkage, cross-sectional	Proximity buffers (3, 10, 25, 50 miles) to operational coal-fired power plants; no measured PM	USA
Other clinical endpoints					
Ammons NIH-AARP $PM_{2.5}$ NO2 ovarian endometrial cancer cohort	Long-term outdoor air pollution and risk of ovarian and endometrial cancers in a large prospective cohort	*J Expo Sci Environ Epidemiol*	Prospective cohort	Residential $PM_{2.5}$ and NO2 from nationwide spatiotemporal models, time-varying 5-year averages	North America (United States)
Kim district carcinogen emissions thyroid cancer Korea Bayesian PM_{10} null	Association Between District-level Aggregate Carcinogen Emissions and Thyroid Cancer Incidence in Korea: An Ecological Study Using Bayesian Spatial Models.	*J Prev Med Public Health*	Observational - ecological	District PM_{10} as a covariate; primary exposure a carcinogen release indicator	South Korea
Madsen systematic review environmental risk factors acute myeloid leukemia GRADE	Environmental risk factors for acute myeloid leukemia in adults, a systematic review.	*Blood Rev*	Systematic review	Ambient $PM_{2.5}$, black carbon and secondary inorganic aerosol components among other environmental exposures	Global / multi-region
Smith sickle cell disease outdoor air pollution scoping review ED visits	The Effects of Outdoor Air Pollution on Health Among People Living with Sickle Cell Disease: A Scoping Review.	*Curr Environ Health Rep*	Scoping review	$PM_{2.5}$, PM_{10}, NO2 and CO as reported by the included studies	Global / multi-region
Xiao air pollution kidney disease mortality subtypes China case-crossover	Air pollution exposure and death from different subtypes of kidney diseases: A nationwide, individual-level, case-crossover study in China.	*Chin Med J (Engl)*	Time-stratified case-crossover	Modelled daily $PM_{2.5}$ and $PM_{2.5}$-10 at residential address, with O3 and NO2	China (nationwide)
Yang US claims case-control $PM_{2.5}$ sinonasal malignant neoplasm	Fine Particulate Matter Exposure and Sinonasal Malignant Neoplasm.	*JAMA Otolaryngol Head Neck Surg*	Case-control (registry)	$PM_{2.5}$ from NASA 1-km satellite-meteorology-land-use models, averaged to residential MSA over 6 months, 1 and 5 years	USA

Provenance and method

Period **1 – 7 November 2026**, Sunday to Saturday, **seven daily issues**, **151 records**, **151 unique DOIs**, no record without a DOI. This rollup is **aggregation only**: it re-reads `state/corpus/*.json`, `state/metrics.csv` and `state/trials.json` over the period and **queries no literature API**; every record was gated by `check_dois.py` at daily-issue time. Tier mix **20 A, 52 B, 79 C**. The 1–5 November dailies were built on runs of 2–6 November; the 6 and 7 November dailies and this rollup on the **10 November recovery run**. Effect estimates are transcribed verbatim from the daily issues on heterogeneous contrasts and are **not** meta-analysed. Abstracts remain the copyright of their respective publishers.